0001. 108.35 MHz can only be ;

a) an ILS frequency.
b) a VOR frequency.
c) an NDB frequency.
d) an ATC frequency.

Answer ;

a) an ILS frequency.

0002. 3D RNAV fixing gives you ;

a) 2D RNAV plus speed control.
b) 2D RNAV plus time guidance.
c) horizontal and vertical profile guidance.
d) horizontal , vertical profile and time guidance.

Answer ;

c) horizontal and vertical profile guidance.

0003. A 2-dimensional RNAV system has a capability in the ;

a) timing function.
b) horizontal and vertical planes.
c) horizontal plane.
d) vertical plane.

Answer ;

c) horizontal plane.

0004. A 3-D RNAV system has capability in ;

a) a horizontal plane and the vertical plane.
b) a horizontal plane and vertical plane an timing function.
c) a horizontal plane and cruise management system.
d) a horizontal plane and speed management system.

Answer ;

a) a horizontal plane and the vertical plane.

0005. A 3-dimensional RNAV system has capability in ;

a) the horizontal plane and a cruise management system.
b) the horizontal plane , in the vertical plane and a timing function.
c) the horizontal plane and in the vertical plane.
d) the horizontal plane and a speed management system.

Answer ;

c) the horizontal plane and in the vertical plane.

0006. A cumulonimbus cloud in the vicinity of an aeroplane can cause certain navigation systems to give false indications. This is particularly true of the ;

a) ADF.
b) DME.
c) VOR.
d) weather radar.

Answer ;

a) ADF.

0007. A DME is located at MSL. An aircraft passing vertically above the station at flight level FL 360 will obtain a DME range of approximately ;

a) 8 NM.
b) 7 NM.
c) 6 NM.
d) 11 NM.

FL 360 = 36000 feet ,
36.000 : 6080 = 5,921 ≈ 6 NM.

Answer ;

c) 6 NM.

0008. A DME station is located 1000 feet above MSL. An aircraft flying at FL 370 in ISA conditions which is 15 NM away from the DME station , will have a DME reading of ;

a) 15 NM.
b) 17 NM.
c) 16 NM.
d) 14 NM.

37.000 – 1000= 36.000 → 36.000 : 6080 = 5,921 ≈ 6 NM
(DME reading)^2 = $6^2 + 15^2$
(DME reading)^2 = 36 + 225 = 261
DME reading = $\sqrt{261}$ = 16,155 ≈ 16 NM.

Answer ;

c) 16 NM.

0009. A fix obtained by rho-rho navigation is based on information from two ;

a) NDBs
b) VORs
c) VDFs
d) DMEs

Answer ;
d) DMEs

0010. A FMS with only a multiple DME sensor operating shall have a position error , 95% probability , in a non precision approach equal or less than ;

a) 0.5 Nm.
b) 1 Nm.
c) 0.3 Nm.
d) 0.06 Nm.

Answer ;
c) 0.3 Nm.

0011. A frequency of 10 GHz is considered to be the optimum for use in an airborne weather radar system because ;

a) the larger water droplets will give good echoes.
b) static interference is minimised.
c) enables the aircraft to detect clear air turbulence.
d) greater detail can be obtained at the more distant ranges of the smaller water droplets.

Answer ;
a) the larger water droplets will give good echoes.

0012. A frequency of airborne weather radar is ;

a) 9375 MHz.
b) 93.75 MHz.
c) 9375 kHz.
d) 9375 GHz.

Answer ;
a) 9375 MHz.

0013. A ground radar transmitting at a PRF of 1200 pulses/second will have a maximum unambiguous range of approximately ;

a) 27 NM.
b) 270 NM.
c) 67 NM.
d) 135 NM.

Answer ;
c) 67 NM.

Teorik Mesafe = 300.000 km/s : (2 x PRF)
Teorik Mesafe = 300.000 : (2 x 1200)
Teorik Mesafe = 300.000 : 2400 = 125 km
125 : 1,852 = 67,49 ≈ 67 NM

0014. A locator ;

1. is a low powered beacon.
2. is a high powered beacon.
3. has a range of 10 - 25 NM.
4. has a range of 10 - 200 NM.

a) 2 and 3
b) 1 and 3
c) 2 and 4
d) 1 and 4

Answer ;
b) 1 and 3

0015. A locator beacon differs from an NDB with respect to ;

1. operational use.
2. transmission power.
3. presentation in the cockpit.
4. frequency band.

From the above stated differences the following numbers are correct ;

a) 1 and 2.
b) 2 , 3 and 4.
c) 1 and 4.
d) 1 , 2 and 3.

Answer ;
a) 1 and 2.

0016. A MLS without DME-P provides ;

- **a)** A category 3 approach.
- **b)** An approach with a curved path but not staged.
- **c)** A staged approach but not with a curved path.
- **d)** An ILS-like aproach.

Answer ;

d) An ILS-like aproach.

0017. A phantom station (as used in a 2D RNAV-system) is ;

- **a)** a waypoint defined by a radial and a DME-distance from a VOR/DME-station.
- **b)** an existing VOR/DME-station created in the memory of the Navigation Computer Unit of the RNAV-system.
- **c)** a waypoint defined by two DME-distances from two different VOR/DME-stations.
- **d)** a non-existing VOR/DME-station defined by two DME distances.

Answer ;

a) a waypoint defined by a radial and a DME-distance from a VOR/DME-station.

0018. A pilot flying an aircraft at FL 80 , tunes in a VOR which has an elevation of 313 m. Given ISA conditions , what is the maximum theoretical distance at which a pilot might expect to receive the VOR signals ?

- **a)** 100 NM
- **b)** 151 NM
- **c)** 180 NM
- **d)** 120 NM

Teorik olarak maksimum alış menzili (NM) = 1,25 x (√H1 +√H2)
1 metre (m) = 3,2808 feet (ft) , 313 m = 313 x 3,2808 = 1026,89 = 1027 ft
= 1,25 x (√8000 + √1027)
= 1,25 x (89,44 + 32,04) = 1,25 x 121,48 = 151,85 ≈ 151 NM

Answer ;

b) 151 NM

0019. A pilot is flying between two waypoints defined by suitably located VOR/DMEs. Equipped with a simple 2D RNAV system this pilot ;

- **a)** enters relative position between his aircraft and the VOR/DMEs on CDU to calculate the cross track error.
- **b)** reads VOR/DME bearing and distance on CDI or HSI to compute himself the cross track error.
- **c)** reads cross track error and the distance to go on CDI or HSI.
- **d)** must update any altitude change in RNAV system to have correct cross track error.

Answer ;

c) reads cross track error and the distance to go on CDI or HSI.

0020. A Primary radar operates on the principle of ;

- **a)** pulse technique.
- **b)** continuous wave transmission.
- **c)** transponder interrogation.
- **d)** phase comparison.

Answer ;

a) pulse technique.

0021. A radar facility transmitting at a Pulse Recurrence Frequency (PRF) of 1200 pulses/second will have a maximum unambiguous range of approximately ;

- **a)** 135 NM.
- **b)** 27 NM.
- **c)** 270 NM.
- **d)** 69 NM.

Teorik Mesafe = 300.000 km/s : (2 x PRF)
Teorik Mesafe = 300.000 : (2 x 1200)
Teorik Mesafe = 300.000 : 2400 = 125 km
125 : 1,852 = 67,49 ≈ 69 NM

Answer ;

d) 69 NM.

0022. A radio altimeter employing a continuous wave signal would have ;

- **a)** a directional aerial for transmission and an omni-directional aerial for reception.
- **b)** an omni-directional aerial for transmission and directional aerial for reception.
- **c)** a directional aerial for transmission and another one for reception.
- **d)** a directional aerial for both transmission and reception.

Answer ;

c) a directional aerial for transmission and another one for reception.

0023. A radio beacon has an operational range of 10 NM. By what factor should the transmitter power be increased in order to achieve an operational range of 20 NM ?

- **a)** Eight
- **b)** Four
- **c)** Six
- **d)** Two

Answer ;

b) Four

0024. A TCAS II equipped a/c will have mode S because ;

- **a)** true altitude is obtained from mode S.
- **b)** the data link is required to co-ordinate evasive manoeuvres.
- **c)** the data link is required to transmit relative position.
- **d)** mode S transmits a 3D position.

Answer ;

b) the data link is required to co-ordinate evasive manoeuvres.

0025. A typical frequency employed in Distance Measuring Equipment (DME) is ;

- **a)** 10 MHz
- **b)** 100 GHz
- **c)** 100 MHz
- **d)** 1000 MHz

Answer ;

d) 1000 MHz

0026. A VDF may be used ;

- **a)** in combination with radar to solve the 180º ambiguity.
- **b)** in emergency type situations when the aircraft is unable to transmit on VHF.
- **c)** to provide the ATC controller with bearings of aircraft in the absence of radar.
- **d)** in lieu of ILS for precision approach purposes.

Answer ;

c) to provide the ATC controller with bearings of aircraft in the absence of radar.

0027. A VOR and an ADF are co-located. In NIL wind conditions you cross the VOR radial of 240 on a heading of 360°(M). In the vicinity of the station you should read an ADF bearing of ;

- **a)** 240
- **b)** 120
- **c)** 060
- **d)** 300

Answer ;

c) 060

0028. A VOR and an NDB are co-located. An aircraft equipped with an RMI is flying away from the beacons on a radial of 090º through an area where magnetic variation is changing rapidly. Which statement is correct ?

- **a)** The VOR needle moves , the ADF needle does not.
- **b)** Both VOR and NDB needles move.
- **c)** Neither the VOR or the NDB needles move.
- **d)** The ADF needle moves , the VOR needle does not.

Answer ;

d) The ADF needle moves , the VOR needle does not.

0029. A VOR and an NDB are co-located. You cross the VOR radial of 240 on a heading of 360°(M). In the vicinity of the station you should read an ADF bearing of ;

- **a)** 060
- **b)** 120
- **c)** 300
- **d)** 240

Answer ;

a) 060

0030. A VOR and an NDB are located in the same position. Both the VOR- and the ADF-readings are displayed on the RMI. The aircraft is tracking away from the beacons along the 090 radial. The magnetic variation is changing rapidly. Which of the following is correct ?

- **a)** The direction of the ADF pointer will change , the direction of the VOR pointer will not change.
- **b)** The direction of the VOR pointer will change , the direction of the ADF pointer will not change.
- **c)** Both the direction of the ADF pointer and the direction of the VOR pointer will change.
- **d)** Neither the direction of the ADF pointer nor the direction of the VOR pointer will change.

Answer ;

a) The direction of the ADF pointer will change , the direction of the VOR pointer will not change.

0031. A VOR and DME are co-located. You want to identify the DME by listening to the callsign. Having heard the same callsign 4 times in 30 seconds the ;

- **a)** DME callsign is the one with the lower pitch that was broadcast several times.
- **b)** DME callsign was not transmitted , the distance information is sufficient proof of correct operation.
- **c)** VOR and DME callsigns were the same and broadcast with the same pitch.
- **d)** DME callsign is the one with the higher pitch that was broadcast only once.

Answer ;

d) DME callsign is the one with the higher pitch that was broadcast only once.

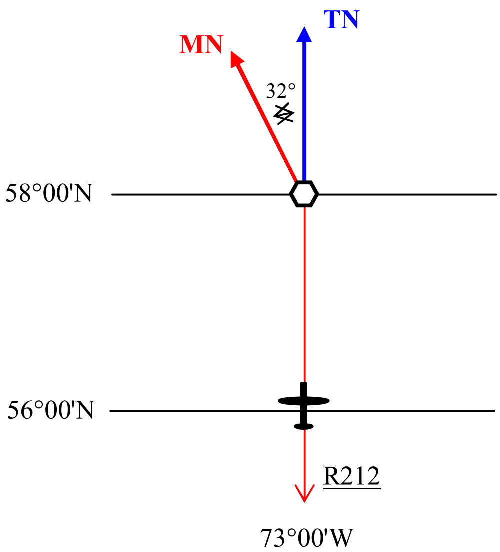
0032. A VOR is sited at position 58°00'N 073°00'W where the magnetic variation equals 32°W. An aircraft is located at position 56°00'N 073°00'W where the magnetic variation equals 28°W. The aircraft is on VOR radial ;

- **a)** 360
- **b)** 180
- **c)** 212
- **d)** 208

Answer ;

c) 212

Çözüm ;



Burada VOR radyali sorulduğu için sadece VOR'daki variation değerini hesaba katacağız. Uçak ve VOR aynı meridyen üzerinde , ve VOR'ın güneyinde VOR'a göre true bearing'i 180°T olduğuna göre ;

VOR radyal = 180° + Variation West
= 180° + 32°
= 212°

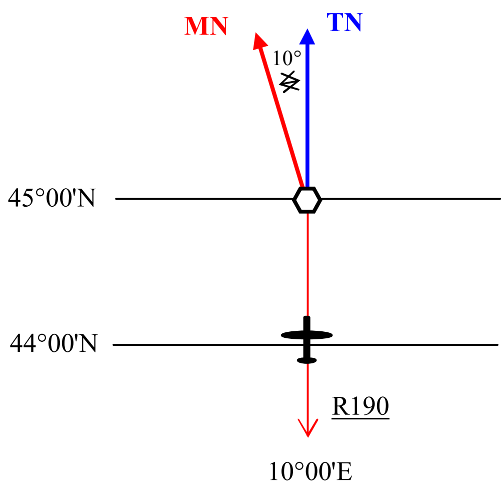
0033. A VOR is sited at position A (45°00'N , 010°00'E). An aircraft is located at position B (44°00'N , 010°00'E). Assuming that the magnetic variation at A is 10°W and at B is 15°W , the aircraft is on VOR radial ;

- **a)** 185°
- **b)** 195°
- **c)** 190°
- **d)** 180°

Answer ;

c) 190°

Çözüm ;



Burada VOR radyali sorulduğu için sadece VOR'daki variation değerini hesaba katacağız. Uçak ve VOR aynı meridyen üzerinde , ve VOR'ın güneyinde VOR'a göre true bearing'i 180°T olduğuna göre ;

VOR radyal = 180° + Variation West
= 180° + 10°
= 190°

0034. A VOR is situated at the far end of a runway on which an aircraft is making an ILS approach. Nav 1 is switched to the localiser frequency and Nav 2 to the VOR frequency. At the moment that the needle of # 1 indicator reaches the outer dot the deflection of the needle of # 2 indicator will be at ;

- **a)** the outer dot.
- **b)** approximately a quarter of the scale.
- **c)** approximately halfway the scale.
- **d)** approximately three quarters of the scale.

Answer ;

b) approximately a quarter of the scale.

0035. A weather radar , set to the 100 NM scale , shows a squall at 50 NM. By changing the scale to 50 NM , the return on the radar screen should ;

- **a)** decrease in area but not change in position on the screen.
- **b)** increase in area and move to the top of the screen.
- **c)** decrease in area and move to the top of the screen.
- **d)** increase in area and appear nearer to the bottom of the screen.

Answer ;

b) increase in area and move to the top of the screen.

0036. According to ICAO 8168 , what is regarded as the maximum safe deviation below the glide path during ILS approach ?

- **a)** Full scale deflection.
- **b)** Half scale deflection.
- **c)** Three quarter scale deflection.
- **d)** One quarter scale deflection.

Answer ;

b) Half scale deflection.

0037. According to ICAO Annex 10 a locator has a range of ;

- **a)** 75 to 150 NM.
- **b)** 75 to 250 NM.
- **c)** 7.5 to 15 NM.
- **d)** 10 to 25 NM.

Answer ;

d) 10 to 25 NM.

0038. According to ICAO Annex 10 , in which frequency band(s) does a locator normally transmit ?

- **a)** HF
- **b)** HF/VHF
- **c)** MF/HF
- **d)** LF/MF

Answer ;

d) LF/MF

0039 According to the ILS coverage area as defined by ICAO Annex 10 , in which of the following situations will the pilot be guaranteed a reliable signal from the localiser ?

- **a)** 19 NM from touchdown inbound and 13° displaced from the localiser centreline.
- **b)** 27 NM from touchdown inbound and 8° displaced from the localiser centreline.
- **c)** 10 NM from touchdown inbound and 38° displaced from the localiser centreline.
- **d)** 20 NM from touchdown inbound and 8° displaced from the localiser centreline.

Answer ;

d) 20 NM from touchdown inbound and 8° displaced from the localiser centreline.

0040. ADF is the abbreviation for ;

- **a)** Automatic Direction Finder.
- **b)** Automatic Direction Finding.
- **c)** Aircraft Direction Finding.
- **d)** Aircraft Directional Finder.

Answer ;

a) Automatic Direction Finder.

0041. Airborne Weather Radar has been primarily developed to detect ;

a) areas of severe clear air turbulence.
b) the kinds of precipitation which are usually accompanied by turbulence.
c) all kinds of frozen precipitation such as hail , snow and graupel.
d) areas of wind shear and severe aircraft icing.

Answer ;
b) the kinds of precipitation which are usually accompanied by turbulence.

0042. Airborne weather radar systems use a wavelength of approximately 3 cm in order to ;

a) detect the smaller cloud formations as well as large.
b) transmit at a higher pulse repetition frequency for extended range.
c) obtain optimum use of the Cosecant squared beam.
d) detect the larger water droplets.

Answer ;
d) detect the larger water droplets.

0043. Airborne weather radars are generally based on the use of ;

a) secondary radar in the VHF band.
b) primary radar in the UHF band.
c) secondary radar in the SHF band.
d) primary radar in the SHF band.

Answer ;
d) primary radar in the SHF band.

0044. All ILS marker beacons transmit at ;

a) it depends on the modulating frequency.
b) 150 MHz.
c) 90 MHz.
d) 75 MHz.

Answer ;
d) 75 MHz.

0045. Allocated frequencies for NDB are ;

a) 1900 kHz to 17500 kHz.
b) 190 kHz to 1750 kHz.
c) 19 Hz to 17500 Hz.
d) 1.90 kHz to 17.50 kHz.

Answer ;
b) 190 kHz to 1750 kHz.

0046. Almanac data stored in the receiver of the satellite navigation system NAVSTAR/GPS is used for the ;

a) recognition whether Selective Availability (SA) is operative.
b) assignment of received PRN-codes (Pseudo Random Noise) to the appropriate satellite.
c) correction of receiver clock error.
d) fast identification of received signals coming from visible satellites.

Answer ;
d) fast identification of received signals coming from visible satellites.

0047. An ADF provides the aircraft with bearing information with respect to a ground station. To do this , the ground station emits a signal pattern which is ;

a) frequency modulated at 30 Hertz.
b) a beam rotating at 30 Hertz.
c) unidirectional.
d) omnidirectional.

Answer ;
d) omnidirectional.

0048. An aeroplane flies over position A which is due North of a VOR station sited at position B. The magnetic variation at A is 18°W , and at B is 10°W. What radial from B is the aircraft on ?

a) 018°
b) 350°
c) 342°
d) 010°

Answer ;
d) 010°

0049. An aircraft at 6.400 ft will be able to receive a VOR groundstation at 100 ft above MSL at an approximate maximum range of ;

a) 123 NM.
b) 113 NM.
c) 90 NM.
d) 98 NM.

Teorik olarak maksimum alış menzili (NM) = 1,25 x (√H1 +√H2)
= 1,25 x (√6400 + √100)
= 1,25 x (80 + 10) = 1,25 x 90 = 112,5 ≈ 113 NM

Answer ;
b) 113 NM.

0050. An aircraft at FL 100 should be able to receive a VOR groundstation at 100 ft above MSL at an approximate maximum range of ;

a) 130 NM.
b) 137 NM.
c) 145 NM.
d) 123 NM.

Teorik olarak maksimum alış menzili (NM) = 1,25 x (√H1 +√H2)
= 1,25 x (√10000 + √100)
= 1,25 x (100 + 10) = 1,25 x 110 = 137,5 ≈ 137 NM

Answer ;
b) 137 NM.

0051. An aircraft at FL 300 , with a groundspeed of 300 kt , is about pass overhead a DME station at MSL. The DME receiver is capable of determining ground-speed. One minute before the overhead , DME speed and distance indications are respectively ;

a) less than 300 kt and 7 NM.
b) 300 kt and 7 NM.
c) 300 kt and 5 NM.
d) less than 300 kt and 5 NM.

Answer ;
a) less than 300 kt and 7 NM.

0052. An aircraft at FL 410 is passing overhead a DME station at mean sea level. The DME-indicates approximately ;

a) 6.1 NM
b) 6.8 NM
c) 6.8 km
d) 6.1 km

FL 410 = 41000 feet ,
41.000 : 6080 = 6,743 ≈ 6.8 NM

Answer ;
b) 6.8 NM

0053. An aircraft DME receiver does not lock on to its own transmissions reflected from the ground because ;

a) the pulse recurrence rates are varied.
b) DME transmits twin pulses.
c) they are not on the receiver frequency.
d) DME uses the UHF band.

Answer ;
c) they are not on the receiver frequency.

0054. An aircraft has a Magnetic Heading of 290º and is on VOR radial 280. Which value has to be selected on the OBS to get a TO indication and the CDI centred ?

a) 110
b) 100
c) 290
d) 280

Answer ;
b) 100

0055. An aircraft is 100 NM from a VOR facility. Assuming no error when using a deviation indicator where 1 dot = 2° deviation , how many dots deviation from the centre line of the instrument will represent the limits of the airway boundary ? (Assume that the airway is 10 NM wide)

a) 4.5
b) 6.0
c) 1.5
d) 3.0

Track Error Angle (TEA) = (Distance Off-Track x 60) : Distance Along Track formülünü kullanarak ;
TEA = (5 x 60) : 100 = 300 : 100 = 3° olarak bulunur. Her 1 dot 2°lik sapmayı temsil ettiğine göre 3°lik sapma 1.5 dot'a karşılık gelir.

Answer ;
c) 1.5

0056. An aircraft is flying a 3° glidepath and experiences a reduction in groundspeed from 150 kt at the outer marker to 120 kt over the threshold. The effect of this change in groundspeed on the aircraft's rate of descent will be a decrease of approximately ;

- **a)** 100 FT/MIN.
- **b)** 250 FT/MIN.
- **c)** 150 FT/MIN.
- **d)** 50 FT/MIN.

ROD (Rate Of Descent) = ө (glidepath angle) x ground speed (NM/min) x 101,3
150 kt , dakikada (150 : 60) = 2,5 ile ROD = 3 x 2,5 x 101,3 = 759,75 ≈ 760 fpm
120 kt , dakikada (120 : 60) = 2 ile ROD = 3 x 2 x 101,3 = 607,8 ≈ 610 fpm
760 – 610 = 150 FT/MIN

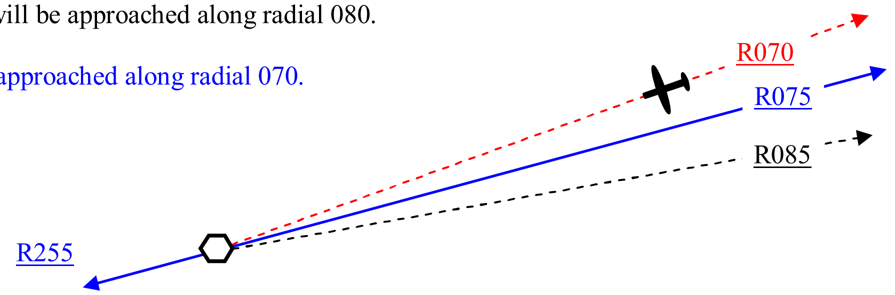
Answer ;
c) 150 FT/MIN.

0057. An aircraft is flying a heading of 245° towards a VOR at FL300. The HSI displays a "selected course" of 255º with a TO-indication. The variation at the VOR is 15ºE. Variation at the aircraft position is 16°E and the deviation is +1°. When the pilot keeps the CDI on the left inner dot on a display with two dots on either side ;

- **a)** the aircraft will pass north of the VOR.
- **b)** the aircraft will pass south of the VOR.
- **c)** the VOR will be approached along radial 070.
- **d)** the VOR will be approached along radial 080.

Answer ;
c) the VOR will be approached along radial 070.

Çözüm ;



0058. An aircraft is flying directly overhead a DME station at FL410. The indicated range will be approximately ;

- **a)** 8.1 nm.
- **b)** 6.8 km.
- **c)** 6.8 nm.
- **d)** 8.1 km.

FL 410 = 41000 feet ,
41.000 : 6080 = 6,743 ≈ 6.8 nm.

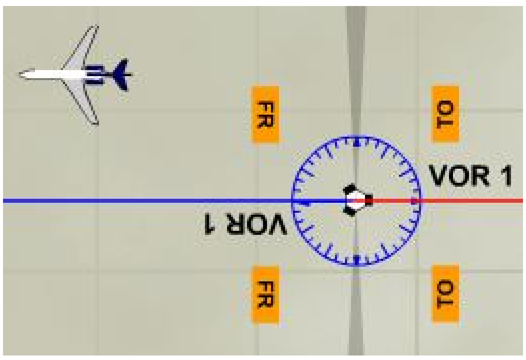
Answer ;
c) 6.8 nm.

0059. An aircraft is flying on a heading of 270°(M). The VOR OBS is also set to 270° with the full left deflection and FROM flag displayed. In which sector is the aircraft from the VOR ground station ?

- **a)** SW
- **b)** SE
- **c)** NE
- **d)** NW

Answer ;
d) NW

Çözüm ;



0060. (Refer to Image 28)

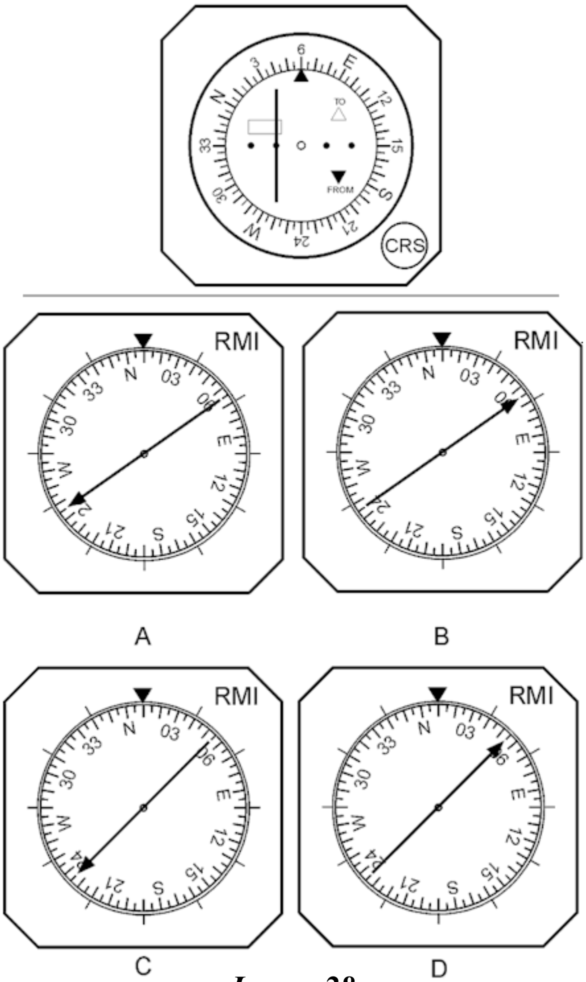


Image 28

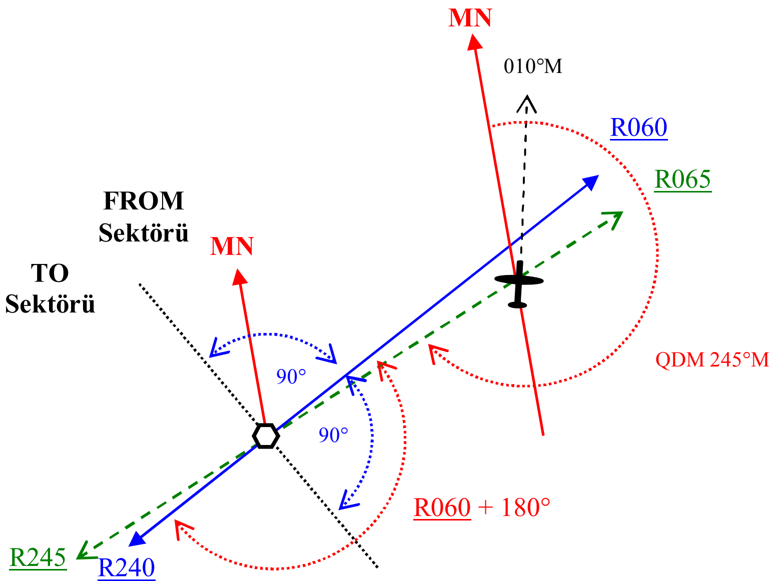
An aircraft is flying on a MH of 010°. ; The magnetic variation at the VOR is 10°W and at the aircraft 12°W. Which VOR-RMI corresponds to the indications on the CDI shown on the Annex ?

- **a)** Figure D
- **b)** Figure C
- **c)** Figure A
- **d)** Figure B

Answer ;

c) Figure A

Çözüm ;



Uçağın CDI'ında (Course Deviation Indicator) Course 060° seçilidir ve FROM göstergesi ile birlikte CDI'ın ibresi seçilen course'un önlenmesi için uçağı sola doğru çağırmaktadır , indikatör'e göre seçili olan radyal uçağın solunda ve 5°lik bir sapma (2 dot , her bir dot 5°) göstermektedir. Burada VOR'daki Variation değeri dikkate alınır.
060 radyalin 180° aksi tarafındaki radyal 240 radyalidir. CDI 1 dot sapma (5°) gösterdiğine göre uçak halihazırda 065 radyalini katetmektedir , yani bulunduğu pozisyondan VOR istasyonuna doğru QDM'i 245°M'dir. Bu QDM'in , RMI'da gösterimi de ibrenin başının 245°i göstermesi şeklinde olacaktır. Aynı anda RMI uçuş başı olarak da 010°yi gösterecektir.
Cevap ; Figure A

0061. (Refer to Image 26)

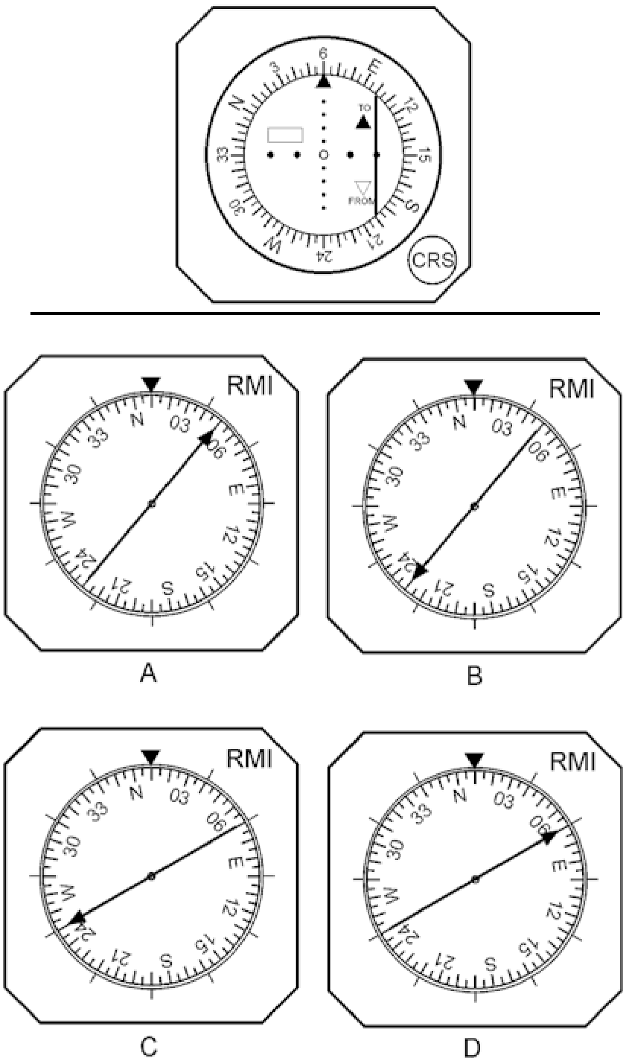


Image 26

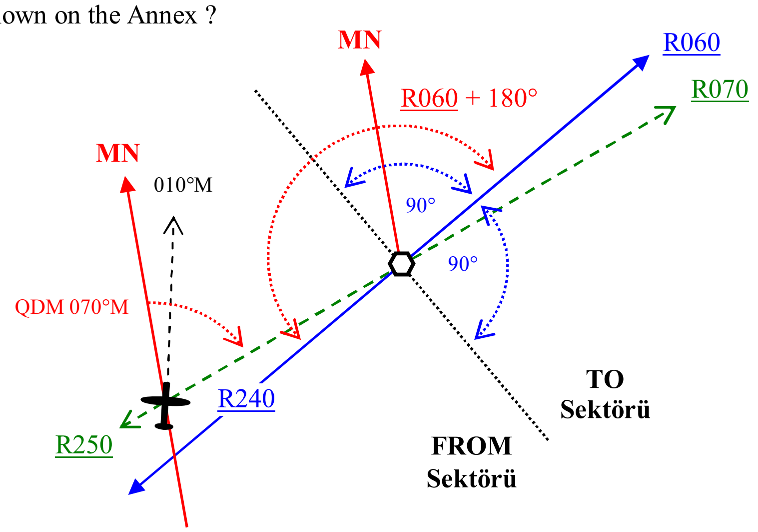
An aircraft is flying on a MH of 010°. ; The magnetic variation at the VOR is 10°W and at the aircraft 12°W. Which VOR-RMI corresponds to the indications on the CDI shown on the Annex ?

- **a)** Figure B
- **b)** Figure D
- **c)** Figure A
- **d)** Figure C

Answer ;

b) Figure D

Çözüm ;



Uçağın CDI'ında (Course Deviation Indicator) Course 060° seçilidir ve TO göstergesi ile birlikte CDI'ın ibresi seçilen course'un önlenmesi için uçağı sağa doğru çağırmaktadır , indikatör'e göre seçili olan radyal uçağın sağında ve 10°lik bir sapma (2 dot , her bir dot 5°) göstermektedir. Burada VOR'daki Variation değeri dikkate alınır.
060 radyalin 180° aksi tarafındaki radyal 240 radyalidir. CDI 2 dot sapma (10°) gösterdiğine göre uçak halihazırda 250 radyalini katetmektedir , yani bulunduğu pozisyondan VOR istasyonuna doğru QDM'i 070°M'dir. Bu QDM'in , RMI'da gösterimi de ibrenin başının 070°i göstermesi şeklinde olacaktır. Aynı anda RMI uçuş başı olarak da 010°yi gösterecektir.
Cevap ; Figure D

0062. (Refer to Image 31)

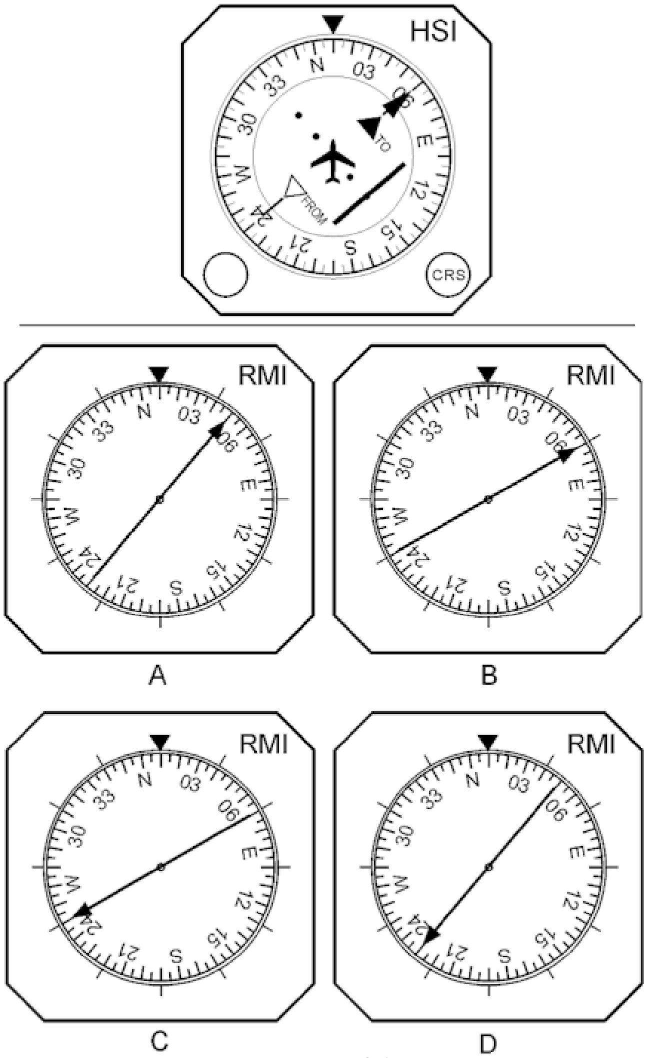


Image 31

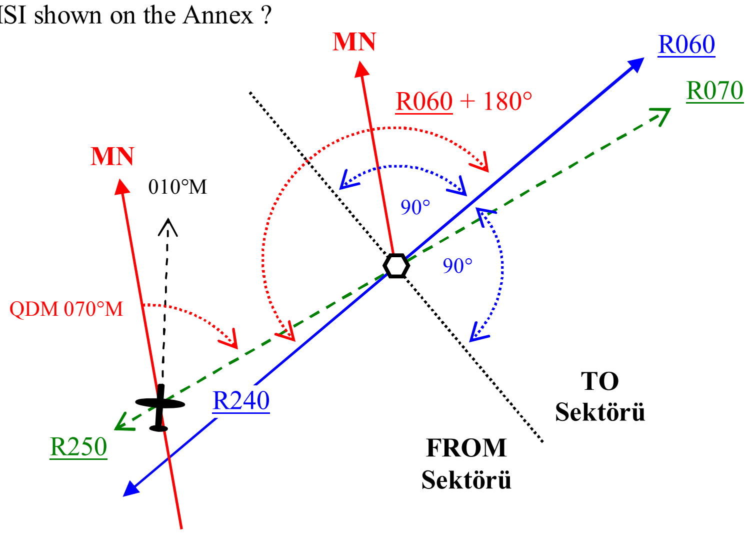
An aircraft is flying on a MH of 010°. The magnetic variation at the VOR is 10°W and at the aircraft 12°W. Which VOR-RMI corresponds to the indications on the HSI shown on the Annex ?

a) Figure A
b) Figure B
c) Figure C
d) Figure D

Answer ;

b) Figure B

Çözüm ;



Uçağın HSI'ında (Horizontal Situation Indicator) Course 060° seçilidir ve TO göstergesi ile birlikte HSI'ın ibresi seçilen course'un önlenmesi için uçağı sağa doğru çağırmaktadır , indikatör'e göre seçili olan radyal uçağın sağında ve 10°lik bir sapma (2 dot , her bir dot 5°) göstermektedir. Burada VOR'daki Variation değeri dikkate alınır.
060 radyalin 180° aksi tarafındaki radyal 240 radyalidir. HSI 2 dot sapma (10°) gösterdiğine göre uçak halihazırda 250 radyalini katetmektedir , yani bulunduğu pozisyondan VOR istasyonuna doğru QDM'i 070°M'dir. Bu QDM'in , RMI'da gösterimi de ibrenin başının 070°i göstermesi şeklinde olacaktır. Aynı anda RMI uçuş başı olarak da 010°yi gösterecektir.
Cevap ; Figure B

0063. (Refer to Image 34)

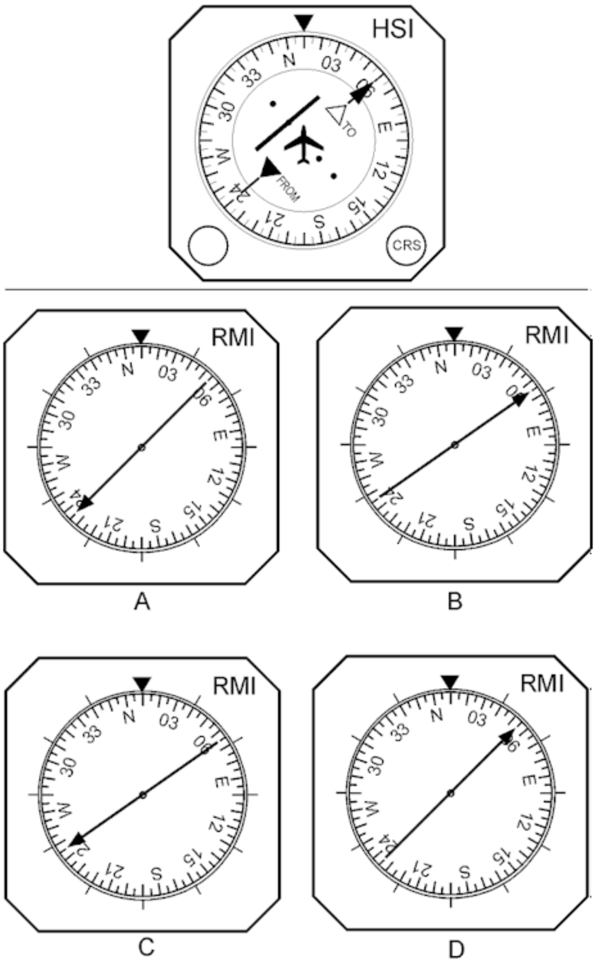


Image 34

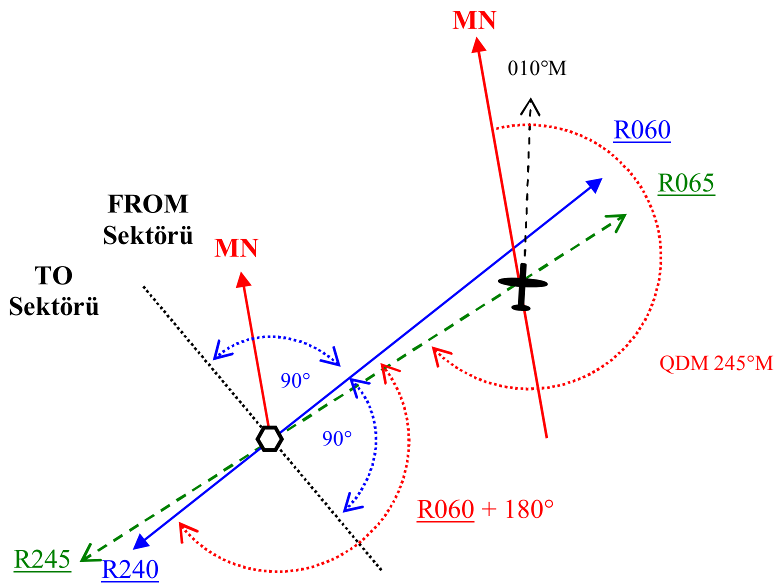
An aircraft is flying on a MH of 010°. The magnetic variation at the VOR is 10°W and at the aircraft 12°W. Which VOR-RMI corresponds to the indications on the HSI shown on the Annex ?

a) Figure B
b) Figure A
c) Figure D
d) Figure C

Answer ;

d) Figure C

Çözüm ;



Uçağın HSI'ında (Horizontal Situation Indicator) Course 060° seçilidir ve FROM göstergesi ile birlikte HSI'ın ibresi seçilen course'un önlenmesi için uçağı sola doğru çağırmaktadır , indikatör'e göre seçili olan radyal uçağın solunda ve 5°lik bir sapma (2 dot , her bir dot 5°) göstermektedir. Burada VOR'daki Variation değeri dikkate alınır.
060 radyalin 180° aksi tarafındaki radyal 240 radyalidir. HSI 1 dot sapma (5°) gösterdiğine göre uçak halihazırda 065 radyalini katetmektedir , yani bulunduğu pozisyondan VOR istasyonuna doğru QDM'i 245°M'dir. Bu QDM'in , RMI'da gösterimi de ibrenin başının 245°i göstermesi şeklinde olacaktır. Aynı anda RMI uçuş başı olarak da 010°yi gösterecektir.
Cevap ; Figure C

0064. (Refer to Image 21)

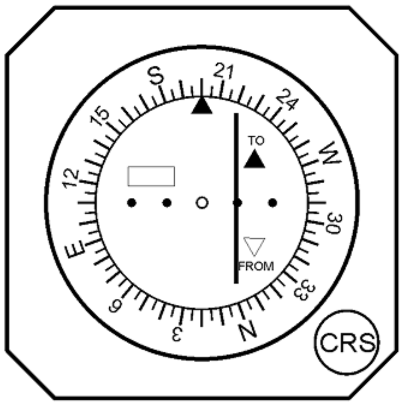


Image 21

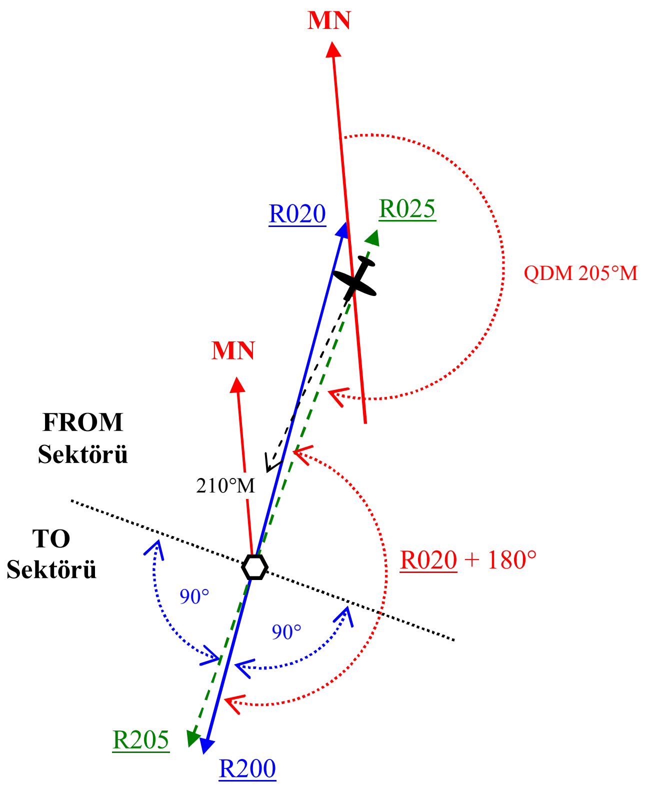
An aircraft is flying on a MH of 210°. The magnetic variation at the VOR is 5°W and at the aircraft 10°W. According to the CDI shown at the Annex the aircraft is on radial ;

a) 015
b) 205
c) 195
d) 025

Answer ;

d) 025

Çözüm ;



Uçağın CDI'ında (Course Deviation Indicator) Course 200° seçilidir ve TO göstergesi ile birlikte CDI'ın ibresi seçilen course'un önlenmesi için uçağı sağa doğru çağırmaktadır , indikatör'e göre seçili olan radyal uçağın sağında ve 5°lik bir sapma (2 dot , her bir dot 5°) göstermektedir. Burada VOR'daki Variation değeri dikkate alınır.
200 radyalin 180° aksi tarafındaki radyal 020 radyalidir. CDI 1 dot sapma (5°) gösterdiğine göre uçak halihazırda 025 radyalini katetmektedir. Cevap ; 025

0065. (Refer to Image 20)

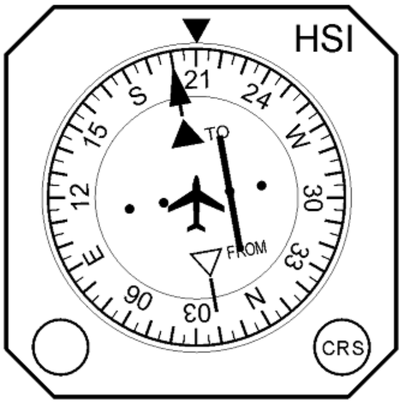


Image 20

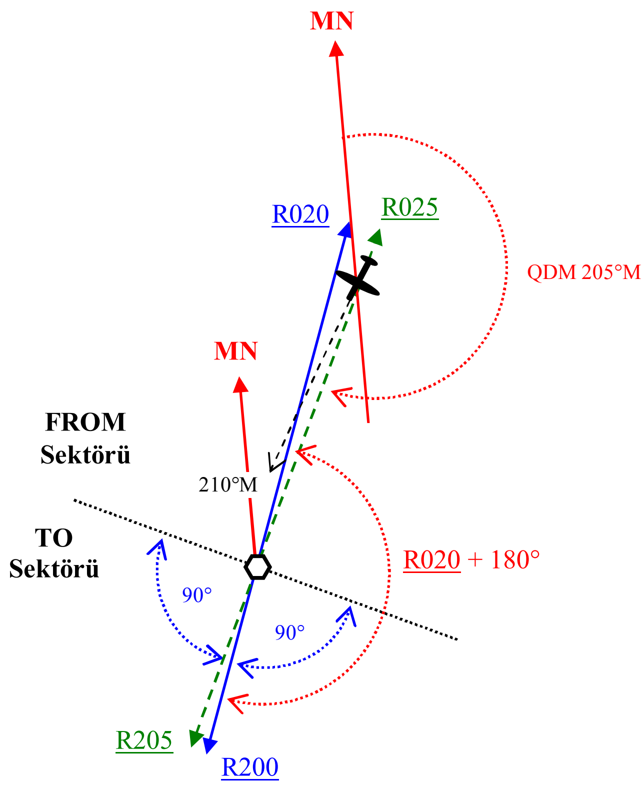
An aircraft is flying on a MH of 210°. The magnetic variation at the VOR is 5°W and at the aircraft 10°W. According to the HSI shown at the Annex the aircraft is on radial ;

a) 015
b) 025
c) 205
d) 195

Answer ;

b) 025

Çözüm ;



Uçağın HSI'ında (Horizontal Situation Indicator) Course 200° seçilidir ve TO göstergesi ile birlikte HSI'ın ibresi seçilen course'un önlenmesi için uçağı sağa doğru çağırmaktadır , indikatör'e göre seçili olan radyal uçağın sağında ve 5°lik bir sapma (2 dot , her bir dot 5°) göstermektedir. Burada VOR'daki Variation değeri dikkate alınır.
200 radyalin 180° aksi tarafındaki radyal 020 radyalidir. HSI 1 dot sapma (5°) gösterdiğine göre uçak halihazırda 025 radyalini katetmektedir. Cevap ; 025

0066. (Refer to Image 25)

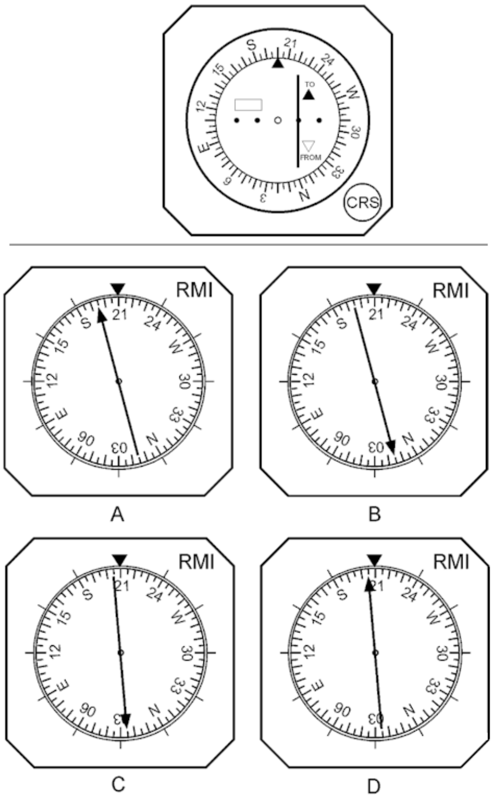


Image 25

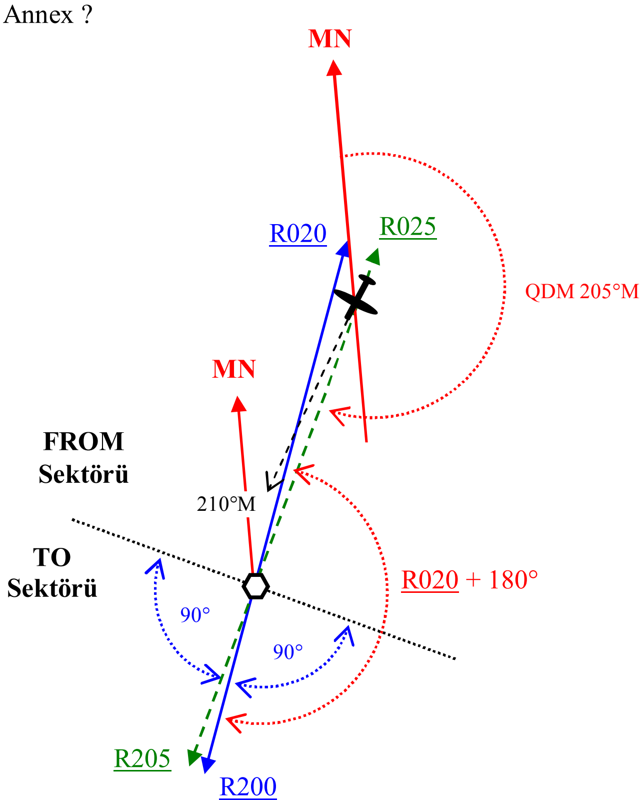
An aircraft is flying on a MH of 210°. The magnetic variation at the VOR is 5°W and at the aircraft 10°W. Which VOR-RMI corresponds to the indications on the CDI shown on the Annex ?

a) Figure C
b) Figure D
c) Figure A
d) Figure B

Answer ;

b) Figure D

Çözüm ;



Uçağın CDI'ında (Course Deviation Indicator) Course 200° seçilidir ve TO göstergesi ile birlikte CDI'ın ibresi seçilen course'un önlenmesi için uçağı sağa doğru çağırmaktadır , indikatör'e göre seçili olan radyal uçağın sağında ve 5°lik bir sapma (2 dot , her bir dot 5°) göstermektedir. Burada VOR'daki Variation değeri dikkate alınır.
200 radyalin 180° aksi tarafındaki radyal 020 radyalidir. CDI 1 dot sapma (5°) gösterdiğine göre uçak halihazırda 025 radyalini katetmektedir , yani bulunduğu pozisyondan VOR istasyonuna doğru QDM'i 205°M'dir. Bu QDM'in , RMI'da gösterimi de ibrenin başının 205°i göstermesi şeklinde olacaktır. Aynı anda RMI uçuş başı olarak da 210°yi gösterecektir.
Cevap ; Figure D

0067. (Refer to Image 30)

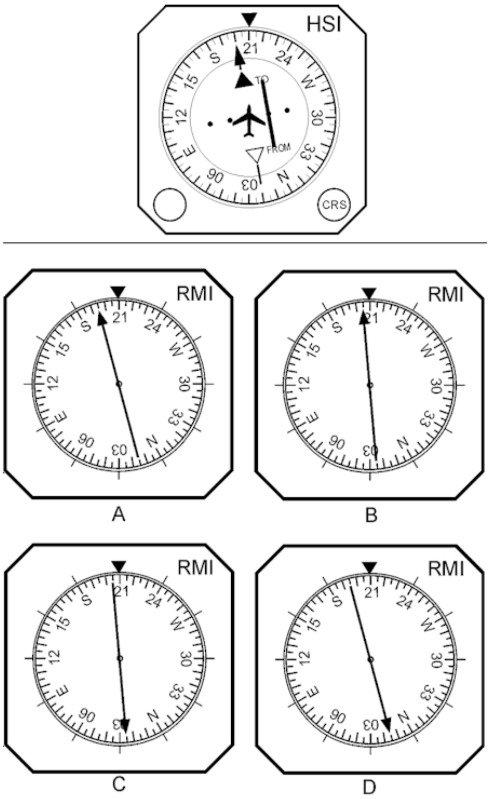


Image 30

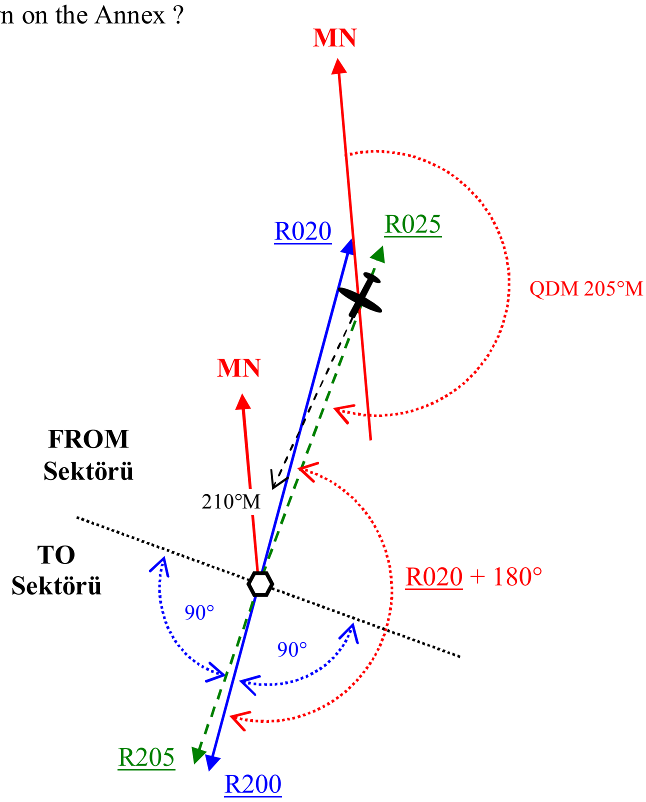
An aircraft is flying on a MH of 210°. The magnetic variation at the VOR is 5°W and at the aircraft 10°W. Which VOR-RMI corresponds to the indications on the HSI shown on the Annex ?

- **a)** Figure A
- **b)** Figure B
- **c)** Figure D
- **d)** Figure C

Answer ;

b) Figure B

Çözüm ;



Uçağın HSI'ında (Horizontal Situation Indicator) Course 200° seçilidir ve TO göstergesi ile birlikte CDI'ın ibresi seçilen course'un önlenmesi için uçağı sağa doğru çağırmaktadır , indikatör'e göre seçili olan radyal uçağın sağında ve 5°lik bir sapma (2 dot , her bir dot 5°) göstermektedir. Burada VOR'daki Variation değeri dikkate alınır.
200 radyalin 180° aksi tarafındaki radyal 020 radyalidir. CDI 1 dot sapma (5°) gösterdiğine göre uçak halihazırda 025 radyalini katetmektedir , yani bulunduğu pozisyondan VOR istasyonuna doğru QDM'i 205°M'dir. Bu QDM'in , RMI'da gösterimi de ibrenin başının 205°i göstermesi şeklinde olacaktır. Aynı anda RMI uçuş başı olarak da 210°yi gösterecektir.
Cevap ; Figure B

0068. (Refer to Image 32)

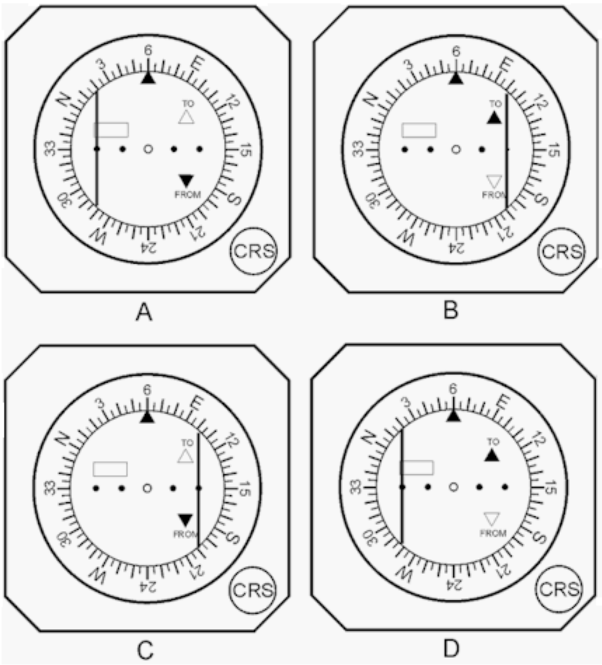


Image 32

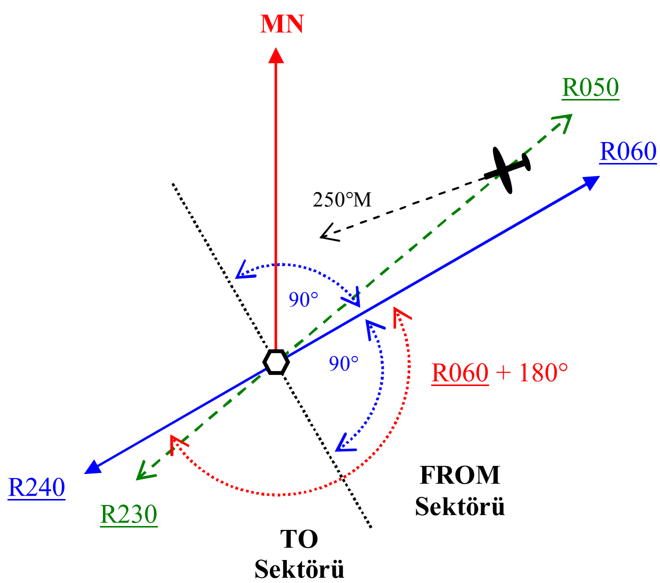
An aircraft is flying on the 050 radial with a MH of 250º. On the CDI the CRS is set to 060. Which CDI shows the correct indications ?

- **a)** Figure C
- **b)** Figure A
- **c)** Figure D
- **d)** Figure B

Answer ;

a) Figure C

Çözüm ;



Uçağın CDI'ında (Course Deviation Indicator) Course 060° seçilidir ve halihazırda 050 radyali geçilmektedir. Burada uçuş başının bir önemi yoktur. Eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile aynı sektörde ise CDI'da görülecek ibare FROM olacaktır. Diğer durumda eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile farklı sektörde ise CDI'da görülecek ibare bu sefer TO olacaktır. Ayrıca CDI üzerinde 10 derecelik sapma görülecektir. Uçağın halihazırda bulunduğu radyal seçmiş olduğu radyal + 180° alanı içinde olmadığından , bu sapmada ibre sağda gözükmelidir (Couse göstergesinin sağı). Cevap ; Figure C

0069. (Refer to Image 29)

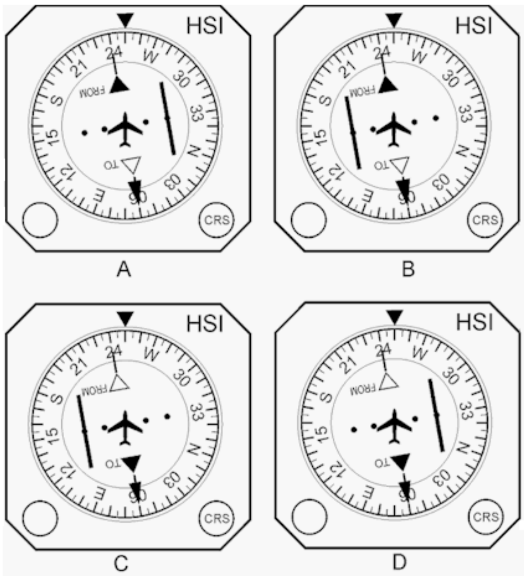


Image 29

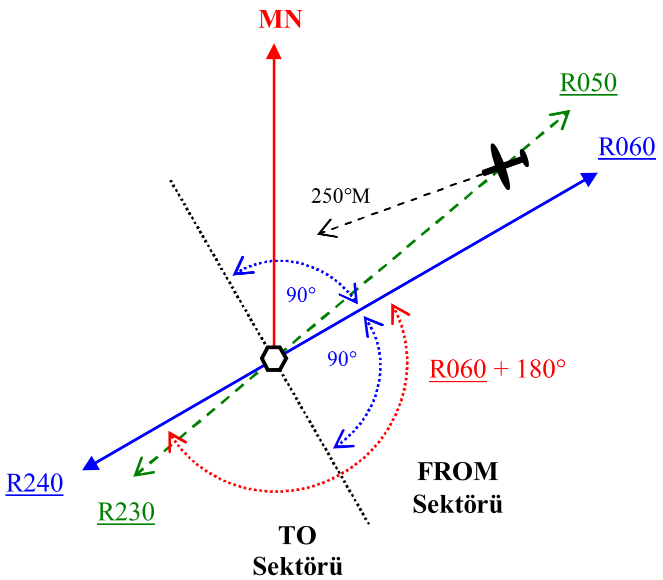
An aircraft is flying on the 050 radial with a MH of 250º. On the HSI the CRS is set to 060. Which HSI shows the correct indications ?

a) Figure A
b) Figure C
c) Figure D
d) Figure B

Answer ;

d) Figure B

Çözüm ;



Uçağın HSI'ında (Horizontal Situation Indicator) Course 060° seçilidir ve halihazırda 050 radyali geçilmektedir. Burada uçuş başının bir önemi yoktur. Eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile aynı sektörde ise HSI'da görülecek ibare FROM olacaktır. Diğer durumda eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile farklı sektörde ise HSI'da görülecek ibare bu sefer TO olacaktır. Ayrıca HSI üzerinde 10 derecelik sapma görülecektir. Uçağın halihazırda bulunduğu radyal seçmiş olduğu radyal + 180° alanı içinde olmadığından , bu sapmada ibre sağda gözükmelidir (Couse göstergesinin sağı). Cevap ; Figure B

0070. (Refer to Image 22)

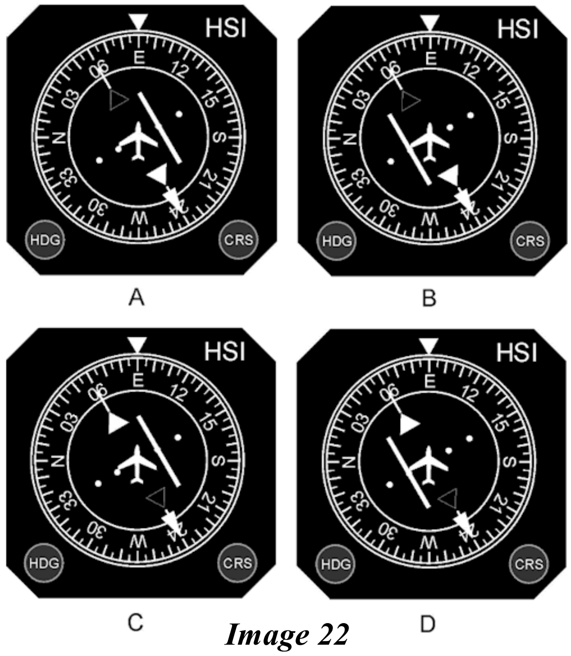


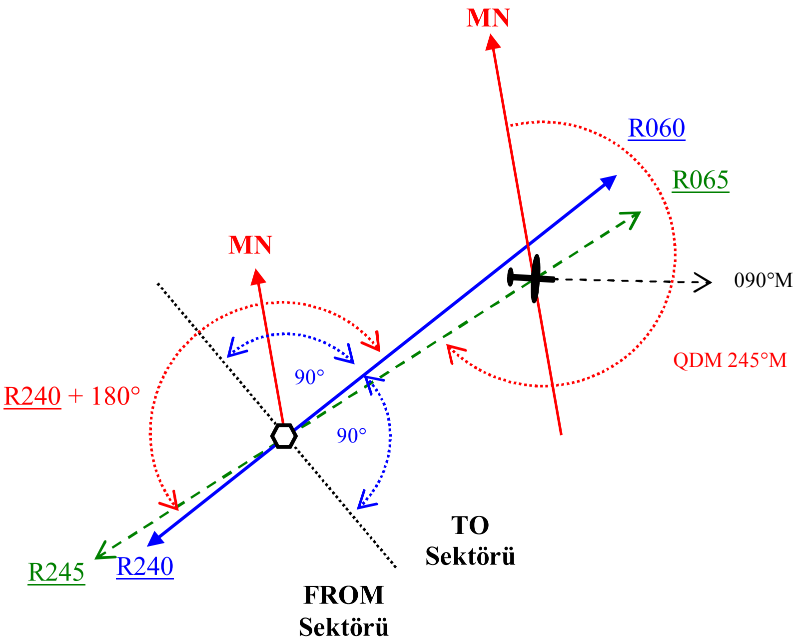
Image 22

An aircraft is on the 065 radial with a heading of 090ºM. The Course Reference Selector (CRS) is set on 240. Which HSI shows the correct indications ?

a) C
b) D
c) A
d) B

Answer ;

d) B



Uçağın HSI'ında (Horizontal Situation Indicator) Course 240° seçilidir ve halihazırda 065 radyali geçilmektedir. Burada uçuş başının bir önemi yoktur. Eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile aynı sektörde ise HSI'da görülecek ibare FROM olacaktır. Diğer durumda eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile farklı sektörde ise HSI'da görülecek ibare bu sefer TO olacaktır. Ayrıca HSI üzerinde 5 derecelik sapma görülecektir. Uçağın halihazırda bulunduğu radyal seçmiş olduğu radyal + 180° alanı içinde olmadığından , bu sapmada ibre sağda gözükmelidir (Couse göstergesinin sağı). Cevap ; B

0071. (Refer to Image 24)

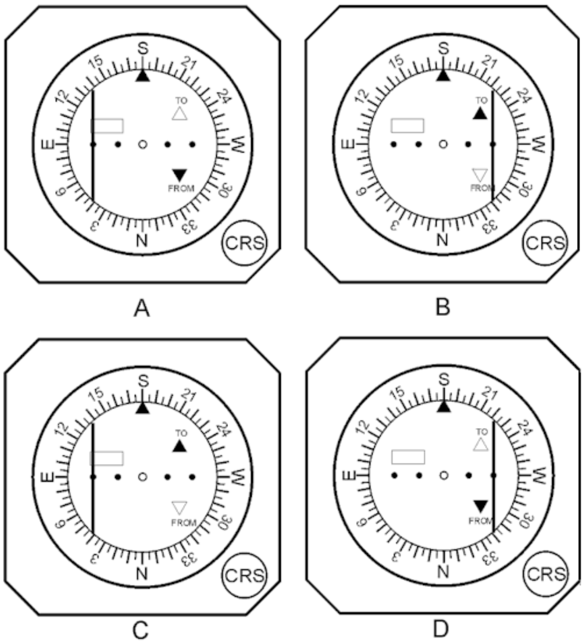


Image 24

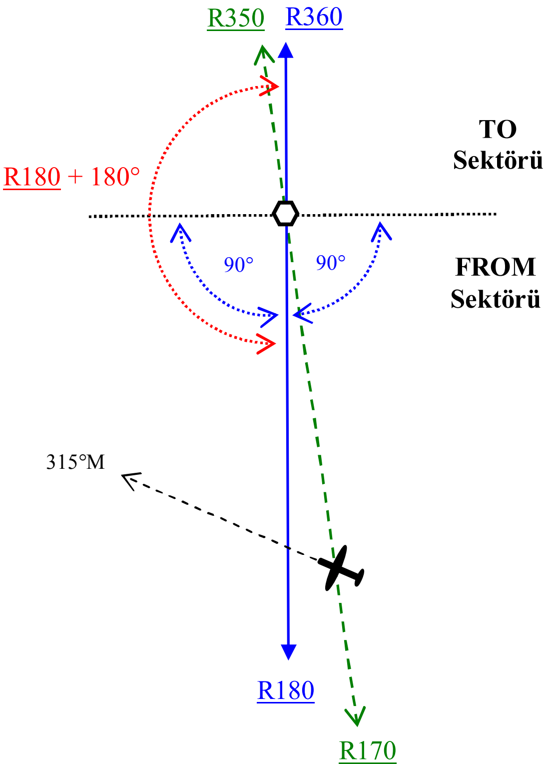
An aircraft is flying on the 170 radial with a MH of 315º. On the CDI the CRS is set to 180. Which CDI shows the correct indications ?

a) Figure A
b) Figure D
c) Figure B
d) Figure C

Answer ;

b) Figure D

Çözüm ;



Uçağın CDI'ında (Course Deviation Indicator) Course 180° seçilidir ve halihazırda 170 radyali geçilmektedir. Burada uçuş başının bir önemi yoktur. Eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile aynı sektörde ise CDI'da görülecek ibare FROM olacaktır. Diğer durumda eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile farklı sektörde ise CDI'da görülecek ibare bu sefer TO olacaktır. Ayrıca CDI üzerinde 10 derecelik sapma görülecektir. Uçağın halihazırda bulunduğu radyal seçmiş olduğu radyal + 180° alanı içinde olmadığından , bu sapmada ibre sağda gözükmelidir (Couse göstergesinin sağı). Cevap ; Figure D

0072. (Refer to Image 33)

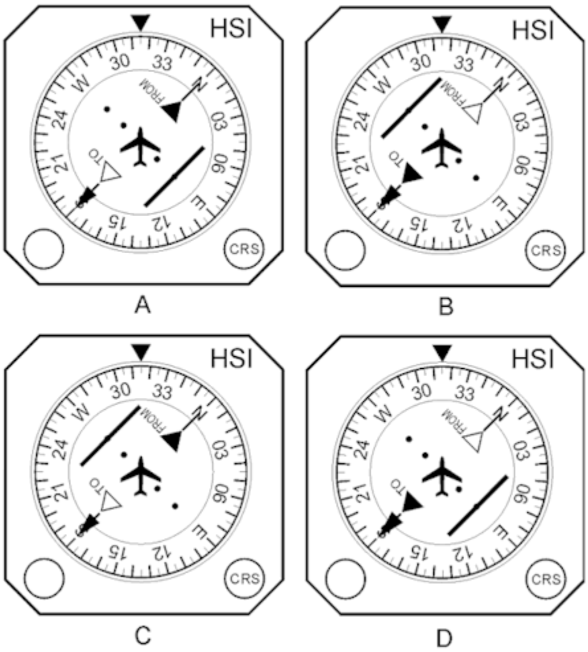


Image 33

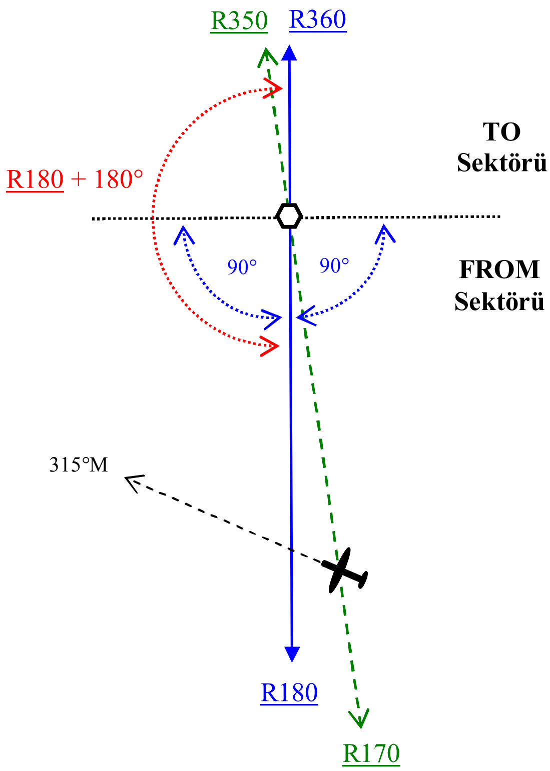
An aircraft is flying on the 170 radial with a MH of 315º. On the HSI the CRS is set to 180. Which HSI shows the correct indications ?

a) Figure A
b) Figure B
c) Figure C
d) Figure D

Answer ;

c) Figure C

Çözüm ;



Uçağın HSI'ında (Horizontal Situation Indicator) Course 180° seçilidir ve halihazırda 170 radyali geçilmektedir. Burada uçuş başının bir önemi yoktur. Eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile aynı sektörde ise HSI'da görülecek ibare FROM olacaktır. Diğer durumda eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile farklı sektörde ise HSI'da görülecek ibare bu sefer TO olacaktır. Ayrıca HSI üzerinde 10 derecelik sapma görülecektir. Uçağın halihazırda bulunduğu radyal seçmiş olduğu radyal + 180° alanı içinde olmadığından , bu sapmada ibre sağda gözükmelidir (Couse göstergesinin sağı). Cevap ; Figure C

0073. (Refer to Image 23)

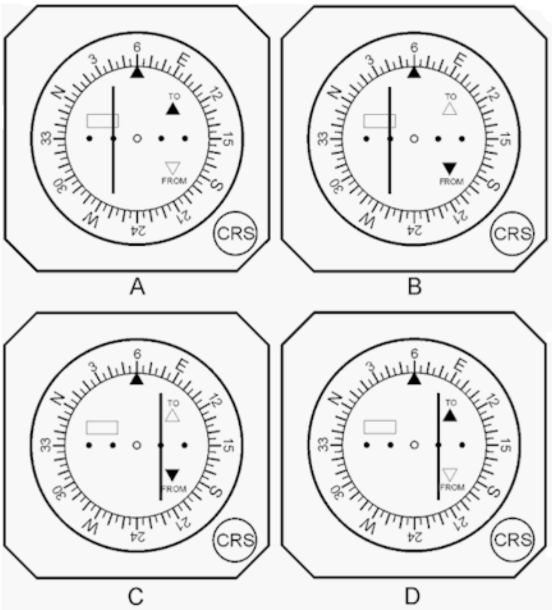


Image 23

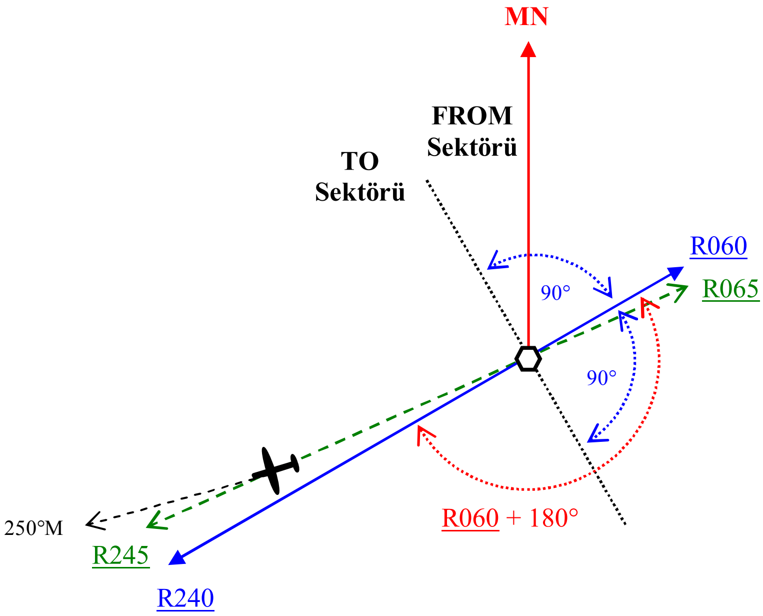
An aircraft is flying on the 245 radial with a MH of 250º. On the CDI the CRS is set to 060. Which CDI shows the correct indications ?

a) Figure D
b) Figure B
c) Figure C
d) Figure A

Answer ;

a) Figure D

Çözüm ;



Uçağın CDI'ında (Course Deviation Indicator) Course 060° seçilidir ve halihazırda 245 radyali geçilmektedir. Burada uçuş başının bir önemi yoktur. Eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile aynı sektörde ise CDI'da görülecek ibare FROM olacaktır. Diğer durumda eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile farklı sektörde ise CDI'da görülecek ibare bu sefer TO olacaktır. Ayrıca CDI üzerinde 5 derecelik sapma görülecektir. Uçağın halihazırda bulunduğu radyal seçmiş olduğu radyal + 180° alanı içinde olmadığından , bu sapmada ibre sağda gözükmelidir (Couse göstergesinin sağı). Cevap ; Figure D

0074. (Refer to Image 27)

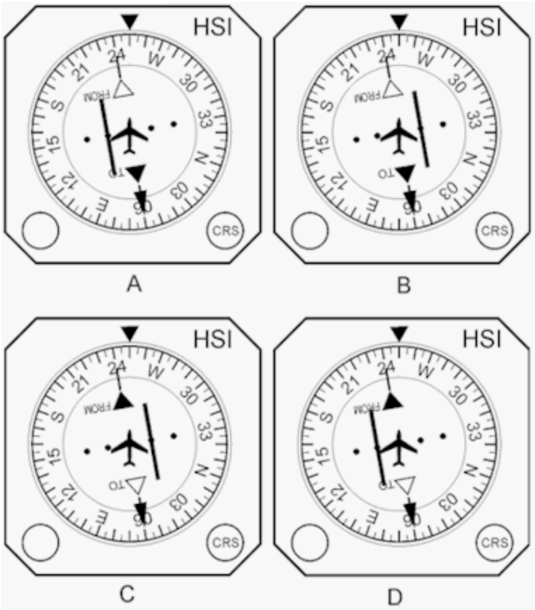


Image 27

An aircraft is flying on the 245 radial with a MH of 250º. On the HSI the CRS is set to 060. Which HSI shows the correct indications ?

a) Figure A
b) Figure B
c) Figure C
d) Figure D

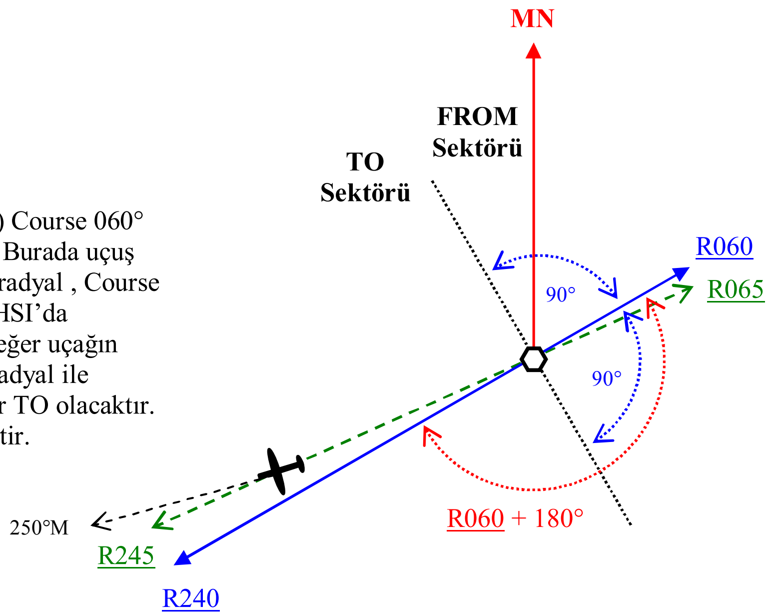
Answer ;

a) Figure A

Çözüm ;

Uçağın HSI'ında (Horizontal Situation Indicator) Course 060° seçilidir ve halihazırda 245 radyali geçilmektedir. Burada uçuş başının bir önemi yoktur. Eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile aynı sektörde ise HSI'da görülecek ibare FROM olacaktır. Diğer durumda eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile farklı sektörde ise CDI'da görülecek ibare bu sefer TO olacaktır. Ayrıca CDI üzerinde 5 derecelik sapma görülecektir. Uçağın halihazırda bulunduğu radyal seçmiş olduğu radyal + 180° alanı içinde olmadığından , bu sapmada ibre sağda gözükmelidir (Couse göstergesinin sağı).

Cevap ; Figure A



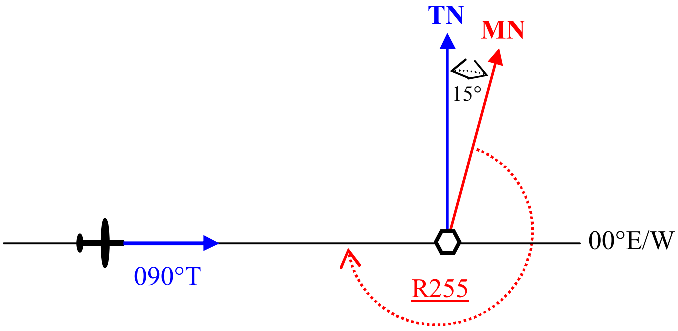
0075. An aircraft is flying on the true track 090° towards a VOR station located near the equator where the magnetic variation is 15°E. The variation at the aircraft position is 8°E. The aircraft is on VOR radial ;

a) 255°
b) 262°
c) 278°
d) 285°

Answer ;

a) 255°

Çözüm ;



Burada uçak 090°T rotada VOR'a doğru uçtuğuna göre enlemi takip ettiğini söyleyebiliriz ve VOR'a göre true bearing'i 270°T'dur. VOR söz konusu olduğuna göre VOR'daki Variation değerini hesaba katacağız. VOR radyalleri manyetik bearing'ler olduğuna göre ; MB = TB – Variation E = 270° – 15° = 255° olarak bulunur.

0076. An aircraft is "homing" to a radio beacon whilst maintaining a relative bearing of zero. If the magnetic heading decreases , the aircraft is experiencing ;

a) left drift.
b) a wind from the west.
c) zero drift.
d) right drift.

Answer ;

d) right drift.

0077. An aircraft is on radial 120 with a magnetic heading of 300° , the track selector (OBS) reads ; 330. The indications on the Course Deviation Indicator (CDI) are "fly" ;

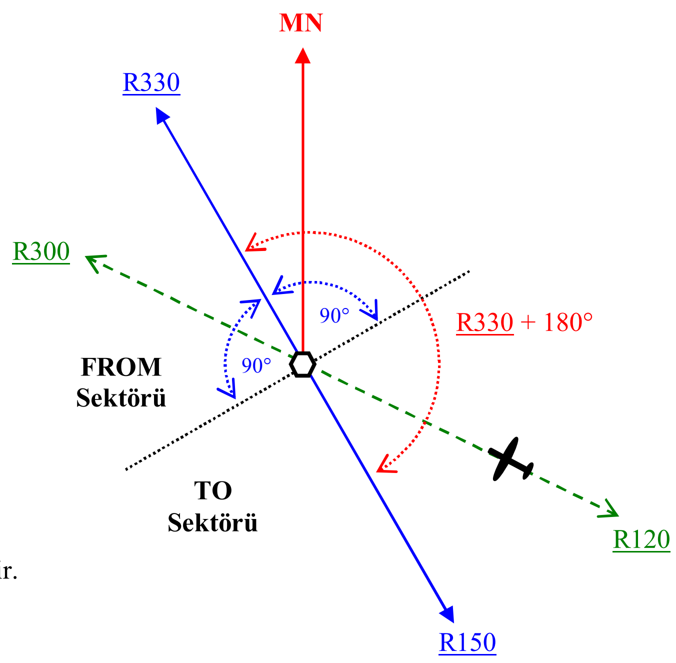
a) right with "TO" showing.
b) left with "TO" showing.
c) right with "FROM" showing.
d) left with "FROM" showing.

Answer ;

b) left with "TO" showing.

Çözüm ;

Uçağın CDI'ında (Course Deviation Indicator) Course 060° seçilidir ve halihazırda 120 radyal üzerindedir. Burada uçuş başının bir önemi yoktur. Eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile aynı sektörde ise CDI'da görülecek ibare FROM olacaktır. Diğer durumda eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile farklı sektörde ise CDI'da görülecek ibare bu sefer TO olacaktır. Uçağın halihazırda bulunduğu radyal seçmiş olduğu radyal + 180° alanı içinde olduğundan bu sapmada ibre Couse göstergesinin solunda gözükecektir.
Cevap ; left with "TO" showing.



0078 An aircraft is on the 120° radial from a VOR station. Course 340° is selected on the HSI (Horizontal Situation Indicator). If the magnetic heading is 070° , the deviation bar relative to the aeroplane model , will be ;

a) in front.
b) left.
c) behind.
d) right.

Answer ;

c) behind.

0079. An aircraft is required to approach a VOR station via the 244° radial. In order to obtain correct sense indications the deviation indicator should be set to ;

a) 244° with the TO flag showing.
b) 064° with the FROM flag showing.
c) 064° with the TO flag showing.
d) 244° with the FROM flag showing.

Answer ;

c) 064° with the TO flag showing.

0080. An aircraft is required to approach a VOR via the 104° radial. Which of the following settings should be made on the VOR/ILS deviation indicator ?

a) 104° with the TO flag showing.
b) 284° with the FROM flag showing.
c) 284° with the TO flag showing.
d) 104° with the FROM flag showing.

Answer ;

c) 284° with the TO flag showing.

0081. An aircraft is situated at 30°N - 005°E with a magnetic variation of 10°W. A VOR is located at 30°N - 013°E with a magnetic variation of 15°W. The aircraft is situated on the VOR radial ;

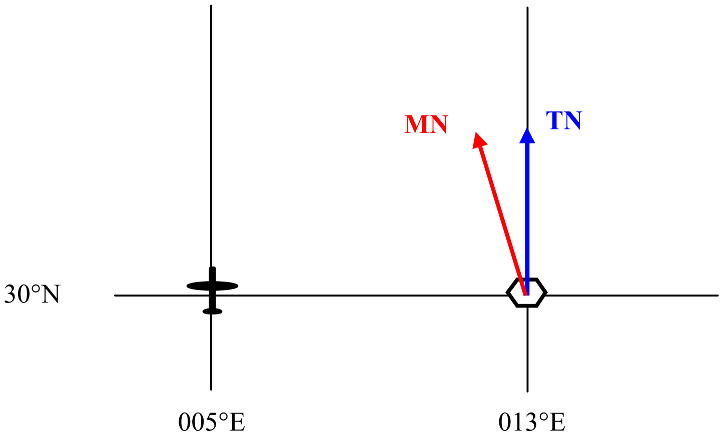
a) 256°
b) 281°
c) 101°
d) 286°

Answer ;
d) 286°

Çözüm ;

Burada VOR radyali sorulduğu için sadece VOR'daki variation değerini hesaba katacağız. Uçak ve VOR aynı enlem üzerinde , ve VOR'ın batısında VOR'a göre true bearing'i 270°T olduğuna göre ;

VOR radyal = 270° + Variation West
= 270° + 15°
= 285° ~ 286°



0082. An aircraft , on a heading of 180°M is on a bearing of 270°M from a VOR. The bearing you should select on the OMNI bearing selector to centralise the VOR/ILS left/right deviation needle with a TO indication is ;

a) 090°
b) 270°
c) 360°
d) 180°

Answer ;
a) 090°

0083. An aircraft passes overhead a DME station at 12.000 feet above the station. At that time , the DME reading will be ;

a) approximately 2 NM.
b) fluctuating and not significant.
c) FLAG/OFF , the aircraft is within the cone of silence.
d) 0 NM.

12.000 feet ,
12.000 : 6080 = 1,973 ≈ 2 NM

Answer ;
a) approximately 2 NM.

0084. An aircraft tracking to intercept the Instrument Landing System (ILS) localiser inbound on the approach side , outside the published ILS coverage angle ;

a) only glide path information is available.
b) will receive signals without identification coding.
c) may receive false course indications.
d) can expect signals to give correct indications.

Answer ;
c) may receive false course indications.

0085. An airway 10 NM wide is to be defined by two VORs each having a resultant bearing accuracy of plus or minus 5.5°. In order to ensure accurate track guidance within the airway limits the maximum distance apart for the transmitter is approximately ;

a) 105 NM
b) 165 NM
c) 50 NM
d) 210 NM

Distance = (Width of the airway : tan 5.5°) x 2
Üçgen hesabından dolayı burada yolun genişliği 5 NM olarak alınmalıdır.
Distance = (5 : 0,096) x 2
Distance = 52,083 x 2 = 104,166 ≈ 105 NM

Answer ;
a) 105 NM

0086. An apparent increase in the transmitted frequency which is proportional to the transmitter velocity will occur when ;

a) the receiver moves towards the transmitter.
b) the transmitter moves towards the receiver.
c) the transmitter moves away from the receiver.
d) both transmitter and receiver move towards each other.

Answer ;
b) the transmitter moves towards the receiver.

0087. An ILS marker beacon operates in the ;

- **a)** LF/MF band.
- **b)** VHF band.
- **c)** HF band.
- **d)** UHF band.

Answer ;

b) VHF band.

0088. An ILS receiver ;

- **a)** compares the difference in frequency of the two transmitted signals.
- **b)** measures the phase rotation of the two transmitted signals.
- **c)** measures the difference in depth of modulation of the two transmitted signals.
- **d)** measures the phase difference between the two transmitted signals.

Answer ;

c) measures the difference in depth of modulation of the two transmitted signals.

0089. An NDB is on a relative bearing of 316° from an aircraft. Given ;

Compass heading 270°
At aircraft deviation 2°W , Variation 30°E
At station Variation 28°E

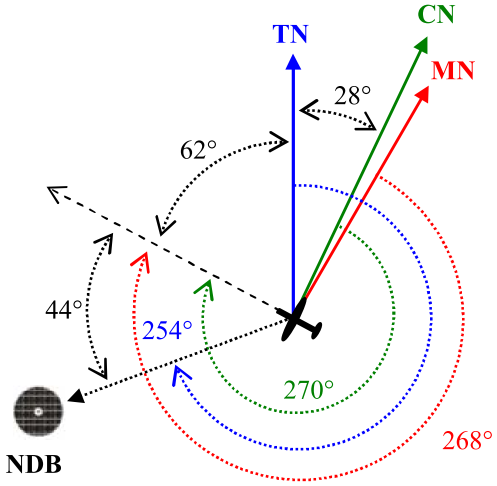
Calculate the true bearing of the NDB from the aircraft ;

- **a)** 254°
- **b)** 074°
- **c)** 072°
- **d)** 252°

Answer ;

a) 254°

Çözüm ;



MH = CH – Deviation W = 270° – 2° = 268°
TH = MH + Variation E = 268° + 30° = 298°
NDB'nin uçağa göre Relative bearing'i 316° ise ,
TB = RB + TH = 316° + 298° = 614°
614° – 360° = 254° olarak bulunur.

0090. An NDB transmits a signal pattern in the horizontal plane which is ;

- **a)** bi-lobal circular.
- **b)** a cardioid balanced at 30 Hz.
- **c)** a beam rotating at 30 Hz.
- **d)** omnidirectional.

Answer ;

d) omnidirectional.

0091. An OBS is set to 048 with a TO flag showing. The VOR deviation bar is showing almost full right deflection. Approximately what radial are you on ?

- **a)** 058
- **b)** 218
- **c)** 238
- **d)** 038

Answer ;

c) 238

Eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile farklı sektörde ise HSI veya CDI'da görülecek ibare TO olacaktır. Yani uçak 048 + 180 = 228 radyalin bulunduğu taraftadır. OBS'nin tam sağa , ki bu 10°dir , sapma göstermesi için uçağın Seçilen radyal (048) + 180 (228°) alanının dışında olması gereklidir. Cevap ; 238

0092. An Omni-bearing selector (OBS) shows full deflection to the left when within range of a serviceable VOR. What angular deviation are you from the selected radial ?

- **a)** less than 10°
- **b)** 10° or more
- **c)** 2.5 or more
- **d)** 1.5° or more

Answer ;

b) 10° or more

0093. An RMI indicates aircraft heading and bearing. To convert the RMI bearings of NDBs and VORs to true bearings the correct combination for the application of magnetic variation is ;

- **a)** NDB : beacon position , VOR : aircraft position
- **b)** NDB : beacon position , VOR : beacon position
- **c)** NDB : aircraft position , VOR : aircraft position
- **d)** NDB : aircraft position , VOR : beacon position

Answer ;

d) NDB : aircraft position , VOR : beacon position

0094. An RMI slaved to a remote indicating compass has gone unserviceable and is locked on to a reading of 090°. The tail of the VOR pointer shows 135°. The available information from the VOR is ;

- **a)** Radial unknown , relative bearing 225°.
- **b)** Radial 315° , relative bearing unknown.
- **c)** Radial unknown , relative bearing 045°.
- **d)** Radial 135° , relative bearing unknown.

Answer ;

d) Radial 135° , relative bearing unknown.

0095. Apart from radials and distances from VOR/DME stations , what information is required by the VOR/DME Area Navigation computer in order to calculate the wind ?

- **a)** Heading from the aircraft compass system and true airspeed from the air data computer.
- **b)** Heading from the aircraft compass system.
- **c)** Vertical speed from the air data computer.
- **d)** True airspeed from the air data computer.

Answer ;

a) Heading from the aircraft compass system and true airspeed from the air data computer.

0096. Area Navigation is a method of navigation that permits aircraft operation on any desired flight path within the limits of the capability of self contained systems. Which of the following systems is such a self contained system ?

- **a)** Inertial Reference System
- **b)** VOR/DME - system
- **c)** Global Positioning System
- **d)** DME/DME - system

Answer ;

a) Inertial Reference System

0097. Assuming a five dot display on either side of the CDI on the ILS localiser cockpit display , what does each of the dots represent approximately ;

- **a)** 1.5 degrees
- **b)** 0.5 degrees
- **c)** 2.0 degrees
- **d)** 2.5 degrees

Answer ;

b) 0.5 degrees

0098. Assuming a five dot display on either side of the ILS localiser cockpit display , what is the angular displacement of the aircraft from the localiser centreline when the CDI is deflected 2 dots to the right ?

- **a)** 1.0° to the left.
- **b)** 2.0° to the left.
- **c)** 1.0° to the right.
- **d)** 2.0° to the right.

Answer ;

a) 1.0° to the left.

0099. Assuming sufficient transmission power , the maximum range of a ground radar with a pulse repetition frequency of 450 pulses per second is ; (Given : velocity of light is 300 000 km/second)

- **a)** 1333 km
- **b)** 666 km
- **c)** 333 km
- **d)** 150 km

Teorik Mesafe = 300.000 km/s : (2 x PRF)
Teorik Mesafe = 300.000 : (2 x 450)
Teorik Mesafe = 300.000 : 900 = 333,3 ≈ 333 km

Answer ;

c) 333 km

0100. (Refer to Image 1)

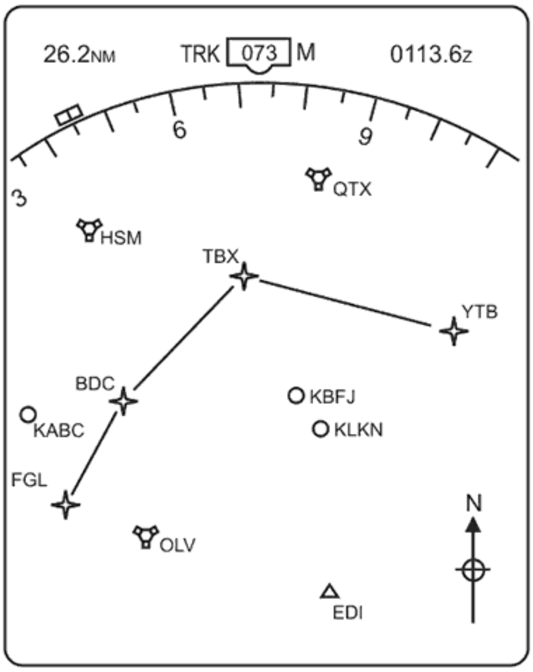


Image 1

At reference. The letters QTX and adjacent symbol indicate a ;

- **a)** TACAN
- **b)** Airport
- **c)** VOR
- **d)** VOR/DME/VORTAC

Answer ;

d) VOR/DME/VORTAC

0101. (Refer to Image 2)

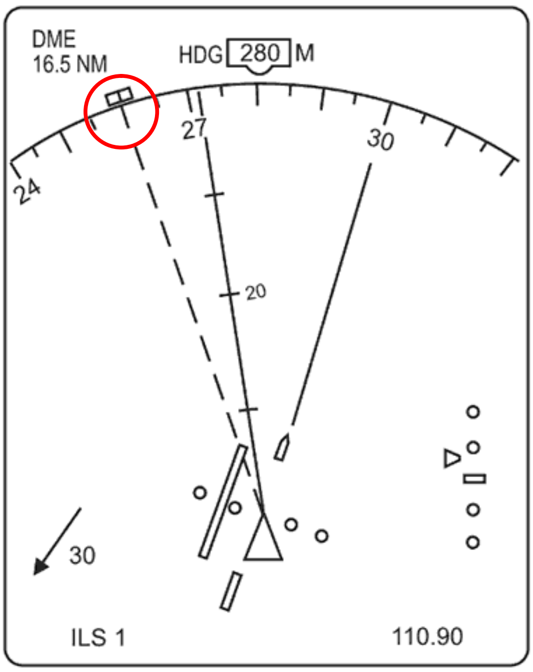


Image 2

At reference. What is the heading bug selected to ?

- **a)** 280º(M)
- **b)** 260º(M)
- **c)** 272º(M)
- **d)** 300º(M)

Answer ;

b) 260º (M)

0102. At what approximate height above the WGS-84 ellipsoid are NAVSTAR/GPS satellites circling the earth ?

a) 10900 km
b) 36000 km
c) 19500 km
d) 20200 km

Answer ;
d) 20200 km

0103. Basic RNAV requires a track-keeping accuracy of ;

a) +/- 5 NM or better throughout the flight.
b) +/- 5 NM or better for 95% of the flight time.
c) +/- 3 NM or better for 90% of the flight time.
d) +/- 2 NM or better for 75% of the flight time.

Answer ;
b) +/- 5 NM or better for 95% of the flight time.

0104. "Beam bends" in the ILS approach path are ;

a) slight curves that can be followed by large aircraft.
b) curves in the normal approach procedure because of "noise abatement".
c) curved approaches made with the aid of the instrument landing system.
d) curves in the glide path that are visible on the indicator , but change too fast to be followed by large aircraft.

Answer ;
a) slight curves that can be followed by large aircraft.

0105. Benefits of Area Navigation include ;

1. Shorter flight distance.
2. Reduction in fuel and flight time.
3. No radio contact within RNAV airspace.
4. Reduction in the number of ground training facilities.
5. Pilot choice of vertical and horizontal separations.

The combination regrouping all the correct statements is ;

a) 1 , 2 , 4 and 5.
b) 2 , 4 and 5.
c) 3 , 4 and 5.
d) 1 , 2 and 4.

Answer ;
d) 1 , 2 and 4.

0106. By selecting one VHF frequency , in the range of 108 to 112 MHz , on the NAV receiver ;

a) theta-theta information from enroute VOR's can be obtained.
b) rho-rho information from an ILS/DME can be obtained.
c) rho-theta information from enroute VOR/DME station can be obtained.
d) rho-theta information from a terminal VOR/DME can be obtained.

Answer ;
d) rho-theta information from a terminal VOR/DME can be obtained.

0107. By which of the following flight deck equipment are the waypoints inserted into an FMS RNAV-system by the pilot ?

a) Symbol generator.
b) Control Display Unit.
c) Navigation display.
d) Course deviation indicator.

Answer ;
b) Control Display Unit.

0108. Classify the marker from lower aural frequency to higher aural frequency ;

1. Inner marker (if available)
2. Middle marker
3. Outer marker

a) 2 - 1 - 3
b) 1 - 2 - 3
c) 3 - 2 - 1
d) 1 - 3 - 2

Answer ;
c) 3 - 2 - 1

0109. Comparing a parabolic reflector with a flat plate antenna of the same size ;

a) the parabolic reflector generates less side lobes than the flat plate antenna.
b) the parabolic reflector has a considerably smaller beam width.
c) the flat plate antenna has a considerably smaller beam width.
d) the flat plate antenna generates less side lobes than the parabolic reflector.

Answer ;
d) the flat plate antenna generates less side lobes than the parabolic reflector.

0110. Complete the following statement. Aircraft Surface movement Radar operates on frequencies in the (i) .... band employing an antenna that rotates at approximately (ii) .... revolutions per minute , it is (iii) .... possible to determine the type of aircraft from the return on the radar screen.

a) (i) SHF , (ii) 60 , (iii) sometimes.
b) (i) SHF , (ii) 10 , (iii) always.
c) (i) EHF , (ii) 30 , (iii) never.
d) (i) EHF , (ii) 100 , (iii) never.

Answer ;
a) (i) SHF , (ii) 60 , (iii) sometimes.

0111. Concerning ADF and NDB ;

a) ADF is a ground equipment and NDB can be a ground equipment or an airborne equipment.
b) NDB is a locator and ADF is an enroute nav-aid.
c) NDB is a ground equipment , and ADF is an airborne equipment.
d) ADF is a civilian equipment whereas NDB is a military equipment used by civilians too.

Answer ;
c) NDB is a ground equipment , and ADF is an airborne equipment.

0112. Concerning FMC databases ;

a) company database and FMC databases can not be used at the same time.
b) the navigation database may be customized for the specific airline operations.
c) only the performance database may be customized by the company.
d) only the navigation database may be customized by the company's cost index strategy.

Answer ;
b) the navigation database may be customized for the specific airline operations.

0113. Concerning the glidepath principle of operation in an ILS system , the needle of the indicator is centred when the difference in depth of modulation (DDM) is ;

a) maximum.
b) less than 90 Hz.
c) null.
d) 150 Hz or 90 Hz.

Answer ;
c) null.

0114. Concerning the glidepath principle of operation in an ILS , the difference in depth of modulation (DDM) will ;

a) increase with displacement above or below the glidepath.
b) increase with displacement above the glidepath and decrease with displacement below the glidepath.
c) decrease with linear displacement above or below the glidepath.
d) increase from center position to half full scale of the needle of the indicator and decrease until full scale of the needle.

Answer ;
a) increase with displacement above or below the glidepath.

0115. Concerning the localiser principle of operation in an ILS system , the difference in depth of modulation (DDM) ;

1. decreases with respect to the angular displacement from the centerline.
2. increases with right displacement from the centerline.
3. decreases with left displacement from the centerline.
4. increases linearly with displacement from the centreline.

The combination regrouping all the correct statements is ;

a) 1 and 4.
b) 2 and 3.
c) 1 and 3.
d) 2 and 4.

Answer ;
d) 2 and 4.

0116. Concerning the localiser principle of operation in an ILS system , the difference in depth of modulation (DDM) ;

- **a)** increases linearly with respect to the angular displacement from the centerline.
- **b)** decreases proportionally to the angular displacement from the centerline.
- **c)** decreases with respect to the angular displacement from the centerline.
- **d)** increases linearly with respect to the distance from the centerline.

Answer ;

a) increases linearly with respect to the angular displacement from the centerline.

0117. Concerning the localiser principle of operation in an ILS , the needle of the aircraft indicator is centred when the difference in depth of modulation (DDM) is ;

- **a)** less than 90 Hz.
- **b)** more than 150 Hz.
- **c)** null.
- **d)** maximum.

Answer ;

c) null.

0118. Concerning the NAVSTAR/GPS satellite navigation system , what is the meaning of the term "Receiver Autonomous Integrity Monitoring" (RAIM) ?

- **a)** It is a method whereby a receiver ensures the integrity of the Pseudo Random Noise (PRN) code transmitted by the satellites.
- **b)** It is a technique whereby the receivers of the world-wide distributed monitor stations (ground segment) automatically determines the integrity of the navigation message.
- **c)** It is a technique by which a receiver ensures the integrity of the navigation information.
- **d)** It is the ability of the GPS satellites to check the integrity of the data transmitted by the monitoring stations of the ground segment.

Answer ;

c) It is a technique by which a receiver ensures the integrity of the navigation information.

0119. Considering a primary radar system , what kind of aerials are used ?

- **a)** An omnidirectional antenna for transmitting , and a directional antenna for receiving.
- **b)** One directional antenna for transmitting and one for receiving.
- **c)** A directional antenna for transmitting , and an omnidirectional antenna for receiving.
- **d)** One directional antenna both for transmitting and for receiving.

Answer ;

d) One directional antenna both for transmitting and for receiving.

0120. Distance Measuring Equipment (DME) operates in the ;

- **a)** SHF band and uses frequency modulation techniques.
- **b)** UHF band and uses one frequency.
- **c)** VHF band and uses the principle of phase comparison.
- **d)** UHF band and uses two frequencies.

Answer ;

d) UHF band and uses two frequencies.

0121. DME channels operate in the frequency band which includes ;

- **a)** 600 MHz
- **b)** 300 MHz
- **c)** 110 MHz
- **d)** 1000 MHz

Answer ;

d) 1000 MHz

0122. DME channels utilise frequencies of approximately ;

- **a)** 600 MHz
- **b)** 110 MHz
- **c)** 1000 MHz
- **d)** 300 MHz

Answer ;

c) 1000 MHz

0123. Due to "Doppler" effect an apparent decrease in the transmitted frequency , which is proportional to the transmitter's velocity , will occur when ;

- **a)** the transmitter moves away from the receiver.
- **b)** the transmitter moves toward the receiver.
- **c)** there is no relative movement between the transmitter and the receiver.
- **d)** the transmitter and receiver move towards each other.

Answer ;

a) the transmitter moves away from the receiver.

0124. During a flight at FL 210 , a pilot does not receive any DME distance indication from a DME station located approximately 220 NM away. The reason for this is that the ;

- **a)** altitude is too high.
- **b)** aeroplane is below the "line of sight" altitude.
- **c)** range of a DME system is always less than 200 NM.
- **d)** aeroplane is circling around the station.

Answer ;

b) aeroplane is below the "line of sight" altitude.

0125. Echoes that do not change in distance from the antenna (relative speed zero) of a ground radar with a Moving Target Indicator (MTI) are dangerous for the following reason ;

- **a)** The moving target indicator (MTI) eliminates such echoes.
- **b)** The frequency of the transmitted radar signals changes too much by the Doppler effect.
- **c)** The MTI does not recognise this as a moving object due to the blind speed.
- **d)** The radar is not able to display two echoes moving in the same direction.

Answer ;

a) The moving target indicator (MTI) eliminates such echoes.

0126. EGNOS (European Geostationary Navigation Overlay System) is a form of ;

- **a)** Wide Area Differential GPS (WADGPS).
- **b)** stand-alone Global Navigation Satellite System (GNSS).
- **c)** Local Area Augmentation System (LAAS).
- **d)** Local Area Differential GPS (LADGPS).

Answer ;

a) Wide Area Differential GPS (WADGPS).

0127. Erratic indications may be experienced when flying towards a basic VOR/DME based Area Navigation System "Phantom Station" ;

- **a)** when the Phantom Station is out of range.
- **b)** when in the cone of silence overhead the Phantom Station.
- **c)** because , under adverse conditions (relative bearing to the Phantom Station other than 180°/360°) it takes the computer more time to calculate the necessary information.
- **d)** when operating at low altitudes close to the limit of reception range from the reference station.

Answer ;

d) when operating at low altitudes close to the limit of reception range from the reference station.

0128. Errors caused by the effect of coastal refraction on bearings at lower altitudes are maximum when the NDB is ;

- **a)** inland and the bearing crosses the coast at an acute angle.
- **b)** near the coast and the bearing crosses the coast at an acute angle.
- **c)** inland and the bearing crosses the coast at right angles.
- **d)** near the coast and the bearing crosses the coast at right angles.

Answer ;

a) inland and the bearing crosses the coast at an acute angle.

0129. Every 10 kt decrease in groundspeed , on a 3° ILS glidepath , will require an approximate ;

- **a)** increase in the aircraft's rate of descent of 100 FT/MIN.
- **b)** decrease in the aircraft's rate of descent of 50 FT/MIN.
- **c)** increase in the aircraft's rate of descent of 50 FT/MIN.
- **d)** decrease in the aircraft's rate of descent of 100 FT/MIN.

Answer ;

b) decrease in the aircraft's rate of descent of 50 FT/MIN.

0130. Factors liable to affect most NDB/ADF system performance and reliability include ;

a) static interference - station interference - latitude error.
b) static interference - night effect - absence of failure warning system.
c) height error - station interference - mountain effect.
d) coastal refraction - lane slip - mountain effect.

Answer ;
b) static interference - night effect - absence of failure warning system.

0131. False beams on the ILS glidepath are ;

a) only found below the correct glide slope.
b) only found above the correct glide slope.
c) only found if flying the backbeam ILS approach.
d) only found if more than 10º left of right of localiser centrline.

Answer ;
b) only found above the correct glide slope.

0132. For a conventional DME facility "Beacon Saturation" will occur whenever the number of aircraft interrogations exceeds ;

a) 80
b) 60
c) 100
d) 200

Answer ;
c) 100

0133. For any given circumstances , in order to double the effective range of a primary radar the power output must be increased by a factor of ;

a) 16
b) 8
c) 4
d) 2

Answer ;
a) 16

0134. For what reason is a mask angle set-up in a GPS-receiver ?

a) To deny the receiver the use of GPS-satellites with an elevation less than the mask angle.
b) To make the receiver select GPS-satellites wih a certain optimum elevation which is about equal to the mask angle.
c) To select from all visible GPS-satellites the satellite-configuration which results in the smallest GDOP.
d) To eliminate the reception of the signal from GPS-satellites with an elevation about 90º.

Answer ;
a) To deny the receiver the use of GPS-satellites with an elevation less than the mask angle.

0135. From which of the following combination of navigational sources provide enough information to the RNAV equipment to calculate the wind vector ;

a) GPS and Compass output.
b) IRS and GPS.
c) IRS and air data computer.
d) Compass system and IRS.

Answer ;
c) IRS and air data computer.

0136. Full deflection on a glide slope indicator indicates that the aircraft is ;

a) 2.5° above or below the correct glide path.
b) 0.5° above or below the correct glide path.
c) 0.7° above or below the correct glide path.
d) 1.25° above or below the correct glide path.

Answer ;
c) 0.7° above or below the correct glide path.

0137. Full scale deflection of the localiser needle indicates that the aircraft is approximately ;

a) 10° offset from the localiser centreline.
b) 5° offset from the localiser centreline.
c) 2.5° offset from the localiser centreline.
d) 1.25° offset from the localiser centreline.

Answer ;
c) 2.5° offset from the localiser centreline.

0138. Garbling in mode A and C may occur ;

- **a)** when two or more aircraft are in approximately the same direction from the interrogator with a difference in slant range of less than 1.7 NM.
- **b)** when two or more aircraft are in approximately the same direction from the ground station with a difference in slant range of more than 1.7 NM.
- **c)** with two or more aircraft in different directions from the ground station , at the same altitude and a slant range of less than 1.7 NM.
- **d)** with two or more aircraft in different directions from the interrogator , at the same altitude and a slant range of more than 1.7 NM.

Answer ;

a) when two or more aircraft are in approximately the same direction from the interrogator with a difference in slant range of less than 1.7 NM.

0139. Given ;

Aircraft position 34°15'N 098°E , magnetic variation 28°W , FL 280.
PTC VOR/DME position 36°12'N 098°E , magnetic variation 13°E.

In order to read the most accurate ground speed given by the DME receiver from his present position , the pilot must fly on which PTC Radial ?

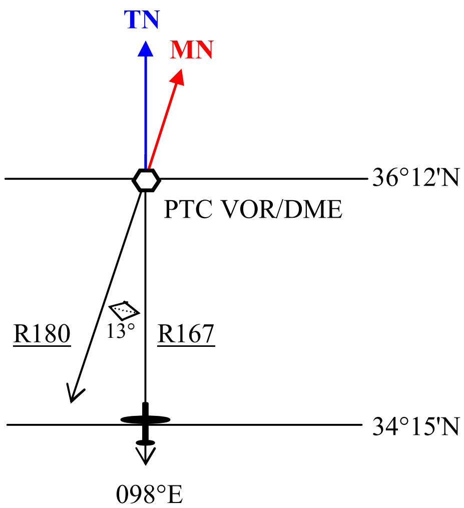
- **a)** 167°
- **b)** Aircraft will not receive DME information from PTC due to the line sight rule.
- **c)** 332°
- **d)** 193°

Answer ;

a) 167°

Çözüm ;

Burada uçağın seviyesi ve bulunduğu yerdeki variation değerinin hesaplamalarda bir önemi yoktur. VOR söz konusu olduğundan VOR'ın bulunduğu yerdeki variation değerini (13°E) hesaba katacağız. Uçak ve PTC VOR/DME aynı meridyen üzerinde yer almaktadır. Variation değeri 13°E olduğuna göre MN , TN'un doğusundadır , dolayısıyla uçağın bulunacağı radyalde 13°lik bir farklılıkta olmalıdır. Uçağın VOR'a göre true bearing'i 180 ve de VOR radyalleri Manyetik bearing'ler olduğuna göre ;
MB = TB – Variation East = 180 – 13 = 167° olarak bulunur. Yani ;
Uçak doğru bir yer sürati bilgisi almak için 167 radyalinde VOR'a doğru uçmalıdır.



0140. Given ;

Aircraft position 52°09'S 024°E , magnetic variation 14°W , FL 310.
BIT VOR/DME position 54°42'S 024°E , magnetic variation 14°E.

In order to read the most accurate ground speed given by the DME receiver from his present position , the pilot must follow which BIT radial ?

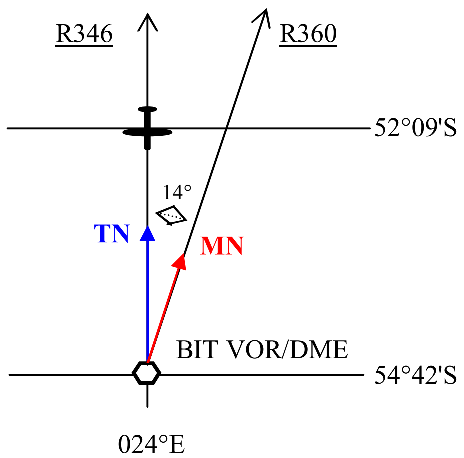
- **a)** 014°
- **b)** 194°
- **c)** 166°
- **d)** 346°

Answer ;

d) 346°

Çözüm ;

Burada uçağın seviyesi ve bulunduğu yerdeki variation değerinin hesaplamalarda bir önemi yoktur. VOR söz konusu olduğundan VOR'ın bulunduğu yerdeki variation değerini (14°E) hesaba katacağız. Uçak ve BIT VOR/DME aynı meridyen üzerinde yer almaktadır. Variation değeri 14°E olduğuna göre MN , TN'un doğusundadır , dolayısıyla uçağın bulunacağı radyalde 14°lik bir farklılıkta olmalıdır. Uçağın VOR'a göre true bearing'i 360 ve de VOR radyalleri Manyetik bearing'ler olduğuna göre ;
MB = TB – Variation East = 360 – 14 = 346° olarak bulunur. Yani ;
Uçak doğru bir yer sürati bilgisi almak için 346 radyalinde VOR'a doğru uçmalıdır.



0141. Given ;

Course Deviation Indicator (CDI) for a VOR is selected to 090°
From/To indicator indicates "TO"
CDI needle is deflected halfway to the right

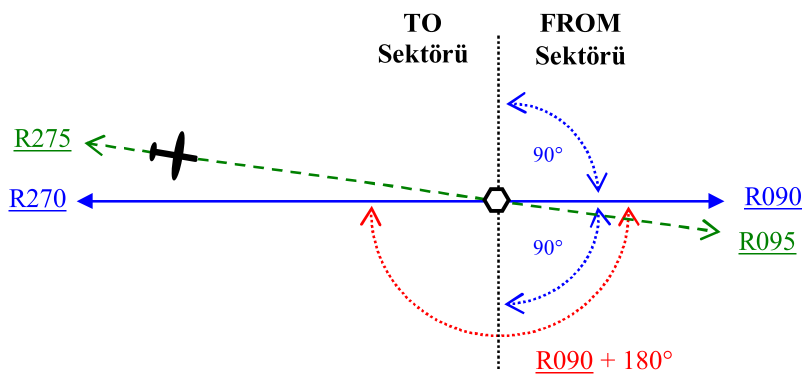
On what radial is the aircraft ?

- **a)** 265
- **b)** 095
- **c)** 275
- **d)** 085

Answer ;

c) 275

Çözüm ;



Uçağın CDI'ında (Course Deviation Indicator) Course 090° seçilidir. Burada uçuş başının bir önemi yoktur. Eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile aynı sektörde ise CDI'da görülecek ibare FROM olacaktır. Diğer durumda eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile farklı sektörde ise CDI'da görülecek ibare bu sefer TO olacaktır. CDI üzerinde görünen sapma yarı ölçektir , yani 5 derecelik sapma görülecektir ve bu ibrenin sağda gözükmesi (Couse göstergesinin sağı) için halihazırda bulunduğu radyalin , seçmiş olduğu radyal + 180° alanı içinde olmaması gereklidir. Cevap ; 275 radyal.

0142. Given ;

Aircraft heading 160°(M)
Aircraft is on radial 240° from a VOR
Selected course on HSI is 250°

The HSI indications are deviation bar ;

- **a)** ahead of the aeroplane symbol with the TO flag showing.
- **b)** behind the aeroplane symbol with the TO flag showing.
- **c)** behind the aeroplane symbol with the FROM flag showing.
- **d)** ahead of the aeroplane symbol with the FROM flag showing.

Answer ;

c) behind the aeroplane symbol with the FROM flag showing.

0143. Given ;

Magnetic heading 280°
VOR radial 090°

What bearing should be selected on the omni-bearing selector in order to centralise the VOR deviation needle with a "TO" indication ?

- **a)** 090°
- **b)** 280°
- **c)** 100°
- **d)** 270°

Answer ;

d) 270°

0144. Given ;

W/V (T) 230 / 20 kt , Var. 6E , TAS 80 kt

What relative bearing from an NDB should be maintained in order to achieve an outbound course of 257°(M) from overhead the beacon ?

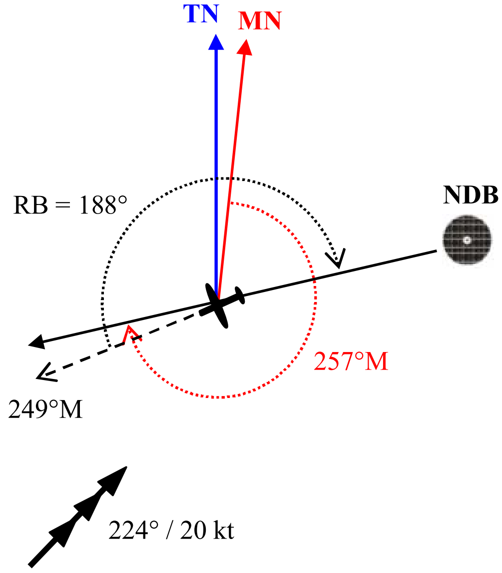
- **a)** 172
- **b)** 188
- **c)** 352
- **d)** 008

Answer ;

b) 188

Çözüm ;

Soruda uçak NDB'den of 257°M rotasında uzaklaşırken bu hattı muhafaza edebilmek için NDB'nin uçağa göre Relative bearing'inin ne olması gerektiği sorulmaktadır. Manyetik Kuzeye göre olan değerler üzerinden işlem yapacağımızdan 230°T olan rüzgar yönü Manyetik değer olarak = 230°T – Variation E = 230°T – 6 = 224°M'dir.
Drift'i hesaplamak için ; DR kompitüründe Grommet'in 90'ı (TAS) gösterdiği durumda rüzgar yönü olan 224'ü True Index'e hizalayın. Rüzgar hızı olan 20 knot'ı Grommet'in altına doğru 20 birim olarak işaretleyin. Manyetik course olan 257'yi True Index'e hizaladığınızda işaretlediğimiz nokta 8°R Drift göstermektedir , yani rotanın muhafazası için sola 8 derecelik baş değişikliği yapılmış olmalıdır.
MH = Manyetik rota – Drift = 257° – 8° = 249°M olur.
Rüzgar faktörü olmasaydı uçak NDB'den uzaklaşırken NDB'nin uçağa göre Relative bearing'i 180° olacaktı , ancak rüzgar nedeniyle oluşan Drift sebebiyle oluşan sola 8 derecelik Manyetik baş farkı sonucu RB = 180 + 8 = 188 olur.



0145. Given ;

VOR station position N61° E025° , variation 13°E
Estimated position of an aircraft N59° E025° , variation 20°E

What VOR radial is the aircraft on ?

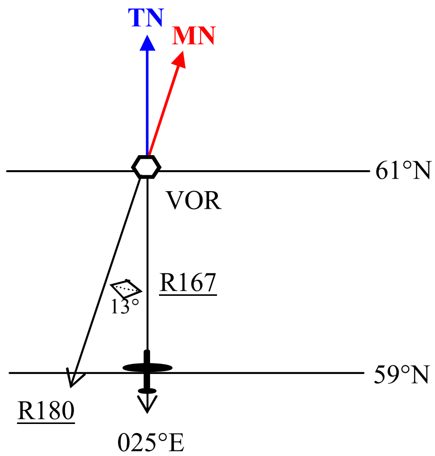
- **a)** 167°
- **b)** 347°
- **c)** 193°
- **d)** 160°

Answer ;

a) 167°

Çözüm ;

VOR söz konusu olduğundan VOR'ın bulunduğu yerdeki variation değerini (13°E) hesaba katacağız. Uçak ve VORaynı meridyen üzerinde yer almaktadır. Variation değeri 13°E olduğuna göre MN , TN'un doğusundadır , dolayısıyla uçağın bulunacağı radyalde 13°lik bir farklılıkta olmalıdır. Uçağın VOR'a göre true bearing'i 180 ve de VOR radyalleri Manyetik bearing'ler olduğuna göre ;
MB = TB – Variation East = 180 – 13 = 167° olarak bulunur.



0146. GPS satellites transmit on two L-band frequencies with different types of signals. Which of these are generally available for use by civil aviation ?

- **a)** L1-coarse acquisition (C/A) with selected availability (S/A).
- **b)** L2-coarse acquisition (C/A).
- **c)** L1-precise (P).
- **d)** L2-for communications purpose.

Answer ;

a) L1-coarse acquisition (C/A) with selected availability (S/A).

0147. GPS satellites transmit on two L-band frequencies with different types of signals. Which of these are generally available for use by civil aviation ?

- **a)** L2 - coarse acquisition (C/A).
- **b)** L2 - precise (P).
- **c)** L1 - precise (P).
- **d)** L1 - coarse acquisition (C/A).

Answer ;

d) L1 - coarse acquisition (C/A).

0148. GPS system satellites transmit their signals on two carrier waves 1575 MHz and 1227 MHz and supply two possible codes accessible according to user (civil or military). Commercial aviation uses ;

- **a)** the two carrier waves and one public code.
- **b)** only the 1575 MHz carrier wave and two codes.
- **c)** only the 1227 MHz carrier wave and one code.unique
- **d)** only the 1575 MHz carrier wave and one code.

Answer ;

d) only the 1575 MHz carrier wave and one code.

0149. How can a DME-interrogator distinguish between its own reply pulse-pairs and the reply pulse-pairs of other aircraft in the area , using the same DME-station ?

- **a)** The Pulse Repetition Frequency of the pulse-pairs transmitted by the interrogator varies , for each interrogator , in a unique rhythm.
- **b)** The time-interval between both pulses of consecutive pulse-pairs transmitted by the interrogator varies , for each interrogator , in a unique pattern.
- **c)** The DME-transponder uses a slightly different , randomly varying , delay for each interrogating aircraft.
- **d)** On the Y-channel the time-interval between the pulses of an interrogator.pulse-pair is 36 msec and of a transponder pulse-pair 30 msec.

Answer ;

a) The Pulse Repetition Frequency of the pulse-pairs transmitted by the interrogator varies , for each interrogator , in a unique rhythm.

0150. How does a Mode S interrogator identify aircraft ?

- **a)** a 24 bit identifier , giving over 16 million combinations.
- **b)** the four letter SELCAL code.
- **c)** the P3 pulse.
- **d)** an eight bit identifier , which gives 212 unique combinations.

Answer ;

a) a 24 bit identifier , giving over 16 million combinations.

0151. How does a NAVSTAR/GPS satellite navigation system receiver recognise which of the received signals belongs to which satellite ?

- **a)** The receiver detects the direction from which the signals are received and compares this information with the calculated positions of the satellites.
- **b)** Each satellite transmits its signal on a separate frequency.
- **c)** The Doppler shift is unique to each satellite.
- **d)** Each satellite transmits its signal , on common frequencies , with an individual Pseudo Random Noise code.

Answer ;

d) Each satellite transmits its signal , on common frequencies , with an individual Pseudo Random Noise code.

0152. How does a receiver of the NAVSTAR/GPS satellite navigation system determine the elevation and azimuth data of a satellite relative to the location of the antenna ?

- **a)** The data is based on the direction to the satellite determined at the location of the antenna.
- **b)** It calculates it by using Almanac data transmitted by the satellites.
- **c)** The data is determined by the satellite and transmitted together with the navigation message.
- **d)** The data is stored in the receiver together with the Pseudo Random Noise (PRN) code.

Answer ;

b) It calculates it by using Almanac data transmitted by the satellites.

0153. How does a VOR/DME Area Navigation system obtain DME information ?

- **a)** The VOR/DME Area Navigation system has its own VHF NAV tuner and the system itself tunes the DME stations providing the best angular position lines.
- **b)** The VOR/DME Area Navigation system uses whatever stations are tuned on the aircraft’s normal VHF NAV selector.
- **c)** The VOR/DME Area Navigation System has its own VHF NAV tuner and it always tunes the DME stations closest to the aircraft position.
- **d)** The pilot tunes the closest VOR/DME stations within range on the VOR/DME Area navigation control panel.

Answer ;

a) The VOR/DME Area Navigation system has its own VHF NAV tuner and the system itself tunes the DME stations providing the best angular position lines.

0154. How does a VOR/DME Area Navigation system selects the DME-stations to be used for positioning ?

a) The VOR/DME Area Navigation system uses whatever stations are tuned on the aircraft's normal VHF NAV selector.
b) The pilot tunes the closest VOR/DME stations within range on the VOR/DME Area navigation control panel.
c) The VOR/DME Area Navigation System has its own VHF NAV tuner and it always tunes the DME stations closest to the aircraft position.
d) The VOR/DME Area Navigation system has its own NAV tuner and the system itself tunes the DME stations providing the most accurate position.

Answer ;
d) The VOR/DME Area Navigation system has its own NAV tuner and the system itself tunes the DME stations providing the most accurate position.

0155. How does the Electronic Flight Instrument System display of a B737-400 respond to the failure of a VHF navigation (VOR) receiver ?

a) The deviation bar and/or pointer change colour to red and flash intermittently.
b) The pointer flashes and a VOR 1 or 2 failure warning bar appears.
c) It removes the associated magenta deviation bar and/or pointer from the display.
d) The pointer rotates around the display and a VOR 1 or 2 failure warning bar appears.

Answer ;
c) It removes the associated magenta deviation bar and/or pointer from the display.

0156. How long does it take a NAVSTAR/GPS satellite to orbit the earth ?

a) Approximately 12 hours (1/2 of a sidereal day).
b) Approximately 24 hours (one sidereal day).
c) 365 days because the satellites are located in a geostationary orbit.
d) 12 days.

Answer ;
a) Approximately 12 hours (1/2 of a sidereal day).

0157. How many clocks are installed in each NAVSTAR GPS satellite ?

a) 3
b) 1
c) 2
d) 4

Answer ;
d) 4

0158. How many operational satellites are required for Full Operational Capability (FOC) of the satellite navigation system NAVSTAR/GPS ?

a) 24
b) 12
c) 30
d) 18

Answer ;
a) 24

0159. How many satellites form the nominal NAVSTAR GPS constellation ?

a) 36
b) 24
c) 12
d) 6

Answer ;
b) 24

0160. ICAO Annex 11 defines Area Navigation (RNAV) as a method of navigation which permits aircraft operation on any desired flight path ;

- **a)** within the coverage of station-referenced navigation aids or within the limits of the capability of self-contained aids , or a combination of these.
- **b)** outside the coverage of station-referenced navigation aids provided that it is equipped with a minimum of two serviceable self-contained navigation aids.
- **c)** outside the coverage of station-referenced navigation aids provided that it is equipped with a minimum of one serviceable self-contained navigation aid.
- **d)** within the coverage of station-referenced navigation aids provided that it is equipped with a minimum of one serviceable self-contained navigation aid.

Answer ;

a) within the coverage of station-referenced navigation aids or within the limits of the capability of self-contained aids , or a combination of these.

0161. ICAO specifications are that range errors indicated by Distance Measuring Equipment (DME) should not exceed ;

- **a)** + or – 0.5 NM or 3% of the distance measured whichever is the greater.
- **b)** + or – 0.25 NM plus 1.25% of the distance measured.
- **c)** + or – 0.25 NM plus 3% of the distance measured up to a maximum of 5 NM.
- **d)** + or – 1.25 NM plus 0.25% of the distance measured.

Answer ;

b) + or – 0.25 NM plus 1.25% of the distance measured.

0162. If a failed RMI rose is stuck on 090° and the ADF pointer indicates 225° , the relative bearing to the station will be ;

- **a)** 135°.
- **b)** 315°.
- **c)** 225°.
- **d)** Impossible to read , due to the RMI failure.

Answer ;

a) 135°.

0163. If an aircraft flies along a VOR radial it will follow a ;

- **a)** line of constant bearing.
- **b)** great circle track.
- **c)** constant magnetic track.
- **d)** rhumbline track.

Answer ;

b) great circle track.

0164. If the (angular) displacement of an aircraft (with respect to the localiser centerline) doubles (e.g. from 1° dot 2°) the measured Difference in Depth of Modulation ;

- **a)** doubles.
- **b)** increases fourfold.
- **c)** remains unchanged.
- **d)** is halved.

Answer ;

a) doubles.

0165. If VOR bearing information is used beyond the published protection range , errors could be caused by ;

- **a)** sky wave interference from distant transmitters on the same frequency.
- **b)** interference from other transmitters.
- **c)** sky wave interference from the same transmitter.
- **d)** noise from precipitation static exceeding the signal strength of the transmitter.

Answer ;

b) interference from other transmitters.

0166. If you are flying a back course ILS , you are flying a ;

- **a)** non precision approach on the reciprocal runway of the precision approach runway.
- **b)** non precision approach on the precision approach end of the runway.
- **c)** precision approach on the reciprocal runway of the precision approach runway.
- **d)** precision approach on the precision approach runway.

Answer ;

a) non precision approach on the reciprocal runway of the precision approach runway.

0167. Ignoring pulse length and fly-back , a radar facility designed to have a maximum unambiguous range of 50 km will have a PRF (pulses per second) of ;

a) 3000
b) 330
c) 6000
d) 167

Teorik Mesafe = 300.000 km/s : (2 x PRF)
50 = 300.000 : (2 x PRF)
PRF = 300.000 : (2 x 50) = 300.000 : 100 = 3000 pps

Answer ;
a) 3000

0168. Ignoring pulse length , the maximum pulse repetition frequency (PRF) that can be used by a primary radar facility to detect targets unambiguously to a range of 200 NM is ; (pps = pulses per second)

a) 405 pps
b) 375 pps
c) 308 pps
d) 782 pps

200 x 1,852 = 370,4 km
Teorik Mesafe = 300.000 km/s : (2 x PRF)
370,4 = 300.000 : (2 x PRF)
PRF = 300.000 : (2 x 370,4) = 300.000 : 740,8 = 404,96 ≈ 405 pps

Answer ;
a) 405 pps

0169. ILS produces ;

a) a radiation pattern which is amplitude modulated by a 90 Hz and a 150 Hz signal.
b) two lobes modulated in frequency by a 90 Hz and a 150 Hz signal.
c) a radiation pattern which is amplitude modulated by the VHF fequency of the ILS.
d) a 90 Hz lobe and a 150 Hz lobe which are amplitude modulated by the VHF frequency of the ILS.

Answer ;
a) a radiation pattern which is amplitude modulated by a 90 Hz and a 150 Hz signal.

0170. ILS transmitters use the ;

a) VHF band only.
b) VHF , UHF and HF bands.
c) UHF and VHF bands.
d) UHF band only.

Answer ;
c) UHF and VHF bands.

0171. In a 2D RNAV system you have entered the DME and VOR data for two waypoints. What do you use to work out the cross track errors when en-route from one to the other ?

a) Use the autmatically computed values on the CDI/HSI.
b) The pilot is presented with VOR/DME information which must then be correlated to the pressure instruments to determine the effect of altitude.
c) The pilot takes the VOR and DME information and computes it himself.
d) The pilot will have to calculate the wind and apply it to the VOR/DME information.

Answer ;
a) Use the autmatically computed values on the CDI/HSI.

0172. In a primary pulse radar you have ;

a) a directional aerial for transmission and another one for reception.
b) an omni-directional aerial for transmission and directional aerial for reception.
c) a directional aerial for transmission and omni-directional aerial for reception.
d) a directional aerial for both transmission and reception.

Answer ;
d) a directional aerial for both transmission and reception.

0173. In a primary radar using pulse technique , pulse length determines ;

a) maximum measurable range.
b) target discrimination.
c) minimum measurable range.
d) beam width.

Answer ;
c) minimum measurable range.

0174. In a primary radar using pulse technique , pulse recurrence frequency (PRF)/pulse recurrence rate (PRR) determines ;

a) target discrimination.
b) minimum range.
c) beam width.
d) maximum theoretical range.

Answer ;

d) maximum theoretical range.

0175. In a primary radar using pulse technique , the ability to discriminate between targets in azimuth is a factor of ;

a) aerial rotation rate.
b) beam width.
c) Pulse Recurrence Rate (PRR).
d) pulse length.

Answer ;

b) beam width.

0176. In a Satellite-Assisted Navigation System (GNSS/GPS) , a fix is obtained by ;

a) measuring the time taken for a minimum number of satellite's transmissions , in known positions , to reach the aircraft's receiver.
b) measuring the time taken for an aircraft's transmissions to travel to a number of satellites , in known positions , and return to the aircraft's receiver.
c) measuring the pulse lengths of signals received from a minimum number of satellites received in a specific sequential order.
d) the aircraft's receiver measuring the phase angle of signals received from a number of satellites in known positions.

Answer ;

a) measuring the time taken for a minimum number of satellite's transmissions , in known positions , to reach the aircraft's receiver.

0177. In a Satellite-Assisted Navigation system (GNSS/GPS) , a position line is obtained by ;

a) timing the period that is taken for a satellite's transmission to reach the aircraft's receiver.
b) the aircraft's receiver measuring the time difference between signals received from a minimum number of satellites.
c) timing the period that is taken for a transmission from the aircraft's transmitter/receiver to reach and return from a satellite in a known position.
d) the aircraft's receiver measuring the phase angle of the signal received from a satellite in a known position.

Answer ;

a) timing the period that is taken for a satellite's transmission to reach the aircraft's receiver.

0178. In accordance with Doc 8168 , a pilot flying an NDB approach must achieve a tracking accuracy within .... of the published approach track.

a) +/-2°
b) +/-10°
c) +/-2.5°
d) +/-5°

Answer ;

d) +/-5°

0179. In accordance with ICAO Annex 10 the GPS NAVSTAR position accuracy in SPS should be for 95% of the time ;

a) 13 metres horizontally.
b) 5 metres vertically.
c) 30 metres horizontally.
d) 22 metres 3-D.

Answer ;

a) 13 metres horizontally.

0180. In accordance with ICAO Annex 10 the NAVSTAR/GPS global average 95% position accuracy in SPS should be ;

a) 13 m horizontally
b) 5 m vertically
c) 30 m horizontally
d) 22 m in 3D

Answer ;

a) 13 m horizontally

0181. In Airborne Weather Radar (AWR) , the main factors which determine whether a cloud will be detected are ;

a) rotational speed of radar scanner , range from cloud.
b) range from cloud , wavelength/frequency used.
c) size of the water drops , wavelength/frequency used.
d) size of the water drops , diameter of radar scanner.

Answer ;
c) size of the water drops , wavelength/frequency used.

0182. In an Airborne Weather Radar that has a colour cathode ray tube (CRT) increasing severity of rain and turbulence is generally shown by a change of colour from ;

a) green to red to black.
b) yellow to orange to red.
c) yellow to amber to blue.
d) green to yellow to red.

Answer ;
d) green to yellow to red.

0183. In an Airborne Weather Radar the areas of greatest turbulence are usually indicated on the screen by ;

a) colour zones of green and yellow.
b) areas which are coloured black.
c) colour zones of red and magenta.
d) blank areas where there is no colour.

Answer ;
c) colour zones of red and magenta.

0184. In an Electronic Flight Instrument System (EFIS) data relating primarily to navigation in the FMC is provided by ;

a) GPS
Aircraft Weather Radar
Navigation radios
b) Inertial Reference Systems
Aircraft Weather Radar
Navigation radios
c) Inertial Reference Systems
Navigation radios
Terrain Collision Alerting System
d) Navigation radios
GPS
Inertial Reference Systems

Answer ;
d) Navigation radios
GPS
Inertial Reference Systems

0185. In an ILS , concerning the localiser principle of operation , the difference in depth of modulation (DDM) will ;

a) increase with displacement from the centerline.
b) decrease with displacement from the centerline.
c) increase from center position to half full scale of te needle of the indicator and decrease until full scale of the needle.
d) increase with left displacement from the centerline and decrease with right displacement from the centerline.

Answer ;
a) increase with displacement from the centerline.

0186. In an ILS system , concerning the glidepath principle of operation , the difference in depth of modulation (DDM) ;

1. decrease with angular displacement below the glidepath.
2. increase with angular displacement above the glidepath.
3. decrease with angular displacement above the glidepath.
4. increase with angular displacement below the glidepath.

The combination regrouping all the correct statements is ;

a) 1 and 4.
b) 2 and 4.
c) 1 and 3.
d) 2 and 3.

Answer ;
b) 2 and 4.

0187. In civil aviation , the height value computed by the receiver of the satellite navigation system NAVSTAR/GPS is the ;

a) flight level.
b) height above the WGS-84 ellipsoid.
c) geometric height above ground.
d) height above Mean Sea Level (MSL).

Answer ;
b) height above the WGS-84 ellipsoid.

0188. In flight , a pilot can improve the range of his transmission with a VDF operator by ;

a) speaking louder.
b) decreasing altitude.
c) increasing altitude.
d) flying out of clouds.

Answer ;
c) increasing altitude.

0189. In ISA conditions , approximately what is the maximum theoretical range at which an aircraft at FL210 may expect to receive signals from a VOR facility sited 340 feet above mean sea level ?

a) 204 NM
b) 183 NM
c) 245 NM
d) 163 NM

Teorik olarak maksimum alış menzili (NM) = 1,25 x ($\sqrt{H1} + \sqrt{H2}$)
= 1,25 x ($\sqrt{21000} + \sqrt{340}$)
= 1,25 x (144,91 + 18,43) = 1,25 x 163,34 = 204,17 ≈ 204 NM

Answer ;
a) 204 NM

0190. In ISA conditions , what is the maximum theoretical range at which an aircraft at FL80 can expect to obtain bearings from a ground VDF facility sited 325 ft above MSL ?

a) 114 NM
b) 134 NM
c) 107 NM
d) 158 NM

Teorik olarak maksimum alış menzili (NM) = 1,25 x ($\sqrt{H1} + \sqrt{H2}$)
= 1,25 x ($\sqrt{8000} + \sqrt{325}$)
= 1,25 x (89,44 + 18,02) = 1,25 x 107,46 = 134,3 ≈ 134 NM

Answer ;
b) 134 NM

0191. In order to ascertain whether a cloud return on an Aircraft Weather Radar (AWR) is at or above the height of the aircraft , the tilt control should be set to ; (Assume a beam width of 5°)

a) 0°
b) 2.5° up
c) 2.5° down
d) 5° up

Answer ;
b) 2.5° up

0192. In order to carry out an Independent three-dimensional fix , Receiver Autonomous Integrity Monitoring (RAIM) and failure detection and exclusion of any faulty satellite , signal reception is required from a minimum number of how many satellites ?

a) 4
b) 7
c) 6
d) 5

Answer ;
c) 6

0193. In order to enter a phantom waypoint that is designated by a VOR/DME simple RNAV system , the VOR/DME ;

a) does not have to be in range when entered or used.
b) has to be positively identified by one of the pilots.
c) must be in range.
d) does not have to be in range when entered but must be when used.

Answer ;
d) does not have to be in range when entered but must be when used.

0194. In order to indicate an emergency situation , the aircraft Secondary Surveillance Radar (SSR) transponder should be set to ;

a) 7600
b) 7500
c) 7000
d) 7700

Answer ;
d) 7700

0195. In order to indicate radio failure the aircraft SSR transponder should be selected to code ;

a) 7000
b) 7600
c) 7500
d) 7700

Answer ;
b) 7600

0196. In order to indicate unlawful interference with the planned operation of the flight , the aircraft Secondary Surveillance Radar (SSR) transponder should be selected to ;

a) 7000
b) 7600
c) 7500
d) 7700

Answer ;
c) 7500

0197. In order to measure the radial from a VOR , the aircraft VOR receiver ;

a) measures the time difference between sending the interrogation signal and receiving the transponder signal.
b) uses pulse technique to determine the radial.
c) measures the time difference between reception of the two signals transmitted from the ground installation.
d) measures the phase difference between the reference phase and the variable phase of the signal.

Answer ;
d) measures the phase difference between the reference phase and the variable phase of the signal.

0198. In order to obtain an ADF bearing on a system using sense and loop aerials , the ;

a) signal must be received by both the sense and loop aerials.
b) sense aerial must be tuned separately.
c) BFO switch must be selected to “ON”.
d) mode selector should be switched to “loop”.

Answer ;
a) signal must be received by both the sense and loop aerials.

0199. In order to plot a bearing from a VOR station , a pilot needs to know the magnetic variation ;

a) at both the VOR and aircraft.
b) at the half-way point between the aircraft and the station.
c) at the VOR.
d) at the aircraft location.

Answer ;
c) at the VOR.

0200. In relation to Area Navigation Systems (RNAV) , which of the following is an Air Data input ?

a) True airspeed.
b) VOR/DME radial/distance.
c) Doppler drift.
d) Inertial Navigation System (INS) position.

Answer ;
a) True airspeed.

0201. In relation to primary radar , what does the term Pulse Recurrence Frequency signify ?

a) The number of revolutions performed by the radar antenna per minute.
b) The number of pulses transmitted per second.
c) The time between each transmission of pulses.
d) The radar frequency used.

Answer ;
b) The number of pulses transmitted per second.

0202. In relation to radar systems that use pulse technology , the term “Pulse Recurrence Rate (PRR)” signifies the ;
- **a)** delay after which the process re-starts.
- **b)** number of pulses per second.
- **c)** ratio of pulse period to pulse width.
- **d)** the number of cycles per second.

Answer ;
b) number of pulses per second.

0203. In relation to the NAVSTAR/GPS satellite navigation system , “Search the Sky” is a ;
- **a)** continuous procedure performed by the receiver that searches the sky for satellites rising above the horizon.
- **b)** procedure that starts after switching on a receiver if there is no stored satellite data available.
- **c)** procedure performed by the receiver to recognise new satellites becoming operational.
- **d)** continuous process by the ground segment to monitor the GPS satellites.

Answer ;
b) procedure that starts after switching on a receiver if there is no stored satellite data available.

0204. In relation to the NAVSTAR/GPS satellite navigation system , what is involved in the differential technique (D-GPS) ?
- **a)** The difference between signals transmitted on the L1 and L2 frequencies are processed by the receiver to determine an error correction.
- **b)** Receivers from various manufacturers are operated in parallel to reduce the characteristic receiver noise error.
- **c)** Signals from satellites are received by 2 different antennas which are located a fixed distance apart. This enables a suitable receiver on the aircraft to recognise and correct for multipath errors.
- **d)** Fixed ground stations compute position errors and transmit correction data to a suitable receiver on the aircraft.

Answer ;
d) Fixed ground stations compute position errors and transmit correction data to a suitable receiver on the aircraft.

0205. In relation to the satellite navigation system NAVSTAR/GPS , “All in View” is a term used when a receiver ;
- **a)** is receiving the signals of all visible satellites but tracking only those of the 4 with the best geometric coverage.
- **b)** requires the signals of all visible satellites for navigation purposes.
- **c)** is tracking more than the required 4 satellites and can instantly replace any lost signal with another already being monitored.
- **d)** is receiving and tracking the signals of all 24 operational satellites simultaneously.

Answer ;
c) is tracking more than the required 4 satellites and can instantly replace any lost signal with another already being monitored.

0206. In relation to the satellite navigation system NAVSTAR/GPS , the term “inclination” denotes the angle between the ;
- **a)** orbital plane and the earth’s axis.
- **b)** orbital plane and the equatorial plane.
- **c)** horizontal plane at the location of the receiver and the orbital plane of a satellite.
- **d)** horizontal plane at the location of the receiver and the direct line to a satellite.

Answer ;
b) orbital plane and the equatorial plane.

0207. In relation to the satellite navigation system NAVSTAR/GPS , which of the following statements correctly describes the term “Pseudo Random Noise (PRN)” signal ?
- **a)** PRN is the atmospheric jamming that affects the signals transmitted by the satellites.
- **b)** PRN describes the continuous electro-magnetic background noise that exists in space.
- **c)** PRN is a code used for the identification of the satellites and the measurement of the time taken by the signal to reach the receiver.
- **d)** PRN occurs in the receiver. It is caused by the signal from one satellite being received from different directions (multipath effect).

Answer ;
c) PRN is a code used for the identification of the satellites and the measurement of the time taken by the signal to reach the receiver.

0208. In the event of the re-use of Selective Availability , how does this affect , if at all , the navigation accuracy of the NAVSTAR/GPS satellite navigation system ?

- **a)** It degrades accuracy by reducing the number of available satellites.
- **b)** It increases because only signals from satellites in the most suitable geometric constellation are selected by the receiver.
- **c)** It has no influence because , by selecting of the most suitable signals , the computing process in the receiver is quicker.
- **d)** It degrades position accuracy by manipulating satellite signals.

Answer ;

d) It degrades position accuracy by manipulating satellite signals.

0209. In the Flight Management Computer (FMC) of the Flight Management System (FMS) , data relating to aircraft flight envelope computations is stored in the ;

- **a)** navigation database.
- **b)** auto flight computers.
- **c)** performance database.
- **d)** air data computer.

Answer ;

c) performance database.

0210. In the Flight Management Computer (FMC) of the Flight Management System (FMS) , data relating to cruising speeds is stored in the ;

- **a)** air data computer.
- **b)** performance database.
- **c)** auto flight computers.
- **d)** navigation database.

Answer ;

b) performance database.

0211. In the Flight Management Computer (FMC) of the Flight Management System (FMS) , data relating to flight plans is stored in the ;

- **a)** air data database.
- **b)** performance database.
- **c)** auto flight database.
- **d)** navigation database.

Answer ;

d) navigation database.

0212. In the Flight Management Computer (FMC) of the Flight Management System (FMS) , data relating to STARs and SIDs is stored in the ;

- **a)** navigation database.
- **b)** auto flight computers.
- **c)** air data computer.
- **d)** performance database.

Answer ;

a) navigation database.

0213. In the Flight Management Computer (FMC) of the Flight Management System (FMS) , data relating to V_1 , V_R and V_2 speeds is stored in the ;

- **a)** performance database.
- **b)** air data computer.
- **c)** navigation database.
- **d)** auto flight computer.

Answer ;

a) performance database.

0214. In the Flight Management Computer (FMC) of the Flight Management System (FMS) , data relating to waypoints is stored in the ;

- **a)** air data computer.
- **b)** navigation database.
- **c)** performance database.
- **d)** auto flight computers.

Answer ;

b) navigation database.

0215. In the MAPPING MODE the airborne weather radar utilises a ;
- **a)** pencil beam to a maximum range of 60 NM.
- **b)** pencil beam effective from zero to 150 NM.
- **c)** fan shaped beam effective up to a maximum of 50 NM to 60 NM range.
- **d)** fan shaped beam effective up to a range of 150 NM.

Answer ;
c) fan shaped beam effective up to a maximum of 50 NM to 60 NM range.

0216. In the NAVSTAR/GPS satellite navigation system , receiver clock error ;
- **a)** is the biggest part of the total error , it cannot be corrected.
- **b)** can be minimised by synchronisation of the receiver clock with the satellite clocks.
- **c)** is negligible small because of the great accuracy the atomic clocks installed in the satellites.
- **d)** is corrected by using signals from four satellites.

Answer ;
d) is corrected by using signals from four satellites.

0217. In the NAVSTAR/GPS satellite navigation system , re-use of Selective Availability would give the option to artificially degrade the accuracy by ;
- **a)** using a less accurate atomic clock in a satellite for signal processing.
- **b)** shutting off selected satellites.
- **c)** offsetting satellite atomic clocks by a predetermined constant amount.
- **d)** dithering the satellite clock.

Answer ;
d) dithering the satellite clock.

0218. In the NAVSTAR/GPS satellite navigation system , what is the maximum time taken to receive the complete set of almanac data from all satellites ?
- **a)** 12 hours (= period of the satellites orbit).
- **b)** 25 seconds (= 1 second per data frame).
- **c)** 24 seconds (= 1 second per data frame).
- **d)** 12.5 minutes (= 30 seconds per data frame).

Answer ;
d) 12.5 minutes (= 30 seconds per data frame).

0219. In the VOR receiver the radial is determined by measurement of the ;
- **a)** phase of the variable signal.
- **b)** doppler shift on the reference signal.
- **c)** phase difference between the variable signal and the reference signal.
- **d)** time difference between the reception of the variable signal and the reference signal.

Answer ;
c) phase difference between the variable signal and the reference signal.

0220. In weather radar the use of a cosecant beam in “Mapping” mode enables ;
- **a)** a greater radar range to be achieved.
- **b)** scanning of a large ground zone producing echoes whose signals are practically independent of distance.
- **c)** higher definition echoes to be produced giving a clearer picture.
- **d)** better reception of echos on contrasting terrain such as ground to sea.

Answer ;
b) scanning of a large ground zone producing echoes whose signals are practically independent of distance.

0221. In what piece of FMS equipment will the pilot enter the waypoint information for the route ?
- **a)** The Primary Flight Display (PFD).
- **b)** The Symbol Generator.
- **c)** The Navigation Display (ND).
- **d)** The Control Display Unit (CDU).

Answer ;
d) The Control Display Unit (CDU).

0222. In what type of nominal orbit are NAVSTAR GPS satellites placed ?
- **a)** Elliptical
- **b)** Circular
- **c)** Pole to pole
- **d)** Geo-stationary

Answer ;
b) Circular

0223. In which screen modes of an Electronic Horizontal Situation Indicator (EHSI) on a B737-400 will radar returns not be shown ?

a) EXP VOR/ ILS , PLAN and MAP.
b) FULL VOR/ILS , EXP VOR/ILS and PLAN.
c) FULL NAV , FULL VOR/ILS and PLAN.
d) FULL NAV , PLAN and MAP.

Answer ;
c) FULL NAV , FULL VOR/ILS and PLAN.

0224. In which frequency band do most airborne weather radars operate ?

a) UHF
b) EHF
c) VHF
d) SHF

Answer ;
d) SHF

0225. In which frequency band do Satellite-Assisted Navigation systems (GNSS/GPS) provide position information that is available to civil aircraft ?

a) UHF
b) EHF
c) VHF
d) SHF

Answer ;
a) UHF

0226. In which frequency band do VOR transmitters operate ?

a) UHF
b) SHF
c) VHF
d) EHF

Answer ;
c) VHF

0227. In which frequency band does an ILS glide slope transmit ?

a) EHF
b) UHF
c) VHF
d) SHF

Answer ;
b) UHF

0228. In which frequency band does the Microwave Landing System (MLS) operate ?

a) UHF
b) VHF
c) SHF
d) EHF

Answer ;
c) SHF

0229. In which frequency bands are the L1 and L2 frequencies used by the satellite navigation system NAVSTAR/GPS for transmission of the navigation message ?

a) UHF
b) EHF
c) SHF
d) VHF

Answer ;
a) UHF

0230. In which mode of operation does the aircraft weather radar use a cosecant radiation pattern ?

a) CONTOUR
b) MANUAL
c) MAPPING
d) WEATHER

Answer ;
c) MAPPING

0231. In which navigation system does the master station transmit a continuous string of pulses on a frequency close to 100 kHz ?

- **a)** Decca
- **b)** Doppler
- **c)** GPS
- **d)** Loran C

Answer ;

d) Loran C

0232. In which screen modes of an Electronic Horizontal Situation Indicator (EHSI) on a B737-400 will radar returns not be shown ?

- **a)** FULL NAV , PLAN and MAP.
- **b)** FULL NAV , FULL VOR/ILS and PLAN.
- **c)** FULL VOR/ILS , EXP VOR/ILS and PLAN.
- **d)** EXP VOR/ ILS , PLAN and MAP

Answer ;

b) FULL NAV , FULL VOR/ILS and PLAN.

0233. In which situation will speed indications on an airborne Distance Measuring Equipment (DME) most closely represent the groundspeed of an aircraft flying at FL400 ?

- **a)** When tracking directly towards the station at a range of 100 NM or more.
- **b)** When passing abeam the station and within 5 NM of it.
- **c)** When overhead the station, with no change of heading at transit.
- **d)** When tracking directly away from the station at a range of 10 NM.

Answer ;

a) When tracking directly towards the station at a range of 100 NM or more

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0234. Inner marker beacons of an ILS transmit at ;

- **a)** 75 MHz.
- **b)** 150 MHz.
- **c)** 90 Hz.
- **d)** It depends on the modulating frequency.

Answer ;

a) 75 MHz.

0235. Instrument Landing Systems (ILS) Glide Paths provide azimuth coverage (i) ….. ° each side of the localiser centre-line to a distance of (ii) …. NM minimum from the threshold.

- **a)** (i) 8 , (ii) 10
- **b)** (i) 35 , (ii) 25
- **c)** (i) 5 , (ii) 8
- **d)** (i) 25 , (ii) 17

Answer ;

a) (i) 8 , (ii) 10

0236. IRS is a self-contained system because ;

- **a)** the system has a battery back-up which guarantees the well-functioning of he system in case of power-failures.
- **b)** the system calculates the position of the aircraft without reference to externally generated (man-made) signals.
- **c)** the system has the abilty to calculate the aircraft's position with an accuracy comparable to the GPS-position.
- **d)** the system has the abilty to calculate the aircraft's radio position without any reference to ither man-made or natural information.

Answer ;

b) the system calculates the position of the aircraft without reference to externally generated (man-made) signals.

0237. Kalman filtering is used within ;

- **a)** VOR receiver.
- **b)** DME receiver.
- **c)** Navigation computer.
- **d)** Electronic Flight Instrument System (EFIS).

Answer ;

c) Navigation computer.

0238. Locators are ;

a) beacons with a range of 10 to 250 NM.
b) low powered ADFs used for airfield or runway approach.
c) LF/MF NDBs used as an aid for final approach.
d) high powered NDBs used for en route and airways navigation.

Answer ;
c) LF/MF NDBs used as an aid for final approach.

0239. Locators are ;

1. High powered NDBs used for en route and airways navigation.
2. Low powered NDBs used for airfield or runway approach.
3. Beacons with a usually range of 10 to 250 NM.
4. Beacons with a usually range of 10 to 25 NM.

The combination regrouping all the correct statements is ;

a) 2 and 3.
b) 1 and 4.
c) 1 and 3.
d) 2 and 4.

Answer ;
d) 2 and 4.

0240. Loran C coverage is ;

a) confined to certain limited areas of the world.
b) unrestricted between latitudes 70°N and 70°S.
c) global.
d) unrestricted over the oceans and adjacent coastlines but limited over the major continental land masses.

Answer ;
a) confined to certain limited areas of the world.

0241. Middle marker beacons of an ILS transmit at ;

a) 75 MHz.
b) 90 MHz.
c) 1300 Hz.
d) 150 MHz.

Answer ;
a) 75 MHz.

0242. MLS can minimise multi path errors because ;

a) the frequency of MLS is much higher than the frequency of ILS.
b) the transmission reverts to circular polarization when the beam is reflected by stationary objects.
c) MLS has a large beam width than ILS.
d) the transmission can be interrupted to avoid reflection by stationary objects.

Answer ;
d) the transmission can be interrupted to avoid reflection by stationary objects.

0243. MLS installations notified for operation , unless otherwise stated , provide azimuth coverage of ;

a) + or - 40° about the nominal courseline out to a range of 30 NM.
b) + or - 20° about the nominal courseline out to a range of 10 NM.
c) + or - 40° about the nominal courseline out to a range of 20 NM.
d) + or - 20° about the nominal courseline out to a range of 20 NM.

Answer ;
c) + or - 40° about the nominal courseline out to a range of 20 NM.

0244. MLS is primarily being installed at airports where ;

a) meteorological conditions are likely to cause ILS ducting by super refraction.
b) topographical conditions preclude the installation of ILS marker beacons.
c) ILS encounters difficulties because of surrounding buildings and/or the trrain or interference from local music stations.
d) the main approach paths lead over water.

Answer ;
c) ILS encounters difficulties because of surrounding buildings and/or the trrain or interference from local music stations.

0245. MLS not equipped with DME P ;

- **a)** provides the capability for segmented approaches but of not curved approaches.
- **b)** provides basically the same approach capabilities as ILS.
- **c)** provides the capability for curved approaches but not of segmented approaches.
- **d)** provides the capability for CAT 3 approaches.

Answer ;

b) provides basically the same approach capabilities as ILS.

0246. Mode A or C garbling may occur to ;

- **a)** Two or more aircraft in the same direction from the ground station at the same altitude with a difference in slant range of more than 1.7 NM.
- **b)** Two or more aircraft in the same direction from the interrogator with a difference in slant range of less than 1.7 NM.
- **c)** Two or more aircraft different directions from the ground station at the same altitude with a difference in slant range of more than 1.7 NM.
- **d)** Two or more aircraft different directions from the interrogator at the same altitude with a difference in slant range of less than 1.7 NM.

Answer ;

b) Two or more aircraft in the same direction from the interrogator with a difference in slant range of less than 1.7 NM.

0247. NDB is the abbreviation for ;

- **a)** Non Directional Beacon.
- **b)** Night Directional Beacon.
- **c)** Non Directional Bearing.
- **d)** Navigation Director Beacon.

Answer ;

a) Non Directional Beacon.

0248. Night Effect in an ADF may cause ;

- **a)** Fluctuating indications of the needle on the RMI.
- **b)** No bearing error because of the built-in compensator unit.
- **c)** Noise in the received EM-wave , which hardly will be noticed by the pilot looking at the RMI.
- **d)** A constant error in the indicated bearing.

Answer ;

a) Fluctuating indications of the needle on the RMI.

0249. "Night Effect" which causes loss of signal and fading , resulting in bearing errors from NDB transmissions , is due to ;

- **a)** static activity increasing at night particularly in the lower frequency band.
- **b)** the effect of the Aurora Borealis.
- **c)** interference from other transmissions and is maximum at dusk when east of the NDB.
- **d)** skywave distortion of the null position and is maximum at dawn and dusk.

Answer ;

d) skywave distortion of the null position and is maximum at dawn and dusk.

0250. Of what use , if any , is a military TACAN station to civil aviation ?

- **a)** It is of no use to civil aviation.
- **b)** It can provide DME distance.
- **c)** It can provide a magnetic bearing.
- **d)** It can provide a DME distance and magnetic bearing.

Answer ;

b) It can provide DME distance.

0251. On a modern commercial aircraft , the FMS provides ;

- **a)** lateral and vertical navigation , and guidance and performance management.
- **b)** traffic alert information.
- **c)** an air/ground datalink.
- **d)** a 3D area navigation and an air/ground datalink.

Answer ;

a) lateral and vertical navigation , and guidance and performance management.

0252. On an ILS approach , the glidepath needle is fully down. How much deflection does this indicate ?

a) 2.5 º
b) 5 º
c) 10 º
d) 0.7 º

Answer ;
d) 0.7 º

0253. On an ILS approach , the localiser needle is fully over to the left. How much deflection does this indicate ?

a) 5º
b) 2.5 º
c) 0.7 º
d) 10 º

Answer ;
b) 2.5 º

0254. On an ILS approach , when flying overhead the inner marker (if available) the color of the flashing light will be ;

a) green.
b) amber.
c) white.
d) blue.

Answer ;
c) white.

0255. On an ILS approach , when flying overhead the middle marker the color of the flashing light will be ;

a) amber.
b) green.
c) white.
d) blue.

Answer ;
a) amber.

0256. On an ILS approach , when flying overhead the outer marker the color of the flashing light will be ;

a) green.
b) white.
c) blue.
d) amber.

Answer ;
c) blue.

0257. On an RMI the front end of a VOR pointer indicates the ;

a) radial plus 180º.
b) magnetic bearing from the station.
c) magnetic bearing to the station.
d) radial.

Answer ;
a) radial plus 180º.

0258. On final on an ILS approach , you are flying overhead the outer marker. You can expect to be at ;

a) 4 NM from the threshold.
b) 10 NM from the threshold.
c) 1 NM from the threshold.
d) 25 NM from the threshold.

Answer ;
a) 4 NM from the threshold.

0259. On final on ILS approach , at 0.6 NM from the threshold , which marker are you likely to hear ?

a) The outer marker.
b) The middle marker.
c) No markers can be locaed at this distance.
d) The inner marker (if available).

Answer ;
b) The middle marker.

0260. On modern passenger aircraft , the navigation data base usually contains ;

a) aircraft performance data.
b) ATC frequencies.
c) obstacle altitudes.
d) airport reference data.

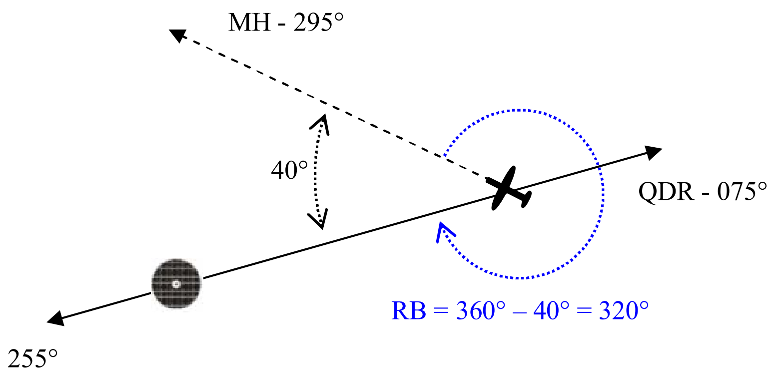
Answer ;
d) airport reference data.

0261. On the QDR of 075° (in the vicinity of the station) with a magnetic heading of 295°, the relative bearing on the ADF indicator is ;

a) 320°
b) 220°
c) 140°
d) 040°

Answer ;
a) 320°

Çözüm ;



0262. On the RMI the tip of a VOR needle indicates 060. With the CRS set on 055 the indications on the HSI are ;

a) FROM , half scale deflection to the right.
b) FROM , half scale deflection to the left.
c) TO , half scale deflection to the right.
d) TO , half scale deflection to the left.

Answer ;
c) TO , half scale deflection to the right.

Çözüm ;

0263. On what carrier frequency does the inner marker transmit ?

a) Same frequency as the glide path.
b) 3000 Hz.
c) 75 MHz.
d) Same frequency as the localizer.

Answer ;
c) 75 MHz.

0264. On what data is a VOR/DME Area Navigation system operating in the dead reckoning mode ?

a) TAS from the Air Data Computer , heading from the aircraft compass.
b) TAS from the Air Data Computer , heading from the aircraft compass , the last computed W/V.
c) Radial from one VOR , distances from two DMEs.
d) Radial from one VOR , distances from two DMEs , TAS from the Air Data computer , heading from the aircraft compass.

Answer ;
b) TAS from the Air Data Computer , heading from the aircraft compass , the last computed W/V.

0265. On which of the following displays are you able to get a direct read-out (no calculation is necessary from the pilot) of the magnetic bearing from the aircraft to the NDB ?

- **a)** Fixed card ADF and RMI.
- **b)** Moving and fixed card ADF.
- **c)** Fixed card ADF only.
- **d)** Moving card ADF and RMI.

Answer ;

d) Moving card ADF and RMI.

0266. On which of the following radar displays is it possible to get an indication of the shape , and to some extent the type , of the aircraft generating the return ?

- **a)** Aerodrome Surface Movement Radar (ASMR).
- **b)** Secondary Surveillance Radar (SSR).
- **c)** Airborne Weather Radar (AWR).
- **d)** Aerodrome Surveillance (approach) Radar.

Answer ;

a) Aerodrome Surface Movement Radar (ASMR).

0267. One of the benefits of RNAV is ;

- **a)** RNAV allows pilots to manage horizontal and vertical separation with other aircrafts without obtaining any ATC clearance.
- **b)** RNAV allows to obtain ATC clearance by HF radio without the requirement to establish any radio contact.
- **c)** RNAV allows aircraft to take a more direct flight path without requiring to fly over ground based facilities.
- **d)** RNAV allows to fly at RNAV flight level with a 500 ft separation.

Answer ;

c) RNAV allows aircraft to take a more direct flight path without requiring to fly over ground based facilities.

0268. One of the possible disturbances of the ILS signal is “scalloping”. Which statement is correct ?

- **a)** Scalloping are rapid changes or bends which can be followed by the aircraft.
- **b)** Scalloping causes rapid indicator changes from side to side of the intended approach path which can not be followed by the aircraft.
- **c)** Scalloping are minor changes or bends which can be followed by the aircraft.
- **d)** Scalloping are major changes or bends in the approach path which can not be followed by the aircraft.

Answer ;

b) Scalloping causes rapid indicator changes from side to side of the intended approach path which can not be followed by the aircraft.

0269. One of the tasks of the control segment of the satellite navigation system NAVSTAR/GPS is to ;

- **a)** monitor the status of the satellites.
- **b)** manufacture and launch the satellites.
- **c)** manipulate the signals of selected satellites to reduce the precision of the position fix.
- **d)** grant and monitor user authorisations.

Answer ;

a) monitor the status of the satellites.

0270. One of the tasks of the GPS control segment is ;

- **a)** to calculate the accuracy of GPS.
- **b)** to calculate and transmit differential corrections to users.
- **c)** to control continuously the motion of GPS-satellites.
- **d)** to detect the malfunctioning of satellites.

Answer ;

d) to detect the malfunctioning of satellites.

0271. One of the tasks of the GPS control segment is ;

- **a)** to calculate and transmit differential corrections to users which are able to receive Wide Area Differential GPS systems.
- **b)** to regulate the transmitted power by the satellites.
- **c)** to determine and send new ephemeris and new satellite clock error data to the GPS-satellites.
- **d)** to control continuously the position and the motion of GPS-satellites.

Answer ;

c) to determine and send new ephemeris and new satellite clock error data to the GPS-satellites.

0272. One of the tasks of the space segment of the satellite navigation system NAVSTAR/GPS is to ;
- **a)** transmit signals to suitable receivers and to monitor the orbital planes autonomously.
- **b)** monitor the satellite's orbits and status.
- **c)** transmit signals which can be used , by suitable receivers , to determine time , position and velocity.
- **d)** compute the user position from the received user messages and to transmit the computed position back to the user segment.

Answer ;
c) transmit signals which can be used , by suitable receivers , to determine time , position and velocity.

0273. One of uses of the VDF service is providing aircraft with ;
- **a)** ground speed.
- **b)** altitude.
- **c)** heading.
- **d)** homing.

Answer ;
d) homing.

0274. Outer marker beacons of an ILS transmit at ;
- **a)** 75 Hz.
- **b)** 90 MHz.
- **c)** 150 Hz.
- **d)** 75 MHz.

Answer ;
d) 75 MHz.

0275. Outer marker transmits on 75 MHz and has an aural frequency of ;
- **a)** 1300 Hz
- **b)** 400 Hz
- **c)** 3000 Hz
- **d)** 2000 Hz

Answer ;
b) 400 Hz

0276. Output from which of the following combination of navigational sources provide enough information to the RNAV - equipment to calculate the wind vector ?
- **a)** Compass System and Inertial Reference System.
- **b)** Inertial Reference System and Air Data Computer.
- **c)** Inertial Reference System and Global Positioning System.
- **d)** Global Positioning System and Compass System.

Answer ;
b) Inertial Reference System and Air Data Computer.

0277. Performing an ILS approach , you will fly overhead the markers in an specific order. This order is ;
- **a)** MM , IM (if available) , OM.
- **b)** OM , MM , IM (if available).
- **c)** IM (if available) , MM , OM.
- **d)** OM , IM (if available) , MM.

Answer ;
b) OM , MM , IM (if available).

0278. Precision RNAV (P-RNAV) requires a track-keeping accuracy of ;
- **a)** ± 8.0 NM for 95% of the flight time.
- **b)** ± 10.0 NM for 95% of the flight time.
- **c)** ± 5.0 NM for 95% of the flight time.
- **d)** ± 1.0 NM for 95% of the flight time.

Answer ;
d) ± 1.0 NM for 95% of the flight time.

0279. Quadrantal errors associated with aircraft Automatic Direction Finding (ADF) equipment are caused by ;
- **a)** skywave/groundwave contamination.
- **b)** misalignment of the loop aerial.
- **c)** signal bending by the aircraft metallic surfaces.
- **d)** signal bending caused by electrical interference from aircraft wiring.

Answer ;
c) signal bending by the aircraft metallic surfaces.

0280. Range of VDF depends on ;

1. Loudness of the voices of the pilot and the operator when transmitting.
2. Power of airborne and ground transmitters.
3. Power of pilot voice when transmitting.
4. Aircraft altitude and ground transmitter elevation.

The combination regrouping all the correct statements is ;

a) 2 and 4.
b) 1.
c) 3 and 4.
d) 2.

Answer ;
a) 2 and 4.

0281. Range of VDF depends on ;

1. Line of sight formula
2. Power of transmitters
3. Intervening high ground.

The combination regrouping all the correct statements is ;

a) 1 and 2.
b) 2.
c) 1 , 2 and 3.
d) 1 and 3.

Answer ;
c) 1 , 2 and 3.

0282. Regarding ILS which of the following is true ?

a) The DME paired with ILS channels are usually zero referenced next to the departure end of the runway.
b) All markers transmit at 75 MHz.
c) The localiser part of frequency band is shared with the DME.
d) The glide path transmitter is located 300 m from the departure end of the runway.

Answer ;
b) All markers transmit at 75 MHz.

0283. Regarding the DME system , which one of the following statements is true ?

a) DME operates in the VHF frequency band.
b) The transponder reply carrier frequency differs by 63 MHz from that of the interrogation signal.
c) When passing overhead the DME station the DME will indicate 0.
d) The DME measures the phase difference between the reference and variable phase signals to calculate the distance.

Answer ;
b) The transponder reply carrier frequency differs by 63 MHz from that of the interrogation signal.

0284. Signal reception is required from a minimum number of satellites that have adequate elevation and suitable geometry in order for a Satellite-Assisted Navigation System (GNSS/GPS) to carry out independent three dimensional operation , Receiver Autonomous Integrity Monitoring (RAIM) and to isolate any faulty satellite and remove it from contributing to the navigation solution. The number of satellites is ;

a) 5
b) 7
c) 4
d) 6

Answer ;
d) 6

0285. Signal reception is required from a minimum number of satellites that have adequate elevation and suitable geometry in order for a Satellite-Assisted Navigation System (GPS) to carry out independent three dimensional operation without the Receiver Autonomous Integrity Monitoring (RAIM) function. The number of satellites is ;

a) 3
b) 4
c) 6
d) 5

Answer ;
b) 4

0286. The addition of DME-P to MLS is necessary to ;

a) support the time referenced scanning beam.
b) obtain three dimensional positions.
c) assure a constant angular velocity of the azimuth and elevation sweep.
d) allow linear approaches.

Answer ;
b) obtain three dimensional positions.

0287. The ADF indication in the cockpit is a ;

a) relative bearing on an RMI.
b) magnetic beaing on a fixed card indicator.
c) relative bearing on a fixed card indicator.
d) true bearing on an RMI.

Answer ;
c) relative bearing on a fixed card indicator.

0288. The advantage of the use of slotted antennas in modern radar technology is to ;

a) eliminate the need for azimuth slaving.
b) have a wide beam and as a consequence better target detection.
c) simultaneously transmit weather and mapping beams.
d) virtually eliminate lateral lobes and as a consequence concentrate more energy in the main beam.

Answer ;
d) virtually eliminate lateral lobes and as a consequence concentrate more energy in the main beam.

0289. The aircraft DME receiver cannot lock on to interrogation signals reflected from the ground because ;

a) DME pulse recurrence rates are varied.
b) aircraft transmitter and DME ground station are transmitting on different frequencies.
c) reflections are subject to doppler frequency shift.
d) DME transmits twin pulses.

Answer ;
b) aircraft transmitter and DME ground station are transmitting on different frequencies.

0290. The aircraft DME receiver is able to accept replies to its own transmissions and reject replies to other aircraft interrogations because ;

a) aircraft interrogation signals and transponder responses are 63 MHz removed from each other.
b) the time interval between pulse pairs is unique to that particular aircraft.
c) pulse pairs are amplitude modulated with the aircraft registration.
d) transmission frequencies are 63 MHz different for each aircraft.

Answer ;
b) the time interval between pulse pairs is unique to that particular aircraft.

0291. The amplitude modulation and the colour of an outer marker (OM) is ;

a) 1300 Hz , blue
b) 3000 Hz , blue
c) 400 Hz , amber
d) 400 Hz , blue

Answer ;
d) 400 Hz , blue

0292. The antenna of modern airborne weather radars are stabilized by means of ;

a) feedback from the antenna accelerometers.
b) mercury switches.
c) inputs from the aircraft's attitude system.
d) artificial gravity switches.

Answer ;
c) inputs from the aircraft's attitude system.

0293. The audio frequency modulation of the inner marker (if available) shall be keyed as follows ;

a) 3 dashes , 3 dots and 3 dashes per second continuously.
b) 2 dashes per second continuously.
c) 6 dots per second continuously.
d) a continuous series of alternate dots and dashes.

Answer ;
c) 6 dots per second continuously.

0294. The audio frequency modulation of the middle marker shall be keyed as follows ;

a) a continuous series of alternate dots and dashes , the dashes keyed at the rate of 2 dashes per second , and the dots at the rate of 6 dots per second.
b) 3 dashes , 3 dots and 3 dashes per second continuously.
c) 2 dashes per second continuously.
d) 6 dots per second continuously.

Answer ;
a) a continuous series of alternate dots and dashes , the dashes keyed at the rate of 2 dashes per second , and the dots at the rate of 6 dots per second.

0295. The audio frequency modulation of the outer marker shall be keyed as follows ;

a) a continuous series of alternate dots and dashes.
b) 6 dots per second continuously.
c) 3 dashes , 3 dots and 3 dashes per second continuously.
d) 2 dashes per second continuously.

Answer ;
d) 2 dashes per second continuously.

0296. The audio modulation of the middle marker is keyed to give ;

a) continuous dashes at a rate of 2 per second.
b) continuous dots at a rate of 6 per second.
c) alternating dots and dashes at a rate of 2 dashes per second and 6 dots per second.
d) a sequence of 3 dashes , 3 dots and 3 dashes every second.

Answer ;
c) alternating dots and dashes at a rate of 2 dashes per second and 6 dots per second.

0297. The azimuth transmitter of a Microwave Landing System (MLS) provides a fan-shaped horizontal approach zone which is usually ;

a) + or - 60° of the runway centre-line.
b) + or - 40° of the runway centre-line.
c) + or - 50° of the runway centre-line.
d) + or - 30° of the runway centre-line.

Answer ;
b) + or - 40° of the runway centre-line.

0298. The back CRS of an ILS may give ;

a) non-precision approach guidance for the reciprocal of the main approach runway.
b) precision approach guidance for the main approach runway.
c) precision approach guidance for the reciprocal of the main approach runway.
d) non-precision approach guidance for the main approach runway.

Answer ;
a) non-precision approach guidance for the reciprocal of the main approach runway.

0299. The basic principle of operation of the ILS is the difference in depth of modulation (DDM) between the 90 Hz and 150 Hz lobes. A DDM of zero indicates ;

1. the higher tone lobe is received at a higher strenght than the lower tone lobe.
2. the exact runway centreline.
3. the aircraft is on the glidepath.
4. a balance between modulations.

a) 1 and 3.
b) 2 , 3 and 4.
c) 2 and 3.
d) 1 , 2 and 4.

Answer ;
b) 2 , 3 and 4.

0300. The basic principle of operation of the ILS is the difference in depth of modulation (DDM) between the two lobes ;

1. if the aircraft strays right , the higher tone lobe will be received at a higher intensity than the lower tone lobe.
2. a DDM of zero indicates the exact runway centreline.
3. the depth of modulation increases away from the centerline.
4. a DDM of zero indicates a balance between modulations.

The combination that regroups all the corrects statements is ;

a) 1 and 3.
b) 2 , 3 and 4.
c) 1 , 2 , 3 and 4.
d) 2 and 3.

Answer ;
c) 1 , 2 , 3 and 4.

0301. The BFO selector on an ADF receiver is used to ;

a) find the loop “null” position.
b) hear the IDENT of some NDB stations radiating a continuous wave signal.
c) stop loop rotation.
d) hear the IDENT and must always be switched ON.

Answer ;
b) hear the IDENT of some NDB stations radiating a continuous wave signal.

0302. The BFO selector switch on the ADF control panel must be in the “on” position to enable the pilot to ;

a) hear the IDENT of NDBs using NON A1A transmissions.
b) adjust the loop to the aural null position.
c) hear the IDENT of NDBs using NON A2A transmissions.
d) stop the loop rotation.

Answer ;
a) hear the IDENT of NDBs using NON A1A transmissions.

0303. The Captain of an aircraft flying at FL100 wishes to obtain weather information at the destination airfield (0 ft) from the airfield’s VOR. Assuming ISA conditions , what is the approximate maximum theoretical range at which it can be expected to obtain this information ?

a) 123 NM
b) 123 km
c) 12.3 NM
d) 1230 km

Teorik olarak maksimum alış menzili (NM) = 1,25 x ($\sqrt{H1} + \sqrt{H2}$)
= 1,25 x ($\sqrt{10000} + \sqrt{0}$)
= 1,25 x (100 + 0) = 1,25 x 100 = 1,25 ≈ 123 NM

Answer ;
a) 123 NM

0304. The code transmitted by a SSR transponder consists of ;

a) amplitude differences.
b) frequency differences.
c) pulses.
d) phase differences.

Answer ;
c) pulses.

0305. The Control and Display Unit (CDU) on an FMS is ;

a) used on ground only to monitor the maintenance procedure.
b) used by the crew to input data into FMC.
c) the system used to update the navigation database.
d) the Autopilot control panel.

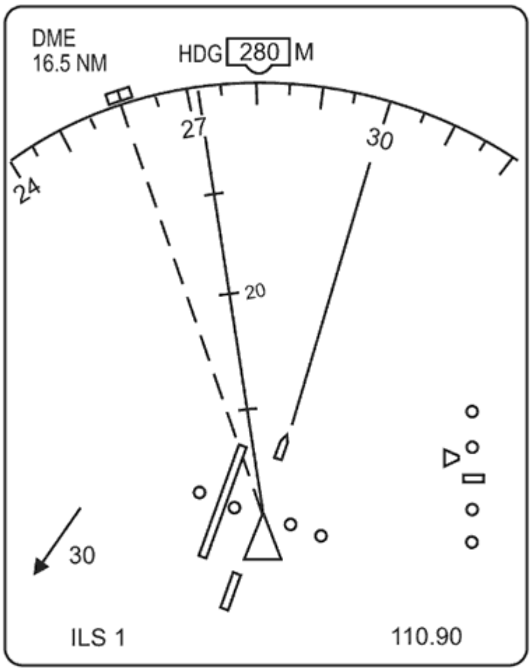
Answer ;
b) used by the crew to input data into FMC.

0306. The design requirements for DME stipulate that , at a range of 100 NM , the maximum systematic error should not exceed ;

a) + or - 0.25 NM
b) + or - 1.25 NM
c) + or - 3 NM
d) + or - 1.5 NM

Answer ;
d) + or - 1.5 NM

0307. (For this question use annex 062-9923A)



Annex 062-9923A

The diagram indicates that the aircraft is to the ;

a) right of the localizer and above the glidepath.
b) left of the localizer and above the glidepath.
c) left of the localizer and below the glidepath.
d) right of the localizer and below the glidepath.

Answer ;
d) right of the localizer and below the glidepath.

0308. The different segments of the satellite navigation system NAVSTAR/GPS are the ;

a) antenna , the receiver and the central control unit (CDU).
b) main control station , the monitoring station and the ground antennas.
c) control , space and user.
d) atomic clock , power supply and transponder.

Answer ;
c) control , space and user.

0309. The distance between a NAVSTAR/GPS satellite and receiver is ;

a) determined by the phase shift of the Pseudo Random Noise code multiplied by the speed of light.
b) calculated , using the WGS-84 reference system , from the known positions of the satellite and the receiver.
c) calculated from the Doppler shift of the known frequencies.
d) determined by the time taken for the signal to arrive from the satellite multiplied by the speed of light.

Answer ;
d) determined by the time taken for the signal to arrive from the satellite multiplied by the speed of light.

0310. The DME (Distance Measuring Equipment) operates within the following frequencies ;

a) 329 to 335 MHz
b) 962 to 1213 kHz.
c) 962 to 1213 MHz
d) 108 to 118 MHz

Answer ;
c) 962 to 1213 MHz

0311. The DME Line Of Position is a circle with radius ;

a) the slant range and centre the DME-station.
b) the ground distance and centre the aircraft.
c) the ground distance and centre the DME-station.
d) the slant range and centre the aircraft.

Answer ;
c) the ground distance and centre the DME-station.

0312. The Doppler Navigation System is based on ;

a) radar principles using frequency shift.
b) radio waves refraction in the ionosphere.
c) phase comparison from ground station transmissions.
d) Doppler VOR (DVOR) Navigation System.

Answer ;
a) radar principles using frequency shift.

0313. The effect of masking the DME antenna of the aircraft from the ground installation is that interruption of reception of DME signals results in ;

a) The airborne installation switching to the memory mode for about 10 to 15 seconds.
b) The ground installation not sending any pulse pairs.
c) The airborne equipment switching directly to the search mode.
d) The signal controlled search circuit (SCS) blanking the DME display.

Answer ;
a) The airborne installation switching to the memory mode for about 10 to 15 seconds.

0314. The Flight Management Computer (FMC) position is ;

a) another source of aircraft position , it is independent of other navigation sources (IRS , Radio , ILS , etc.).
b) the actual position of the aircraft at any point in time.
c) the computed position based on a number of sources (IRS , Radio , ILS , GPS etc.).
d) the same as that given on the No.1 IRS.

Answer ;
c) the computed position based on a number of sources (IRS , Radio , ILS , GPS etc.).

0315. The Flight Management System (FMS) is organised in such a way that ;

a) the pilot is able to modify the navigation database in the FMC between two updates.
b) the navigation database is read only to the pilot.
c) the navigation database of the FMC is valid for one year.
d) the navigation database of the FMC is created by the pilot.

Answer ;
b) the navigation database is read only to the pilot.

0316. The Flight Management System (FMS) is organised in such a way that the pilot can ;

a) modify the database every 14 days.
b) modify the data in the database between two updates.
c) read and write at any time in the database.
d) insert navigation data between two database updates.

Answer ;
d) insert navigation data between two database updates.

0317. The FMS navigation data base usually contains ;

1. airport reference data.
2. ATC frequencies.
3. Company routes.
4. Nav. Aid. frequencies.

The combination regrouping all the correct statements is ;

a) 1 , 2 and 3.
b) 1 , 3 and 4.
c) 1 , 2 , 3 and 4.
d) 1 and 3.

Answer ;
b) 1 , 3 and 4.

0318. The frequency of an SSR ground transmission is ;

a) 1120 +/- 0.6 MHz
b) 1030 +/- 0.2 MHz
c) 1090 +/- 0.3 MHz
d) 1050 +/- 0.5 MHz

Answer ;
b) 1030 +/- 0.2 MHz

0319. The frequency of the amplitude modulation and the colour of an outer marker (OM) light is ;

a) 400 Hz , amber.
b) 3000 Hz , blue.
c) 400 Hz , blue.
d) 1300 Hz , blue.

Answer ;
c) 400 Hz , blue.

0320. The frequency range of a VOR receiver is ;

a) 108 to 111.95 MHz.
b) 108 to 135.95 MHz.
c) 118 to 135.95 MHz.
d) 108 to 117.95 MHz.

Answer ;
d) 108 to 117.95 MHz.

0321. The frequency which corresponds to a wavelength of 12 cm is ;

a) 360 MHz.
b) 2500 MHz.
c) 3600 MHz.
d) 2500 KHz.

Answer ;
b) 2500 MHz.

f (frekans) = C (Işık hızı) : λ (Dalga boyu)
Işık hızı olarak alınan (300 x 10^6) değeri m/sec cinsinden olduğu için ;
12 cm = 0,12 m , ve de ;
f = (300 x 10^6) : (0,12 x 10^6) → f = 300 : 0,12 = 2500 MHz olarak bulunur.

0322. The "gain" control knob of an AWR adjusts ;

a) the receiver sensitivity in order to achieve optimum target acquisition.
b) the power level of the transmitted energy is made dependent on the selected range.
c) the brightness of the display.
d) the automatic gain control of the AWR is activated.

Answer ;
a) the receiver sensitivity in order to achieve optimum target acquisition.

0323. The geometric shape of the reference system for the satellite navigation system NAVSTAR/GPS , defined as WGS 84 , is ;

a) a geoid.
b) a mathematical model that describes the exact shape of the earth.
c) a sphere.
d) an ellipsoid.

Answer ;
d) an ellipsoid.

0324. The GPS control segment consists of ;

a) a Master Control Station , Monitor Stations , Ground Antenna's and Geostationary Satellites.
b) a Master Control Station , Monitor Stations and Ground Antenna's.
c) a Master Control Station , Monitor Stations and Geostationary Satellites.
d) Master Control Stations and Monitor Stations.

Answer ;
b) a Master Control Station , Monitor Stations and Ground Antenna's.

0325. The ground Secondary Surveillance Radar (SSR) equipment incorporates a transmitter and receiver respectively operating in the following frequencies ;

	Transmitter	Receiver
a)	1090 MHz	1090 MHz
b)	1090 MHz	1030 MHz
c)	1030 MHz	1090 MHz
d)	1030 MHz	1030 MHz

Answer ;
c) 1030 MHz 1090 MHz

0326. The heading rose of an HSI is frozen on 200°. Lined up on the ILS of runway 25 , the localizer needle will be ;

a) centred with the "fail" flag showing.
b) centred.
c) left of centre.
d) right of centre.

Answer ;
b) centred.

0327. The heaviest turbulence is likely to be encountered ;

a) in the area of heaviest precipitation.
b) in the very core of a thunderstorm cell.
c) about halfway between two thunderstorm cells.
d) where the area of heaviest precipitation is closest to the edge of the thunderstorm cell.

Answer ;
d) where the area of heaviest precipitation is closest to the edge of the thunderstorm cell.

0328. The ident of a DME in case of collocation with a VOR is as follows ;

a) In case of collocation the DME ident is not necessary if the VOR ident is present.
b) In 40 seconds the DME ident will sound once.
c) The DME ident comes up every 10 seconds at an audio frequency of 1020 Hz.
d) In 30 seconds the DME ident comes up 3 times at an audio frequency of 1350 Hz.

Answer ;
b) In 40 seconds the DME ident will sound once.

0329. The identification of a DME in combination with a collocated VOR is as follows ;

a) The VOR ident and DME ident are the same and no difference can be heard.
b) Every 30 seconds the DME ident will be repeated.
c) The DME ident will repeat three times every period of 30 seconds.
d) In a period of 40 seconds the DME ident will be heard once on an audio frequency of 1350 Hz.

Answer ;
d) In a period of 40 seconds the DME ident will be heard once on an audio frequency of 1350 Hz.

0330. The ILS inner marker (if available) modulation frequency is ;

a) 3000 Hz
b) 240 Hz
c) 1300 Hz
d) 400 Hz

Answer ;
a) 3000 Hz

0331. The ILS marker identified audibly by a series of alternate dots and dashes is the ;

a) locator.
b) outer marker.
c) inner marker.
d) middle marker.

Answer ;
d) middle marker.

0332. The ILS marker identified audibly by a series of dots (6/sec.) is the ;

a) locator.
b) outer marker.
c) inner marker.
d) middle marker.

Answer ;
c) inner marker.

0333. The ILS marker identified audibly by a series of two dashes per second is the ;

a) outer marker.
b) middle marker.
c) locator.
d) inner marker.

Answer ;
a) outer marker.

0334. The ILS marker identified visually by an amber light flashing is the ;

a) middle marker.
b) inner marker.
c) locator.
d) outer marker.

Answer ;
a) middle marker.

0335. The ILS marker identified visually by an blue flashing light is the ;

a) middle marker.
b) inner marker.
c) outer marker.
d) locator.

Answer ;
c) outer marker.

0336. The ILS marker identified visually by a white light flashing is the ;

a) inner marker.
b) outer marker.
c) middle marker.
d) locator.

Answer ;
a) inner marker.

0337. The ILS marker with an aural frequency of 400 Hz is ;

a) locator.
b) middle marker.
c) outer marker.
d) inner marker.

Answer ;
c) outer marker.

0338. The ILS marker with an aural frequency of 1300 Hz is the ;

a) locator.
b) outer marker.
c) inner marker.
d) middle marker.

Answer ;
d) middle marker.

0339. The ILS marker with an aural frequency of 3000 Hz is the ;

a) middle marker.
b) outer marker.
c) centerline marker.
d) inner marker (if available).

Answer ;
d) inner marker (if available).

0340. The ILS marker with the higher aural frequency is the ;

a) inner marker.
b) outer marker.
c) middle marker.
d) locator.

Answer ;
a) inner marker.

0341. The ILS marker with the lower aural frequency is the ;

a) outer marker.
b) inner marker (if available).
c) centreline marker.
d) middle marker.

Answer ;
a) outer marker.

0342. The ILS middle marker modulation frequency is ;

a) 3000 Hz
b) 400 Hz
c) 1300 Hz
d) 800 Hz

Answer ;
c) 1300 Hz

0343. The ILS outer marker modulation frequency is ;

a) 1300 Hz
b) 400 Hz
c) 3000 Hz
d) 1500 Hz

Answer ;

b) 400 Hz

0344. The ILS receiver of an aircraft flying down the exact runway centreline will receive ;

a) no modulated signal because the left and the right lobes cancel each other along the centerline.
b) the same frequency modulated signal from both lobes with the maximum of magnitude.
c) 90 Hz and 150 Hz lobes at equal depth.
d) the maximum magnitude of the difference between the 90 Hz and 150 Hz amplitudes.

Answer ;

c) 90 Hz and 150 Hz lobes at equal depth.

0345. The ILS receiver of an aircraft on approach and flying on the glidepath will receive ;

a) no modulated signal because the two lobes cancel each other along the centerline.
b) the modulation from both lobes at equal depth.
c) the same modulation frequency signals from both lobes with the maximum of magnitude.
d) the maximum magnitude of the difference between the two amplitudes.

Answer ;

b) the modulation from both lobes at equal depth.

0346. The ILS receiver of an aircraft on approach and flying on the right of the exact runway centreline will receive ;

a) the left lobe modulation only.
b) more of the 150 Hz localiser signal than the 90 Hz localiser signal.
c) the modulation from both lobes at equal amplitude.
d) a modulated signal and will shift the localiser needle to the right according to the magnitude of the difference between the two amplitudes.

Answer ;

b) more of the 150 Hz localiser signal than the 90 Hz localiser signal.

0347. The inclination to the equatorial plane of the NAVSTAR/GPS orbits is ;

a) 65°
b) 45°
c) 35°
d) 55°

Answer ;

d) 55°

0348. The indicated range from a DME station is ;

a) 0 when passing overhead the station.
b) ground range.
c) slant range.
d) ground range only if the beacon is co-located with VOR.

Answer ;

c) slant range.

0349. The influence of the ionosphere on the accuracy of the satellite navigation system NAVSTAR/GPS is ;

a) minimised by the receiver using a model of the atmosphere and comparing signals transmitted by the satellites.
b) minimised by computing the average of all signals.
c) negligible.
d) only significant if the satellites are located at a small elevation angle above the horizon.

Answer ;

a) minimised by the receiver using a model of the atmosphere and comparing signals transmitted by the satellites.

0350. The inputs of information used to achieve the RNAV required accuracy may be ;

1. NDB
2. IRS
3. VOR/DME
4. GNSS

The combination regrouping all the correct statements is ;

a) 2 , 3 and 4.
b) 1 , 2 and 3.
c) 2 and 4.
d) 1 , 2 , 3 and 4.

Answer ;
a) 2 , 3 and 4.

0351. The IRS is a self-contained system because ;

a) it operates off its own power supply.
b) it operates independently of navigational aids outside the aircraft.
c) the calculation of the position does not require any software.
d) the system generates a warning in case of a failure.

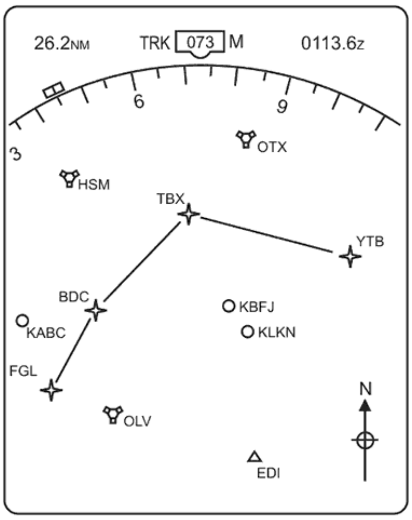
Answer ;
b) it operates independently of navigational aids outside the aircraft.

0352. The ISO-ECHO facility of an airborne weather radar is provided in order to ;

a) inhibit unwanted ground returns.
b) give an indication of cloud tops.
c) extend the mapping range.
d) detect areas of possible severe turbulence in cloud.

Answer ;
d) detect areas of possible severe turbulence in cloud.

0353. (For this question use annex 062-9916A)



Annex 062-9916A

The letters QTX and adjacent symbol indicate a ;

a) Airport
b) TACAN
c) VOR/DME/VORTAC
d) VOR

Answer ;
c) VOR/DME/VORTAC

0354. The localiser transmits in ;

a) the UHF band.
b) the HF band.
c) both UHF and VHF bands.
d) the VHF band.

Answer ;
d) the VHF band.

0355. The localiser transmitters operate in a frequency band between ;

a) 108 MHz and 111.975 MHz.
b) 108 MHz and 117.975 MHz.
c) 329.15 MHz and 335 MHz.
d) 111.975 MHz and 117.975 MHz.

Answer ;
a) 108 MHz and 111.975 MHz.

0356. The main factor which determines the minimum range that can be measured by a pulsed radar is pulse ;

a) repetition rate.
b) length.
c) amplitude.
d) frequency.

Answer ;
b) length.

0357. The main task of the user segment (receiver) of the satellite navigation system NAVSTAR/GPS is to calculate receiver position by ;

a) transmiting signals which , from the time taken, are used to determine the distance to the satellite.
b) monitoring the status of the satellites , determine their positions and to measure the time.
c) monitoring the orbital planes of the satellites.
d) selecting appropriate satellites automatically , to track the signals and to measure the time taken by signals from the satellites to reach the receiver.

Answer ;
d) selecting appropriate satellites automatically , to track the signals and to measure the time taken by signals from the satellites to reach the receiver.

0358. The maximum pulse repetition frequency (PRF) that can be used by a primary radar facility in order to detect targets unambiguously at a range of 50 NM is ; (pps = pulses per second)

a) 3240 pps
b) 1620 pps
c) 610 pps
d) 713 pps

50 NM = 50 x 1,852 = 92,6 km
Teorik Mesafe = 300.000 km/s : (2 x PRF)
92,6 = 300.000 : (2 x PRF)
PRF = 300.000 : (2 x 92,6) = 300.000 : 185,2 =1619,87 ≈ 1620 pps

Answer ;
b) 1620 pps

0359. The maximum range obtainable from an ATC Long Range Surveillance Radar is approximately ;

a) 300-400 NM
b) 100-200 NM
c) 50-100 NM
d) 200-300 NM

Answer ;
d) 200-300 NM

0360. The maximum range of primary radar depends on ;

a) pulse recurrence frequency.
b) frequency.
c) wave length.
d) pulse length.

Answer ;
a) pulse recurrence frequency.

0361. The maximum theoretical range at which an aircraft at FL230 may receive signals from a VOR facility sited at mean sea level is ;

a) 190 NM
b) 170 NM
c) 230 NM
d) 151 NM

Teorik olarak maksimum alış menzili (NM) = 1,25 x (√H1 +√H2)
= 1,25 x (√23000 + √0)
= 1,25 x (151,65 + 0) = 1,25 x 151,65 = 189,56 ≈ 190 NM

Answer ;
a) 190 NM

0362. The MIDDLE MARKER of an Instrument Landing System (ILS) facility is identified audibly and visually by a series of ;

a) alternate dots and dashes and an amber light flashing.
b) dots and a white light flashing.
c) dashes and an amber light flashing.
d) two dashes per second and a blue light flashing.

Answer ;
a) alternate dots and dashes and an amber light flashing.

0363. The middle marker transmits on ;

a) 75 MHz
b) 1300 Hz
c) 75 Hz
d) 1300 MHz

Answer ;
a) 75 MHz

0364. The minimum range of a primary radar , using the pulse technique , is determined by the (i)........................ , the maximum unambiguous range by the (ii)..........................

a) (i) pulse length , (ii) pulse recurrence frequency.
b) (i) pulse length , (ii) length of the timebase.
c) (i) transmission frequency , (ii) pulse recurrence frequency.
d) (i) transmission frequency , (ii) transmitter power output.

Answer ;
a) (i) pulse length , (ii) pulse recurrence frequency.

0365. The MLS frequencies and available channels are ;

a) in the SHF band , 300 kHz frequency separation giving 200 available channels.
b) in the SHF band for the MLS elements and the VHF band for the DME , 100 available channels.
c) in the rane 5060 - 5090 MHz , 200 kHz separation giving 150 available channels.
d) in the VHF and UHF band , 40 available channels.

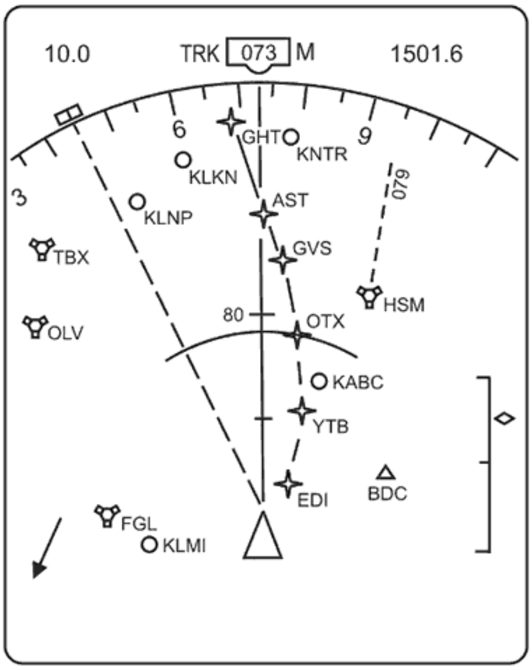
Answer ;
a) in the SHF band , 300 kHz frequency separation giving 200 available channels.

0366. The navigational function of the horizontal situation indicator (HSI) in relation to area navigation systems is ;

a) the indication of the track angle error (TKE).
b) the indication of the cross track distance (XTK).
c) the indication of the wind component.
d) the indication of the RNP.

Answer ;
b) the indication of the cross track distance (XTK).

0367. (For this question use annex 062-9921A)



Annex 062-9921A

The "O" followed by the letters "KABC" indicate ;

- **a)** the destination airport.
- **b)** an off-route airport.
- **c)** a designated alternate airport.
- **d)** an off-route VOR/DME.

Answer ;

b) an off-route airport.

0368. The OBS is set on 048° , TO appears in the window. The needle is close to full right deflection. The VOR radial is approximately ;

- **a)** 218°
- **b)** 038°
- **c)** 238°
- **d)** 058°

Answer ;

c) 238°

0369. The OBS is set to 235°. The indications of the VOR are half full scale deflection left and "to". The aircraft is on the radial ;

- **a)** 060°
- **b)** 230°
- **c)** 050°
- **d)** 240°

Answer ;

c) 050°

0370. The operating principle of a DME is the measurement of the ;

- **a)** frequency change between the emitted wave and reflected wave.
- **b)** time between the transmission and reception of radio pulses.
- **c)** phase difference between emitted wave and reflected wave.
- **d)** frequency of the reflected wave.

Answer ;

b) time between the transmission and reception of radio pulses.

0371. The orbital planes of the satellite navigation system NAVSTAR/GPS are ;

- **a)** inclined 55° to the earth axis.
- **b)** parallel to the equatorial plane.
- **c)** inclined 55° to the equatorial plane.
- **d)** inclined 90° to the equatorial plane.

Answer ;

c) inclined 55° to the equatorial plane.

0372. The OUTER MARKER of an Instrument Landing System (ILS) facility transmits on a frequency of ;

a) 75 MHz and is modulated by alternate dot/dash in morse.
b) 300 MHz and is modulated by morse at two dashes per second.
c) 75 MHz and is modulated by morse at two dashes per second.
d) 200 MHz and is modulated by alternate dot/dash in morse.

Answer ;
c) 75 MHz and is modulated by morse at two dashes per second.

0373. The pencil shaped beam of an airborne weather radar is used in preference to the mapping mode for the determination of ground features ;

a) beyond 50 to 60 NM because more power can be concentrated in the narrower beam.
b) beyond 100 NM because insufficient antenna tilt angle is available with the mapping mode.
c) when approaching coast-lines in polar regions.
d) beyond 150 NM because the wider beam gives better definition.

Answer ;
a) beyond 50 to 60 NM because more power can be concentrated in the narrower beam.

0374. (Refer to Image 35)

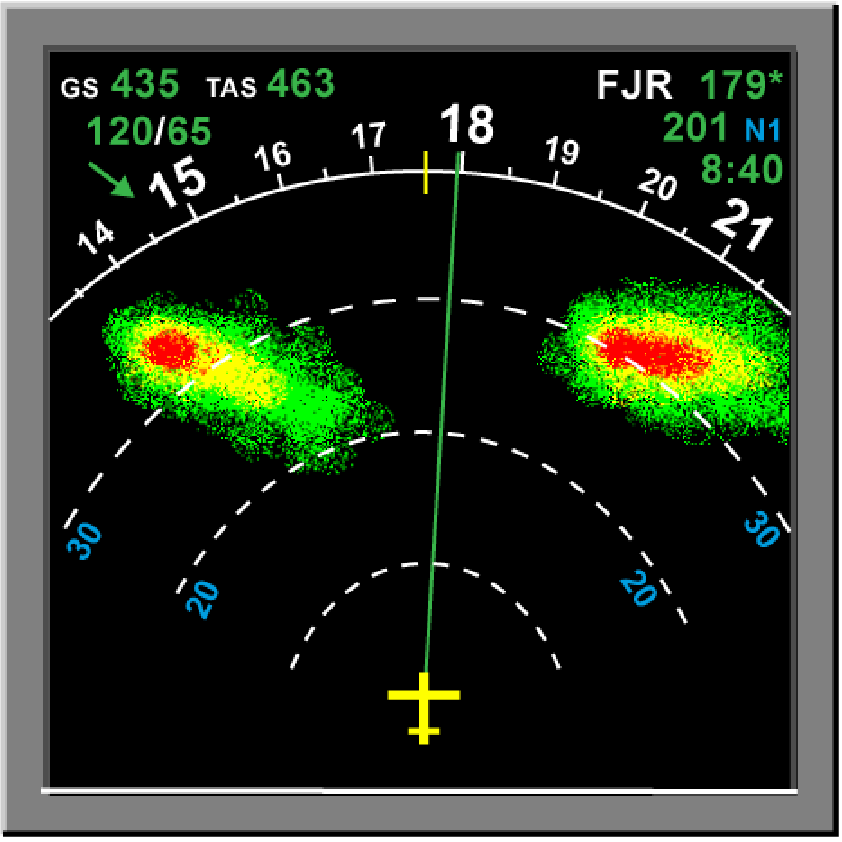


Image 35

The pictures at reference are showing an aircraft's position with respect to some thunderstorm cells and the image the pilot is getting from the radar. To detect this "blind alley" the pilot should ;

a) initiate a steep climb.
b) increase the range setting of his radar.
c) switch over from normal to WX-mode.
d) select the cosecant square beam for better alley detection.

Answer ;
b) increase the range setting of his radar.

0375. The prime factor in determining the maximum unambiguous range of a primary radar is the ;

a) size of parabolic receiver aerial.
b) height of the transmitter above the ground.
c) pulse recurrence rate.
d) power output.

Answer ;
c) pulse recurrence rate.

0376. The principle of operation of an ILS is ;

a) the phase comparison.
b) the difference between the frequencies of the two tones.
c) the difference in depth of modulation.
d) modulation of the ILS frequency by the amplitude of two modulating signal.

Answer ;
c) the difference in depth of modulation.

0377. The principle of operation of an ILS localiser transmitter is based on two overlapping lobes that are transmitted on (i).......... frequencies and carry different (ii)..........

a) (i) the same , (ii) modulation frequencies.
b) (i) different , (ii) phases.
c) (i) the same , (ii) phases.
d) (i) different , (ii) modulation frequencies.

Answer ;
a) (i) the same , (ii) modulation frequencies

0378. The principle used in VOR bearing measurement is ;

a) difference in depth of modulation.
b) phase comparison.
c) beat frequency discrimination.
d) envelope matching.

Answer ;
b) phase comparison.

0379. The quadrantal error of an ADF ;

a) is caused by interference from the sky wave.
b) may be caused by the interference of VOR's within range of the ADF receiver and cannot be compensated for.
c) is caused by the refraction from the aircraft's fuselage and is compensated for.
d) is caused by aircraft magnetism and varies with the deviation as shown on the deviation table.

Answer ;
c) is caused by the refraction from the aircraft's fuselage and is compensated for.

0380. The rate of descent required to maintain a 3.25° glide slope at a groundspeed of 140 kt is approximately ;

a) 850 FT/MIN
b) 760 FT/MIN
c) 700 FT/MIN
d) 670 FT/MIN

Answer ;
b) 760 FT/MIN

ROD (Rate Of Descent) = ɵ (glidepath angle) x ground speed (NM/min) x 101,3
140 kt , dakikada (140 : 60) = 2,3 NM/min
ROD = 3,25 x 2,3 x 101,3 = 757,21 ≈ 760 FT/MIN

0381. The reading of the RMI bearing is 300° at the tip of the needle. The magnetic variation at the DR position is 24°W, the magnetic variation at the NDB is 22°W and the deviation is -2°. The compass heading is 020°. The true bearing is ;

a) 274°
b) 272°
c) 294°
d) 094°

Answer ;
a) 274°

Çözüm ;

0382. The reason for using different frequencies for the airborne and ground equipment of a DME is ;

a) to avoid second trace returns when a DME is more than 200 NM away.
b) to prevent that DME interrogation pulse pairs being received by the aircraft after reflection on the earth surface.
c) that more DME frequencies are available for different beacons.
d) that side lobes can be suppressed by the Side Lobe Suppressor (SLS).

Answer ;
b) to prevent that DME interrogation pulse pairs being received by the aircraft after reflection on the earth surface.

0383. The reason why pre take-off holding areas are sometimes further from the active runway when ILS Category 2 and 3 landing procedures are in progress than during good weather operations is ;

- **a)** heavy precipitation may disturb guidance signals.
- **b)** aircraft manoeuvring near the runway may disturb guidance signals.
- **c)** to increase aircraft separation in very reduced visibility conditions.
- **d)** to increase distance from the runway during offset approach operations.

Answer ;

b) aircraft manoeuvring near the runway may disturb guidance signals.

0384. The reason why the measured distance between a NAVSTAR/GPS satellite navigation system satellite and a receiver is called a "Pseudo-Range" is because the ;

- **a)** calculated range includes receiver clock error.
- **b)** measured distance is based on the Pseudo Random Noise code.
- **c)** calculated range is based on an idealised Keplerian orbit.
- **d)** movement of satellite and receiver during the distance calculation is not taken into account.

Answer ;

a) calculated range includes receiver clock error.

0385. The receiver aerial for a NAVSTAR/GPS system should be mounted ;

- **a)** in the vicinity of the receiver to avoid long transmission lines.
- **b)** under the fuselage in order to receive correction data transmitted by D-GPS stations.
- **c)** on the upper side of the fuselage in the vicinity of the centre of gravity.
- **d)** inside the tail fin to minimise the influence of reflections from the wing and fuselage.

Answer ;

c) on the upper side of the fuselage in the vicinity of the centre of gravity.

0386. The required 24 NAVSTAR/GPS operational satellites are located on ;

- **a)** 6 orbital planes with 4 satellites in each plane.
- **b)** 3 orbital planes with 8 satellites in each plane.
- **c)** 4 orbital planes with 6 satellites in each plane.
- **d)** 6 orbital planes with 3 satellites in each plane plus 6 reserve satellites positioned in a geostationary orbital plane.

Answer ;

a) 6 orbital planes with 4 satellites in each plane.

0387. The selection of code 7500 on an aircraft SSR transponder indicates ;

- **a)** unlawful interference with the planned operation of the flight.
- **b)** radio communication failure.
- **c)** transponder malfunction.
- **d)** an emergency.

Answer ;

a) unlawful interference with the planned operation of the flight.

0388. The selection of code 7600 on an aircraft SSR transponder indicates ;

- **a)** transponder malfunction.
- **b)** unlawful interference with the planned operation of the flight.
- **c)** radio communication failure.
- **d)** an emergency.

Answer ;

c) radio communication failure.

0389. The selection of code 7700 on an aircraft SSR transponder indicates ;

- **a)** unlawful interference with the planned operation of the flight.
- **b)** an emergency.
- **c)** transponder malfunction.
- **d)** radio communication failure.

Answer ;

b) an emergency.

0390. The selection of the DME frequency for a ILS/DME installation is as follows ;

- **a)** The DME frequency is paired with the localizer frequency so only the localiser frequency is set.
- **b)** The pilot has to insert the channel number and also the X or Y.
- **c)** When an ILS/DME is flown then selection of the DME has always to be one manually.
- **d)** The DME frequency has to be selected by the pilot.

Answer ;

a) The DME frequency is paired with the localizer frequency so only the localiser frequency is set.

0391. The skip distance of HF-transmission will increase with ;

a) higher frequency and higher position of the reflecting ionospheric layer.
b) higher frequency and lower position of the reflecting ionospheric layer.
c) lower frequency and higher position of the reflecting ionospheric layer.
d) lower frequency and lower position of the reflecting ionospheric layer.

Answer ;
a) higher frequency and higher position of the reflecting ionospheric layer.

0392. The theoretical maximum range for an Airborne Weather Radar is determined by the ;

a) pulse recurrence frequency.
b) carrier wave frequency.
c) pulse length.
d) beamwidth.

Answer ;
a) pulse recurrence frequency.

0393. The three different markers can be used in the ILS to determine the distance to the ILS touchdown point of the runway as follows ;

a) The inner marker warns the pilot of the last chance to commence the missed approach procedure.
b) The middle marker indicates the position for the decision for a missed approach during a CAT I approach due to bad visibility.
c) The outer marker indicates the position where normally the descent has to commence.
d) The markers are only important in the situation when the Glide Path transmission has ceased.

Answer ;
b) The middle marker indicates the position for the decision for a missed approach during a CAT I approach due to bad visibility.

0394. The three main components of VOR airborne equipment are ;

a) Demodulator , Antenna , CDU.
b) Receiver , Phase comparator , Range gate.
c) Display , Pulse generator , phase comparator.
d) Receiver , Antenna , Display.

Answer ;
d) Receiver , Antenna , Display.

0395. The time taken for the transmission of an interrogation pulse by a Distance Measuring Equipment (DME) to travel to the ground transponder and return to the airborne receiver was 2000 micro-second , including time delay. The slant range from the ground transponder was ;

a) 158 NM
b) 186 NM
c) 296 NM
d) 330 NM

Answer ;
a) 158 NM

Distance (Mesafe) = Speed (Hız) x Time (Zaman) formülünden yola çıkarak ;
Burada hız olarak ışık hızını kullanıyoruz = (300 x 10^{-6}) metre/saniye , zaman olarak da DME’nin gönderdiği 1 pulse’ın gidiş-dönüş süresi olan 2000 micro-saniye’nin yarısını yani 1000 mikro-saniye = (1000 x 10^{-6}) saniye’yi alıyoruz ;
Mesafe = (300 x 10^{-6}) x (1000 x 10^{-6}) = 300 000 metre = 300 000 : 1852 = 161,9 ≈ 158 NM

0396. The transmission of the glide slope beacon is characterised by a ;

a) UHF frequency with a minimum range of 10 NM.
b) 300 to 3000 Hz Amplitude modulation for the ATIS.
c) VHF frequency modulated with a 90 Hz AM and 150 Hz AM navigation signal.
d) UHF carrier frequency with a possible “voce ident”.

Answer ;
a) UHF frequency with a minimum range of 10 NM.

0397. The two main design functions of Secondary Surveillance Radar (SSR) Mode S are ;

a) the elimination of ground to air communications and the introduction of automatic separation between aircraft using TCAS II.
b) collision avoidance using TCAS II and improved long range (HF) communication capability.
c) air to ground and ground to air data link communications and improved ATC aircraft surveillance capability.
d) continuous automatic position reporting using Global Positioning System (GPS) satellites and collision avoidance using TCAS II.

Answer ;
c) air to ground and ground to air data link communications and improved ATC aircraft surveillance capability.

0398. The type of modulation used for the ILS frequency carrier is ;

a) amplitude modulation.
b) dual modulation.
c) frquency modulation.
d) phase modulation.

Answer ;
a) amplitude modulation.

0399. The UHF band is assigned to the ;

1. Locator.
2. Localiser.
3. Outer marker.
4. Glide path.

The combination that regroups all the corrects statements is ;

a) 3 and 4.
b) 4.
c) 1.
d) 1 and 3.

Answer ;
b) 4.

0400. The UHF band is the assigned frequency band of the ;

a) ILS glide path transmitter.
b) ILS localizer transmitter.
c) outer marker beacon.
d) all the 3 ILS marker beacons.

Answer ;
a) ILS glide path transmitter.

0401. The VOR system is limited to about 1° of accuracy. One degree at 200 NM represents a width of ;

a) 2.5 NM
b) 2.0 NM
c) 3.5 NM
d) 3.0 NM

Answer ;
c) 3.5 NM

Track Error Angle (TEA) = (Distance Off-Track x 60) : Distance Along Track formülünü kullanarak ;
1° = (Distance Off-Track x 60) : 200 NM , öğelerin yerini değiştirdiğimizde ;
Distance Off-Track = (200 x 1°) : 60 = 200 : 60 = 3,33 ≈ 3.5 NM

0402. There are two NDBs , one 20 NM inland , and the other 50 NM inland from the coast. Assuming that the error caused by coastal refraction is the same for both propagations , the extent of the error in a position line plotted by an aircraft that is over water will be ;

a) greater from the beacon that is 50 NM inland.
b) greater from the beacon that is 20 NM inland.
c) the same from both beacons when the aircraft is on a relative bearing of 090° and 270°.
d) the same from both beacons when the aircraft is on a relative bearing of 180° and 360°.

Answer ;
a) greater from the beacon that is 50 NM inland.

0403. To improve the detection of precipitation area(s) in e.g thunderstorms , in which the top of the cloud lies at or slightly above the level of flight ;

a) the tilt setting should be lower when the selected range increases.
b) the tilt setting should be higher when the selected range decreases.
c) the tilt setting should be higher when the aircraft climbs to a higher altitude.
d) the tilt setting should be lower when the aircraft climbs to a higher altitude.

Answer ;
d) the tilt setting should be lower when the aircraft climbs to a higher altitude.

0404. To provide a pilot with the position of the aircraft in the absence of radar , ATC must have at its disposal at least ;

a) two VDF's at different locations , able to take bearings simultaneously on the transmitted frequency.
b) two co-located VDF's , able to take bearings simultaneously on the transmitted frequency.
c) one VDF able to take simultaneously bearings on different frequencies.
d) three VDF's at different locations able to take bearings simultaneously on different frequency.

Answer ;
a) two VDF's at different locations , able to take bearings simultaneously on the transmitted frequency.

0405. Transmissions from VOR facilities may be adversely affected by ;

a) night effect.
b) uneven propagation over irregular ground surfaces.
c) quadrantal error.
d) static interference.

Answer ;
b) uneven propagation over irregular ground surfaces.

0406. TVOR is a ;

a) VOR with a limited range used in the terminal area.
b) low power DVOR in the frequency range 112 MHz - 118 MHz.
c) test VOR transmitting such a signal that the reference and variable signal are always in phase.
d) high power VOR in the frequency range 108 MHz - 112 MHz.

Answer ;
a) VOR with a limited range used in the terminal area.

0407. Two aircraft are located on (arbitrary) different radials but at equal distances from a VOR-station. Which statement is true ?

a) At a certain moment of time the phase of the reference signals is equal and the phase of variable signals is unequal for both aircraft.
b) At a certain moment of time , both the phase of the reference signals and of the variable signals are equal for both aircraft.
c) At a certain moment of time , both the phase of the reference signals and of the variable signals are unequal for both aircraft.
d) At a certain moment of time the phase of the reference signals is unequal and the phase of variable signals is equal for both aircraft.

Answer ;
a) At a certain moment of time the phase of the reference signals is equal and the phase of variable signals is unequal for both aircraft.

0408. Two aircraft are located on the same radial but at (arbitrary) different distances from a VOR-station. Which statement is true ?

a) At a certain moment of time , both the phase of the reference signals and of the variable signals are equal for both aircraft.
b) At a certain moment of time , the phase of the reference signals is equal and the phase of variable signals is unequal for both aircraft.
c) At a certain moment of time , both the phase of the reference signals and of the variable signals are unequal for both aircraft.
d) At a certain moment of time , the phase of the reference signals is unequal and the phase of variable signals is equal for both aircraft.

Answer ;
a) At a certain moment of time , both the phase of the reference signals and of the variable signals are equal for both aircraft.

0409. Under JAR-25 colour code rules , features displayed in amber/yellow on an Electronic Flight Instrument System (EFIS) , indicate ;

a) flight envelope and system limits.
b) warnings.
c) cautions , abnormal sources.
d) engaged modes.

Answer ;
c) cautions , abnormal sources.

0410. Under JAR-25 colour code rules , features displayed in cyan/blue on an Electronic Flight Instrument Systems (EFIS) , indicate ;

a) engaged modes.
b) the flight director bar(s).
c) flight envelope and system limits.
d) the sky.

Answer ;
d) the sky.

0411. Under JAR-25 colour code rules , features displayed in red on an Electronic Flight Instrument System (EFIS) , indicate ;

- **a)** warnings , cautions and abnormal sources.
- **b)** warnings , flight envelope and system limits.
- **c)** flight envelope and system limits , engaged modes.
- **d)** cautions and abnormal sources , engaged modes.

Answer ;

b) warnings , flight envelope and system limits.

0412. Under JAR-25 colour code rules for Electronic Flight Instrument Systems (EFIS) , increasing intensity of precipitation are coloured in the order:

- **a)** green , red , magenta , black.
- **b)** green , amber/yellow , red , magenta.
- **c)** black , amber/yellow , magenta , red.
- **d)** amber/yellow , magenta , black.

Answer ;

b) green , amber/yellow , red , magenta.

0413. Under JAR-25 general colour code rules , features displayed in green on an Electronic Flight Information System should indicate ;

- **a)** the earth.
- **b)** the ILS deviation pointer.
- **c)** engaged modes.
- **d)** cautions , abnormal sources.

Answer ;

c) engaged modes.

0414. Under which of the following circumstances does a VOR/DME Area Navigation system switch to Dead Reckoning mode ?

- **a)** VOR/DME Area Navigation computer is receiving neither radial nor distance data information from VOR/DME stations.
- **b)** When "DR" is selected by the pilot.
- **c)** VOR/DME Area Navigation computer is not receiving information from the Air Data Computer.
- **d)** VOR/DME Area Navigation computer is not receiving information from the aircraft compass system.

Answer ;

a) VOR/DME Area Navigation computer is receiving neither radial nor distance data information from VOR/DME stations.

0415. Under which of the following circumstances does a VOR/DME Area Navigation system switch to Dead Reckoning mode ?

- **a)** The system is receiving information from the two DMEs.
- **b)** The system is receiving information from one VOR and one DME.
- **c)** The system is receiving information from one VOR and two DMEs.
- **d)** The system is receiving information from only one VOR.

Answer ;

d) The system is receiving information from only one VOR.

0416. Unless otherwise specified a radial is ;

- **a)** the true great circle direction from the beacon.
- **b)** the magnetic great circle direction from the beacon.
- **c)** the true great circle direction to the beacon.
- **d)** the magnetic great circle direction to the beacon.

Answer ;

b) the magnetic great circle direction from the beacon.

0417. VDF is the abbreviation for ;

- **a)** VHF Direction Finder
- **b)** Very Direct Finder
- **c)** Very High Frequency Deviation Finding Station
- **d)** VDF Direction Finder

Answer ;

a) VHF Direction Finder

0418. VDF measures the bearing of the aircraft with ;

- **a)** reference to true or magnetic north at the station.
- **b)** reference to aircraft relative bearing.
- **c)** reference to true or magnetic north at the aircraft.
- **d)** reference only to magnetic north at the aircraft.

Answer ;

a) reference to true or magnetic north at the station.

0419. What according to ICAO Annex 10 is the range of a locator ?

- **a)** 10 - 25 NM
- **b)** 100 - 300 NM
- **c)** 25 - 50 NM
- **d)** 50 - 100 NM

Answer ;

a) 10 - 25 NM

0420. What actually happens in the ADF receiver when the BFO position is selected ?

- **a)** The BFO circuit imposes a tone onto the carrier wave to make the NDB's ident audible.
- **b)** The BFO circuit is de-activated.
- **c)** The BFO circuit oscillates at an increased frequency in order to allow identification of A2A NDBs.
- **d)** The BFO circuit is activated , and the receiver accepts only A1A modulated signals.

Answer ;

a) The BFO circuit imposes a tone onto the carrier wave to make the NDB's ident audible.

0421. What airborne equipment , if any , is required to be fitted in order that a VDF let-down may be flown ?

- **a)** none
- **b)** VHF radio
- **c)** VOR
- **d)** VOR/DME

Answer ;

b) VHF radio

0422. What approximate rate of descent is required in order to maintain a 3° glidepath at a groundspeed of 90 kt ?

- **a)** 400 FT/MIN
- **b)** 450 FT/MIN
- **c)** 700 FT/MIN
- **d)** 600 FT/MIN

ROD (Rate Of Descent) = ɵ (glidepath angle) x ground speed (NM/min) x 101,3
90 kt , dakikada (90 : 60) = 1,5 NM/min
ROD = 3 x 1,5 x 101,3 = 455,85 ≈ 450 FT/MIN

Answer ;

b) 450 FT/MIN

0423. What approximate rate of descent is required in order to maintain a 3° glide path at a groundspeed of 120 kt ?

- **a)** 600 FT/MIN
- **b)** 550 FT/MIN
- **c)** 950 FT/MIN
- **d)** 800 FT/MIN

ROD (Rate Of Descent) = ɵ (glidepath angle) x ground speed (NM/min) x 101,3
120 kt , dakikada (120 : 60) = 2 NM/min
ROD = 3 x 2 x 101,3 = 607,8 ≈ 600 FT/MIN

Answer ;

a) 600 FT/MIN

0424. What are the effects , if any , of shadowing by parts of the aircraft (e.g. wing) on the reception of signals from NAVSTAR/GPS satellites ?

- **a)** It causes multipath propagation.
- **b)** The signals will be distorted , however the error can be corrected for using an algorithm and information from unaffected signals.
- **c)** It may prevent the reception of signals.
- **d)** It has no influence because high frequency signals are unaffected.

Answer ;

c) It may prevent the reception of signals.

0425. What are the modulation frequencies of the two overlapping lobes that are used on an ILS approach ?

- **a)** 90 Hz 150 Hz.
- **b)** 328 MHz 335 MHz.
- **c)** 75 kHz 135 kHz.
- **d)** 63 MHz 123 MHz.

Answer ;

a) 90 Hz 150 Hz.

0426. What causes the so-called night effect ?

a) A change in the direction of the plane of polarisation due to reflection in the ionosphere.
b) The difference in velocity of the EM-waves over land and over sea , at night.
c) The absence of the surface wave at distances larger than the skip distance.
d) Interference beween the ground and the space wave.

Answer ;
a) A change in the direction of the plane of polarisation due to reflection in the ionosphere.

0427. What datum is used for the Minimum Descent Altitude (MDA) on a non-precision approach when using the NAVSTAR/GPS satellite navigation system ?

a) If using Differential-GPS (D-GPS) the altitude obtained from the D-GPS , otherwise barometric altitude.
b) Barometric altitude.
c) Radar altitude.
d) GPS altitude.

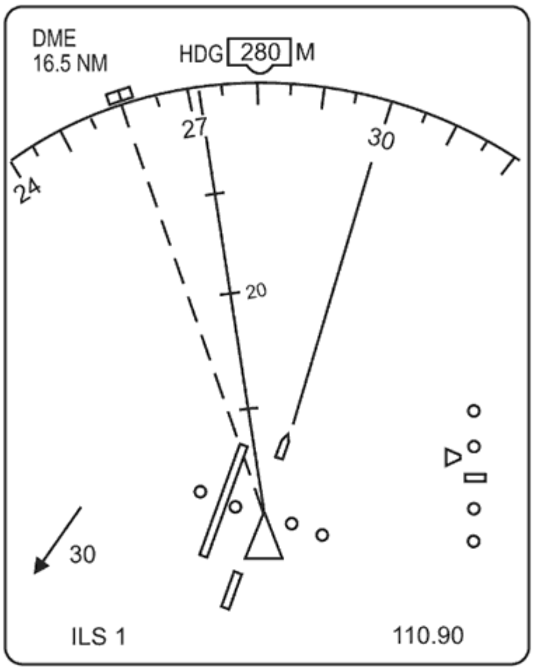
Answer ;
b) Barometric altitude.

0428. What does ADF stand for ?

a) Airborne Direction Finding
b) Airport Direction Finder
c) Aeroplane Direction Finding
d) Automatic Direction Finder

Answer ;
d) Automatic Direction Finder

0429. (For this question use annex 062-9915A)



Annex 062-9915A

What drift is being experienced ?

a) 20° Left
b) 8° Left
c) 12° Right
d) 20° Right

Answer ;
b) 8° Left

0430. What information may be displayed on an ATC radar screen connected only to a primary radar system ?

a) Aircraft position only.
b) Aircraft altitude.
c) Aircraft position , SSR code and altitude.
d) Aircraft position and SSR code.

Answer ;
a) Aircraft position only.

0431. What is a VDF referenced to ?

a) True north at the aircraft.
b) Magnetic north at the aircraft.
c) Relative bearing to the aircraft.
d) Magnetic north at the station.

Answer ;

d) Magnetic north at the station.

0432. What is cross track distance (XTK-distance) in an Area Navigation System ?

a) Distance between air position and planned track.
b) Distance between actual position and great circle track between active waypoints.
c) Distance between air position and great circle track between active waypoints.
d) Distance between actual position and next waypoint.

Answer ;

b) Distance between actual position and great circle track between active waypoints.

0433. What is EGNOS (European Global Navigation Overlay System) ?

a) GLONASS
b) Local Area Differential GPS (LADGPS)
c) Local Area Augmentation System (LAAS)
d) Wide Area Differential GPS (WADGPS)

Answer ;

d) Wide Area Differential GPS (WADGPS)

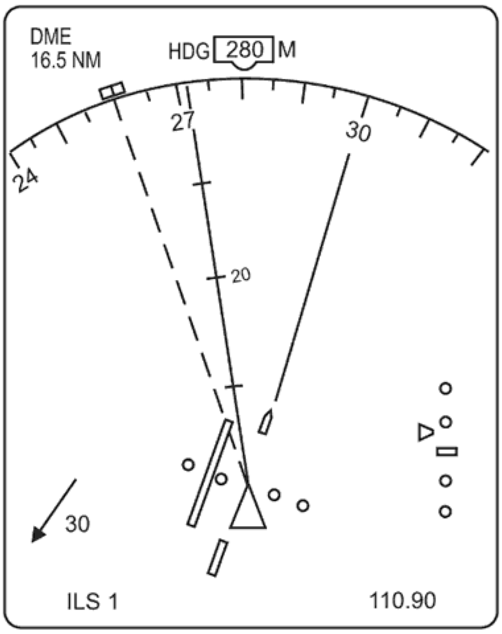
0434. What is measured in order to establish aircraft position in relation to the localizer beam on an ILS ?

a) The difference in depth between the 90 Hz modulation and the 150 Hz modulation.
b) The difference in time between the 90 Hz modulation and the 150 Hz modulation.
c) The bearing to the localizer antenna found by means of a loop antenna.
d) The difference in phase between the 90 Hz modulation and the 150 Hz modulation.

Answer ;

a) The difference in depth between the 90 Hz modulation and the 150 Hz modulation.

0435. (For this question use annex 062-9919A)



Annex 062-9919A

What is the aircraft track ?

a) 260°(M)
b) 272°(M)
c) 280°(M)
d) 300°(M)

Answer ;

b) 272°(M)

0436. What is the approximate angular coverage of reliable navigation information for a 3° ILS glide path out to a minimum distance of 10 NM ?

a) 3° above and below the glide path and 10° each side of the localiser centreline.
b) 0.45° above the horizontal to 1.75° above the glide path and 8° each side of the localiser centreline.
c) 1.35° above the horizontal to 5.25° above the horizontal and 8° each side of the localiser centreline.
d) 0.7° above and below the glide path and 2.5° each side of the localiser centreline.

Answer ;
c) 1.35° above the horizontal to 5.25° above the horizontal and 8° each side of the localiser centreline.

0437. What is the approximate maximum theoretical range at which an aircraft at FL130 could receive information from a VDF facility which is sited 1024 ft above MSL ?

a) 150 NM
b) 180 NM
c) 220 NM
d) 120 NM

Teorik olarak maksimum alış menzili (NM) = 1,25 x ($\sqrt{H1} + \sqrt{H2}$)
= 1,25 x ($\sqrt{13000} + \sqrt{1024}$)
= 1,25 x (114 + 32) = 1,25 x 146 = 182,5 ≈ 180 NM

Answer ;
b) 180 NM

0438. What is the audio frequency of the inner marker ?

a) 3000 Hz
b) 1300 Hz
c) 75 MHz
d) 400 Hz

Answer ;
a) 3000 Hz

0439. What is the colour sequence when passing over an Outer , Middle and Inner Marker beacon ?

a) amber - white - green
b) blue - amber - white
c) blue - green - white
d) white - amber - blue

Answer ;
b) blue - amber - white

0440. What is the cross track deviation (XTK) indicated on an RNAV system ?

a) The distance along a track between two waypoints.
b) The distance between the actual position and the great circle track between two active waypoints.
c) The distance between the air position and the planned track.
d) The distance between the air position and the great circle track between two active waypoints.

Answer ;
b) The distance between the actual position and the great circle track between two active waypoints.

0441. What is the effect of FM broadcast stations that transmit on frequencies just below 108 MHz on the performance of ILS ?

a) These transmissions may interfere with the ILS localizer signal which may lead to erroneous localizer deviation indiction.
b) These transmissions may activate the FM immune filter which results in the appearance of the localizer failure flag.
c) These transmissions may activate the FM immune filter which results in the appearance of the localizer and glide path failure flag.
d) These transmissions may interfere with the ILS localizer and glide path signals which may lead to erroneous deviation indictions.

Answer ;
a) These transmissions may interfere with the ILS localizer signal which may lead to erroneous localizer deviation indiction.

0442. What is the effect of multipath signals (coming from the same aircraft) at the Ground VHF Direction Finder station ?

a) Regardless the difference in distance travelled by these signals , it results in their extinction of the signals at the Ground VHF Direction Finder station.
b) It may results in an increase of the distance at which the Ground VHF Direction Finder station receives signals from the aircraft , if the Ground Station is situated in the skipzone.
c) It reduces the range at which the Ground VHF Direction Finder station receives signals from the aircraft.
d) They may result in bearing errors.

Answer ;
d) They may result in bearing errors.

0443. What is the function of a FM-immune filter ?

- **a)** To make the ILS localizer receiver less susceptible to interference from earth-reflected localizer-signals.
- **b)** To make both the ILS localizer and glide path receiver less susceptible to interference from earth-reflected localizer-signals.
- **c)** To make the ILS localizer receiver less susceptible to interference from commercial FM-stations (radio and television).
- **d)** To make both the ILS localizer and glide path receiver less susceptible to interference from commercial FM-stations (radio and television).

Answer ;

c) To make the ILS localizer receiver less susceptible to interference from commercial FM-stations (radio and television).

0444. What is the function of the control segment in GPS NAVSTAR ?

- **a)** It calculates the aircraft position.
- **b)** To ensure the transmitted data of the satellites is controlled and updated from time to time by ground stations.
- **c)** To monitor and ensure that the transmitted signals are saved and processed to utilise WAAS.
- **d)** To transmit a signal used by a suitable receiver to calculate position.

Answer ;

b) To ensure the transmitted data of the satellites is controlled and updated from time to time by ground stations.

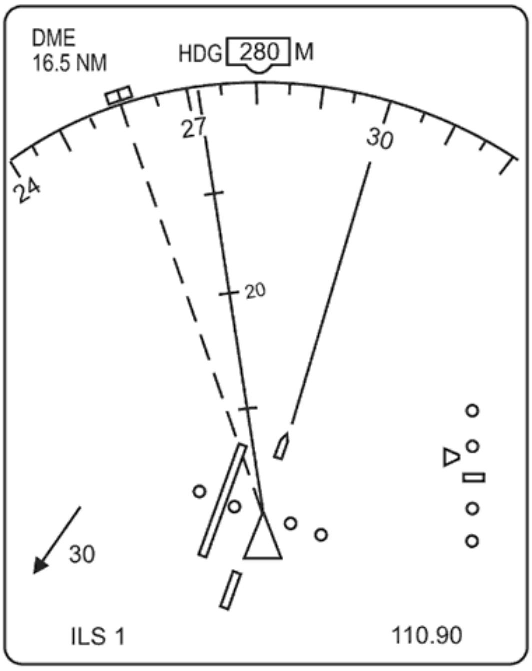
0445. What is the function of the Echo Protection Circuit (EPC) ?

- **a)** The EPC has to detect if pulse-pairs reflected by the ionosphere are interfering with directly received pulse-pairs.
- **b)** The EPC has to detect if unwanted squitter-pulses are interfering with the well-functioning of the interrogator-receiver.
- **c)** The EPC has to detect if the interrogator-receiver has been locked on , by reflected pulse-pairs.
- **d)** The EPC has to protect the transponder-receiver against reflected pulse-pairs.

Answer ;

c) The EPC has to detect if the interrogator-receiver has been locked on , by reflected pulse-pairs.

0446. (For this question use annex 062-9919A)



Annex 062-9919A

What is the heading bug selected to ?

- **a)** 280°(M)
- **b)** 300°(M)
- **c)** 272°(M)
- **d)** 260°(M)

Answer ;

d) 260°(M)

0447. What is the inclination to the equatorial plane of the satellite's orbit in the NAVSTAR GPS constellation ?

a) 45°
b) 65°
c) 35°
d) 55°

Answer ;
d) 55°

0448. What is the maximum latitude of a GPS-satellite ground track ?

a) 55º N/S
b) 67.5º N/S
c) 35º N/S
d) 90º N/S

Answer ;
a) 55º N/S

0449. What is the maximum number of usable Secondary Surveillance Radar (SSR) transponder codes in Mode A ?

a) 4096
b) 3600
c) 760
d) 1000

Answer ;
a) 4096

0450. What is the maximum theoretical range that an aircraft at FL150 can receive signals from a VOR situated 609 feet above MSL ?

a) 184 NM
b) 220 NM
c) 156 NM
d) 147 NM

Teorik olarak maksimum alış menzili (NM) = 1,25 x ($\sqrt{H1} + \sqrt{H2}$)
= 1,25 x ($\sqrt{15000} + \sqrt{609}$)
= 1,25 x (122,47 + 24,67) = 1,25 x 147,14 = 183,9 ≈ 184 NM

Answer ;
a) 184 NM

0451. What is the minimum level that an aircraft , at a range of 113 NM , must fly in order to contact the tower on R/T for a VDF bearing from an airport sited 169 ft above MSL ?

a) FL80
b) FL50
c) FL60
d) FL100

Teorik olarak maksimum alış menzili (NM) = 1,25 x ($\sqrt{H1} + \sqrt{H2}$)
Burada yayın alınabilecek minimum seviyeye X dersek ;
113 = 1,25 x ($\sqrt{X} + \sqrt{169}$) → 113 = 1,25 x ($\sqrt{X}$ + 13)
$\sqrt{X}$ = (113 : 1,25) – 13 → $\sqrt{X}$ = 77,4 → X = (77,4)^2 = 5990,76 ft ≈ FL60

Answer ;
c) FL60

0452. What is the minimum number of NAVSTAR/GPS satellites required to produce an accurate independent 3-D position fix ?

a) 24
b) 4
c) 5
d) 3

Answer ;
b) 4

0453. What is the minimum number of satellites required by a GPS in order to obtain a three dimensional fix ?

a) 4
b) 5
c) 3
d) 6

Answer ;
a) 4

0454. What is the minimum number of satellites required for the NAVSTAR/GPS to carry out two dimensional operation ?

a) 5
b) 4
c) 2
d) 3

Answer ;
d) 3

0455. What is the procedure to be followed if , on a flight under IFR conditions using the NAVSTAR/GPS satellite navigation system , the number of satellites required to maintain the RAIM (Receiver Autonomous Integrity Monitoring) function are not available ?

- **a)** A constant heading and speed must be flown until the required number of satellites are again available.
- **b)** The flight may be continued using other certificated navigation systems.
- **c)** The flight may be continued as planned if at least 4 satellites are available and the pilot monitors the GPS-System manually.
- **d)** The flight has to be continued under VFR conditions.

Answer ;

b) The flight may be continued using other certificated navigation systems.

0456. What is the "Q" code for a magnetic bearing from a VDF station ?

- **a)** "Request QNH".
- **b)** "Request QTE".
- **c)** "Request QDR".
- **d)** "Request QDM".

Answer ;

c) "Request QDR".

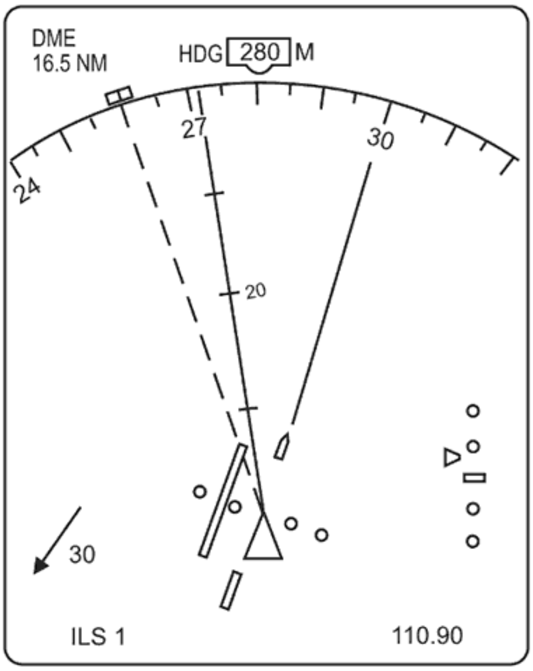
0457. What is the time taken to receive the complete Navigation Message (complete set of data) from all satellites ?

- **a)** 12.5 minutes (= 30 seconds per data frame).
- **b)** 25 seconds (= 1 second per data frame).
- **c)** 24 seconds (= 1 second per data frame).
- **d)** 12 hours (= period of the satellites orbit).

Answer ;

a) 12.5 minutes (= 30 seconds per data frame).

0458. (For this question use annex 062-9918A)



Annex 062-9918A

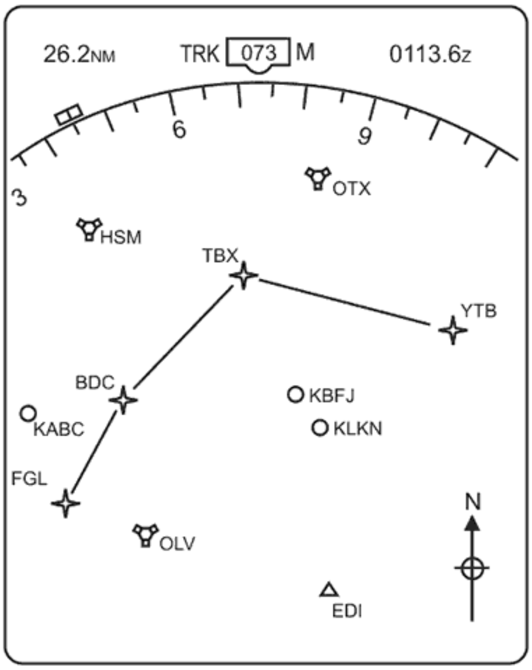
What is the value of the selected course ?

- **a)** 280°(M)
- **b)** 260°(M)
- **c)** 299°(M)
- **d)** 272°(M)

Answer ;

c) 299°(M)

0459. (For this question use annex 062-9916A)



Annex 062-9916A

What is the value of the track from TBX to YTB ?

a) 280°(T)
b) 170°(M)
c) 097°(T)
d) 140°(M)

Answer ;
c) 097°(T)

0460. What is the wavelength of an NDB transmitting on 375 kHZ ?

a) 8 m
b) 8000 m
c) 80 m
d) 800 m

f (frekans) = C (Işık hızı) : λ (Dalga boyu)
Değerimiz kHz olduğu için Işık hızını (300 x 10^3) km/sec olarak alıyoruz ;
375 = (300 x 10^3) : λ → λ = 300.000 : 375 = 800 m olarak bulunur.

Answer ;
d) 800 m

0461. What is true about the FMC databases ?

a) The data can not be customized for the specific airline operations.
b) The navigation database contains the company's cost index strategy.
c) The perfomance database contains aeronautical information for the planned area of operations of the aircraft.
d) The data includes SIDs , STARs and runway approaches.

Answer ;
d) The data includes SIDs , STARs and runway approaches.

0462. What type of clock is used in NAVSTAR GPS satellites ?

a) Mechanical
b) Quartz
c) Laser
d) Atomic

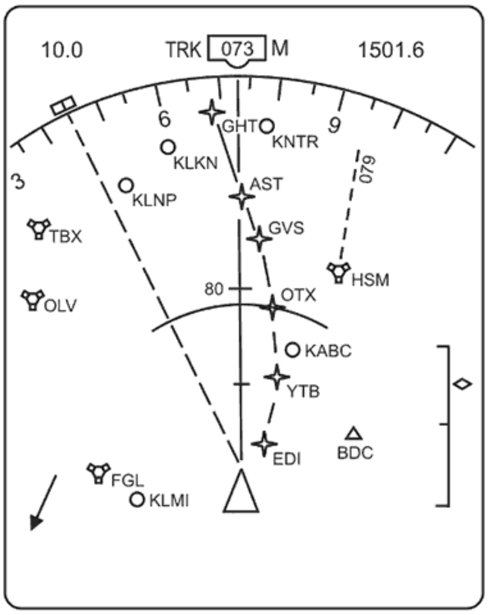
Answer ;
d) Atomic

0463. What type of satellite navigation system NAVSTAR/GPS receiver is most suitable for use on board an aircraft ?

a) Multiplex.
b) Any hand held type.
c) Sequential.
d) Multichannel.

Answer ;
d) Multichannel.

0464. (For this question use annex 062-9917A)



Annex 062-9917A

What wind velocity is indicated ?

- **a)** 105°(M) / 20 KT
- **b)** 285°(M) / 20 KT
- **c)** 030°(M) / 20 KT
- **d)** 255°(M) / 20 KT

Answer ;

a) 105°(M) / 20 KT

0465. When an aircraft is operating its Secondary Surveillance Radar in Mode C an air traffic controller's presentation gives information regarding the aircraft's indicated flight level in increments of ;

- **a)** 250 ft.
- **b)** 150 ft.
- **c)** 100 ft.
- **d)** 200 ft.

Answer ;

c) 100 ft.

0466. When an operator increases the range on a radar-display , in general ;

- **a)** the PRF becomes lower and the pulse length larger.
- **b)** the PRF becomes higher and the pulse length smaller.
- **c)** both the PRF and the pulse length decrease.
- **d)** both the PRF and the pulse length increase.

Answer ;

a) the PRF becomes lower and the pulse length larger.

0467. When entering and using a phantom waypoint in area navigation equipment ;

- **a)** you don't need to be in range of the referenced station to enter or use it.
- **b)** the referenced station must be positively identified by at least 1 pilot.
- **c)** you must be in range of the referenced station to enter or use it.
- **d)** you don't need to be within range of the referenced station to enter the waypoint , but you must be to use it.

Answer ;

d) you don't need to be within range of the referenced station to enter the waypoint , but you must be to use it.

0468. When flying at 6000 feet above ground level , the DME indicates 5 NM. What is the horizontal distance from the aircraft to overhead the DME ?

- **a)** 4.6 NM
- **b)** 5.2 NM
- **c)** 4.3 NM
- **d)** 4.9 NM

$A^2 = B^2 + C^2$ eşitliğine göre ; A = Slant range ve Üçgenin hipotenüsü , B = Uçağın irtifası ve Üçgenin bir kenarı C = Yeryüzündeki mesafe ve Üçgenin diğer kenarı ;
6000 ft = 6000 : 6080 = 0,98684 NM ,
$(5)^2 = (0,98684)^2 + C^2 \rightarrow 25 = 0,97385 + C^2 \rightarrow C^2 = 25 - 0,97385 = 24,02$
$C = \sqrt{24,02} = 4,901 \approx 4.9$ NM

Answer ;

d) 4.9 NM

0469. When Mode C is selected on the aircraft SSR transponder the additional information transmitted is ;

- **a)** altitude based on regional QNH.
- **b)** flight level based on 1013.25 hPa.
- **c)** height based on QFE.
- **d)** aircraft height based on sub-scale setting.

Answer ;

b) flight level based on 1013.25 hPa.

0470. When using airborne weather radar in the mapping mode in polar areas one runs the risk ;

- **a)** of getting a distorted picture because of ice reflection.
- **b)** of aurora borealis (polar light) causing false returns.
- **c)** of underestimating distances because the cold seawater is causing super refraction.
- **d)** of mistaking the edge of coastal ice off shore for the real coastline.

Answer ;

d) of mistaking the edge of coastal ice off shore for the real coastline.

0471. Where , in relation to the runway , is the ILS localiser transmitting aerial normally situated ?

- **a)** On the non-approach end of the runway about 300 m from the runway on the extended centreline.
- **b)** At the approach end of the runway about 300 m from touchdown on the centreline.
- **c)** At the non-approach end about 150 m to one side of the runway and 300 m along the extended centreline.
- **d)** At the approach end about 150 m to one side of the runway and 300 m from touchdown.

Answer ;

a) On the non-approach end of the runway about 300 m from the runway on the extended centreline.

0472. Which answer states the typical distances along the centerline of the runway of the various ILS components ?

- **a)** Localizer transmitter : 300 meter behind end of runway , Glide path transmitter : 300 meter behind threshold , Middle Marker : 1.5 NM from threshold , Outer Marker : 10 NM from threshold.
- **b)** Localizer transmitter : 100 meter behind end of runway , Glide path transmitter : 300 meter behind threshold , Middle Marker : 1000 meter from threshold , Outer Marker : 4 NM from threshold.
- **c)** Localizer transmitter : 100 meter behind end of runway , Glide path transmitter : 100 meter behind threshold , Middle Marker : 1.5 NM from threshold , Outer Marker : 10 NM from threshold.
- **d)** Localizer transmitter : 300 meter behind end of runway , Glide path transmitter : 300 meter behind threshold , Middle Marker : 1000 meter from threshold , Outer Marker : 4 NM from threshold.

Answer ;

d) Localizer transmitter : 300 meter behind end of runway , Glide path transmitter : 300 meter behind threshold , Middle Marker : 1000 meter from threshold , Outer Marker : 4 NM from threshold.

0473. Which combination of characteristics gives best screen picture in a primary search radar ?

- **a)** long pulse length and wide beam.
- **b)** short pulse length and wide beam.
- **c)** long pulse length and narrow beam.
- **d)** short pulse length and narrow beam.

Answer ;

d) short pulse length and narrow beam.

0474. Which component of an Area Navigation System displays the Cross Track Distance ?

- **a)** Attitude display.
- **b)** Radio Magnetic Indicator.
- **c)** Navigation display.
- **d)** DME Indicator.

Answer ;

c) Navigation display.

0475. Which component of the B737-400 Electronic Flight Instrument System generates the visual displays on the EADI and EHSI ?

- **a)** Navigation database.
- **b)** Symbol Generator.
- **c)** Flight Management Computer.
- **d)** Flight Control Computer.

Answer ;

b) Symbol Generator.

0476. Which facility associated with the ILS may be identified by a two-letter identification group ?

a) Locator.
b) Inner marker.
c) Glide path.
d) Outer marker.

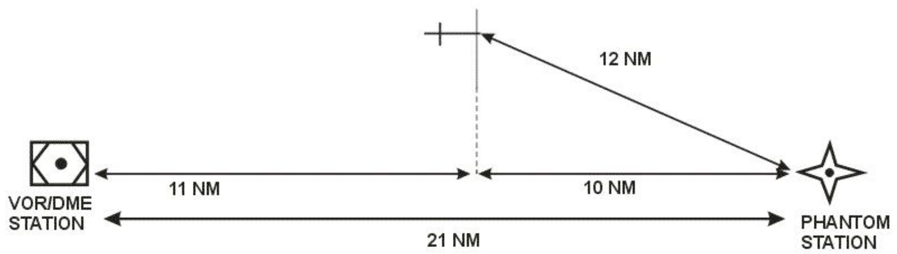
Answer ;
a) Locator.

0477. Which frequency band is used by VOR transmissions ?

a) UHF
b) SHF
c) VHF
d) HF

Answer ;
c) VHF

0478. (For this question use annex 062-12409A)



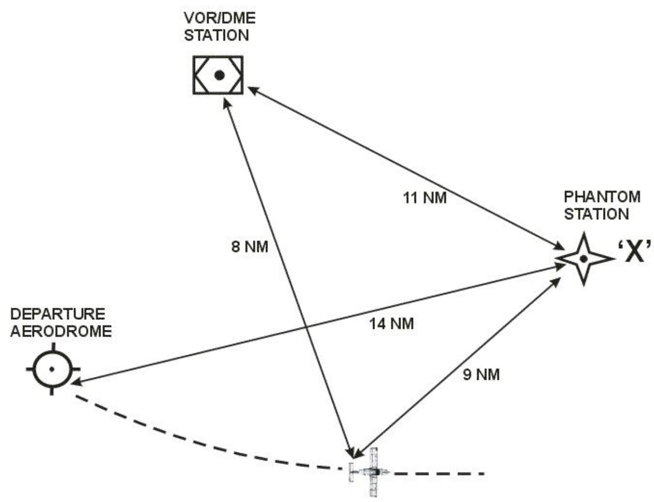
Annex 062-12409A

Which of the distances indicated will be shown on a basic VOR/DME-based Area Navigation Equipment when using a "Phantom Station" ?

a) 11 NM
b) 12 NM
c) 10 NM
d) 21 NM

Answer ;
b) 12 NM

0479. (For this question use annex 062-12410A)



Annex 062-12410A

Which of the distances indicated will be shown on a basic VOR/DME-based Area Navigation Equipment when using a "Phantom Station" at position "X" ?

a) 11 NM
b) 8 NM
c) 14 NM
d) 9 NM

Answer ;
d) 9 NM

0480. (Refer to Image 3)

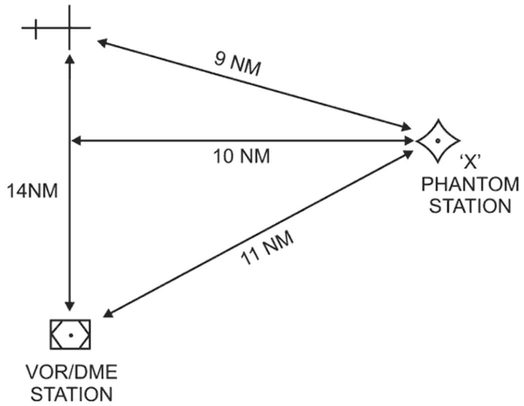


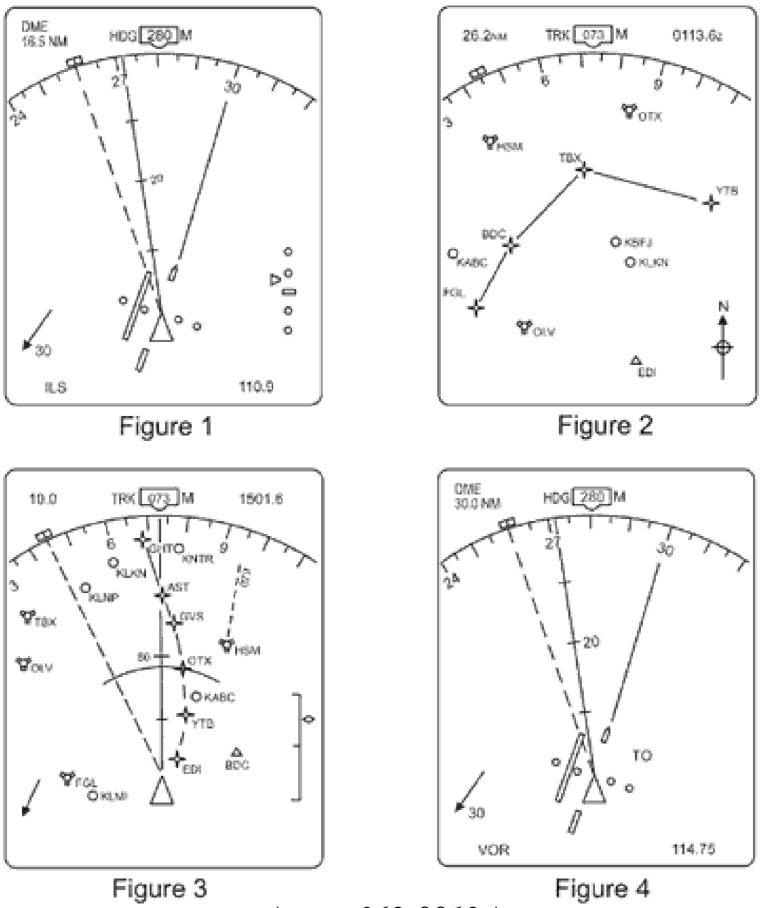
Image 3

Which of the distances indicated will be shown on a basic VOR/DME-based Area Navigation Equipment when using a "Phantom Station" at position "X" ?

a) 14 NM
b) 9 NM
c) 8 NM
d) 11 NM

Answer ;
b) 9 NM

0481. (For this question use annex 062-9912A)



Annex 062-9912A

Which of the figures depicts an Electronic Flight Instrument System (EFIS) display in Expanded (EXP) VOR/ILS mode with a VOR frequency selected ?

a) Figure 4
b) Figure 5
c) Figure 6
d) Figure 1

Answer ;
a) Figure 4

0482. (For this question use annex 062-9909A)

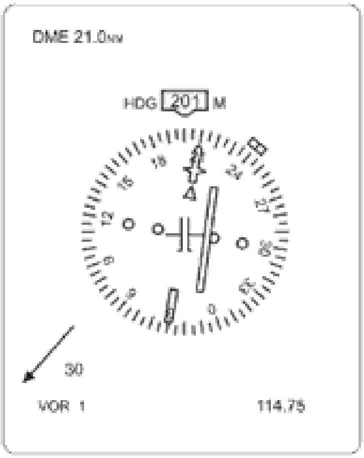


Figure 1

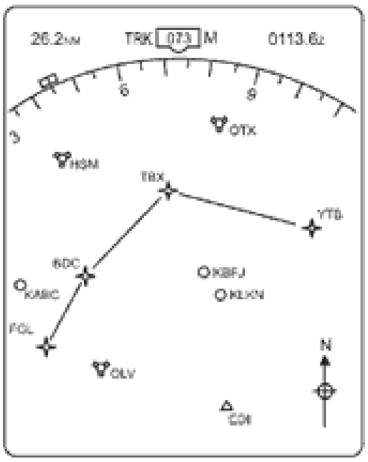


Figure 2

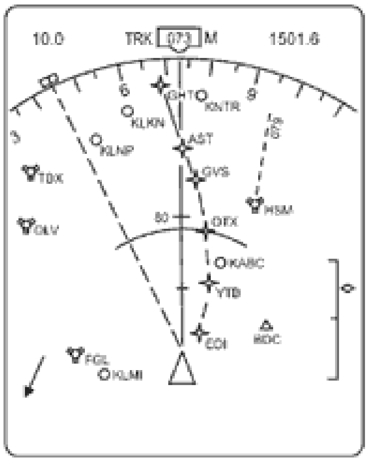


Figure 3

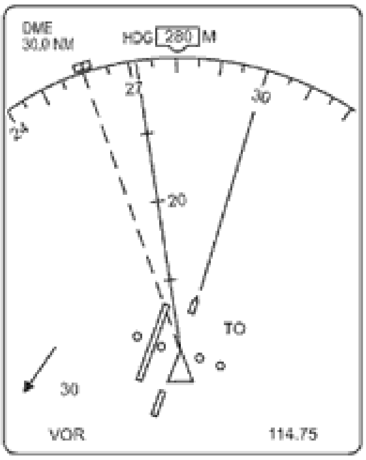


Figure 4

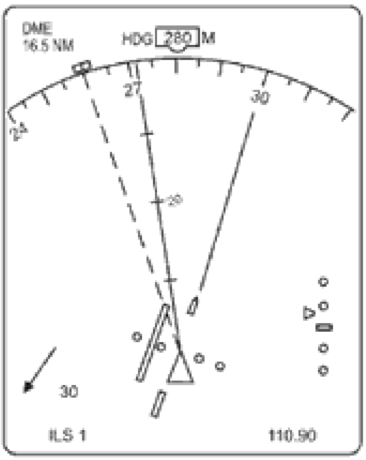


Figure 5

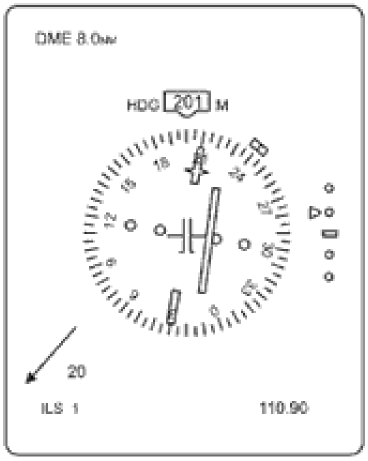


Figure 6

Annex 062-9909A

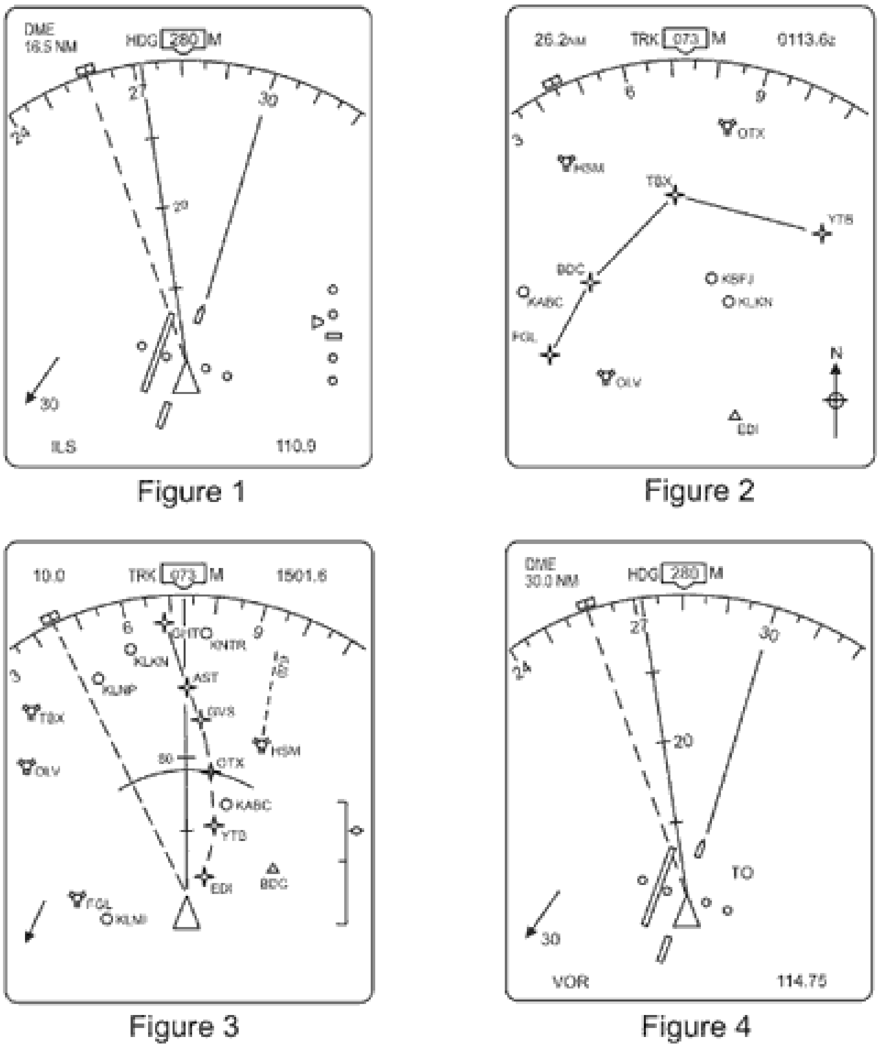
Which of the figures depicts an Electronic Flight Instrument System (EFIS) display in Expanded (EXP) VOR/ILS mode with a VOR frequency selected ?

- **a)** Figure 5
- **b)** Figure 6
- **c)** Figure 1
- **d)** Figure 4

Answer ;

d) Figure 4

0483. (For this question use annex 062-9907A)



Annex 062-9907A

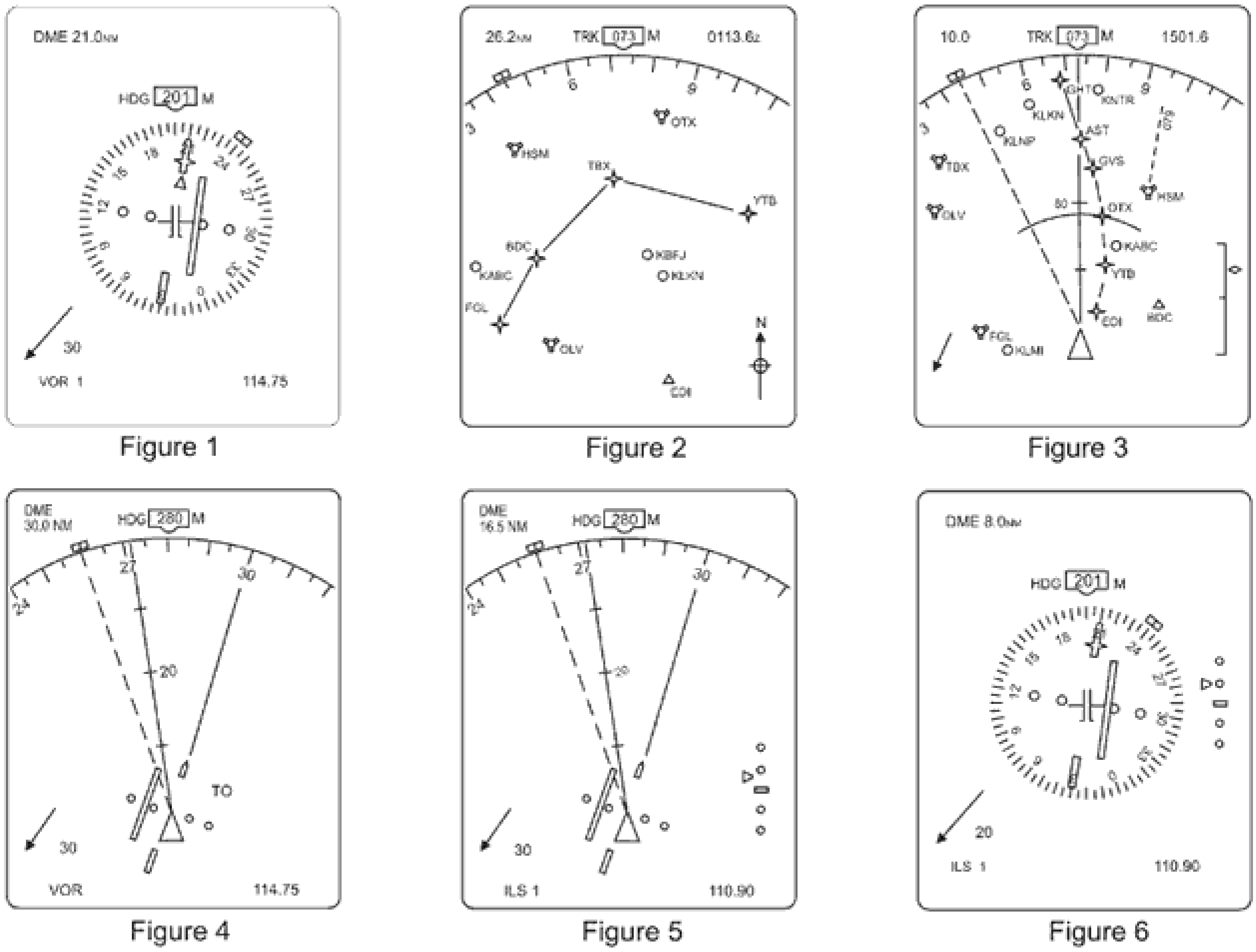
Which of the figures depicts an Electronic Flight Instrument System (EFIS) display in Expanded (EXP) VOR/ILS mode with an ILS frequency selected ?

a) Figure 4
b) Figure 3
c) Figure 2
d) Figure 1

Answer ;

d) Figure 1

0484. (For this question use annex 062-9909A)



Annex 062-9909A

Which of the figures depicts an Electronic Flight Instrument System (EFIS) display in Expanded (EXP) VOR/ILS mode with an ILS frequency selected ?

a) Figure 3
b) Figure 6
c) Figure 2
d) Figure 5

Answer ;
d) Figure 5

0485. (For this question use annex 062-9909A)

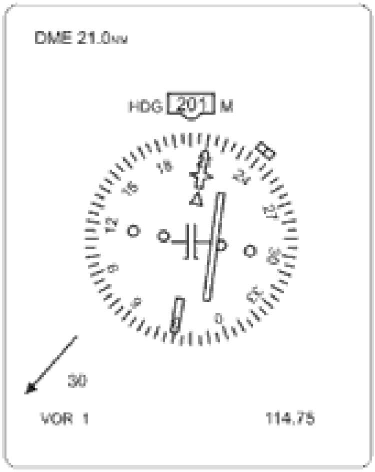


Figure 1

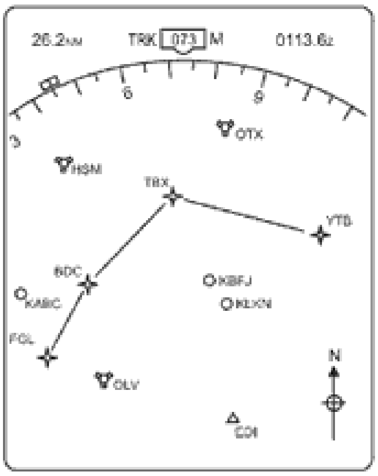


Figure 2

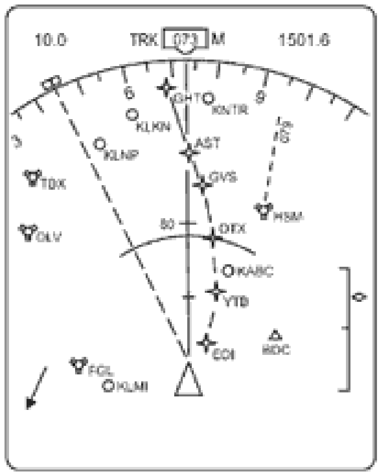


Figure 3

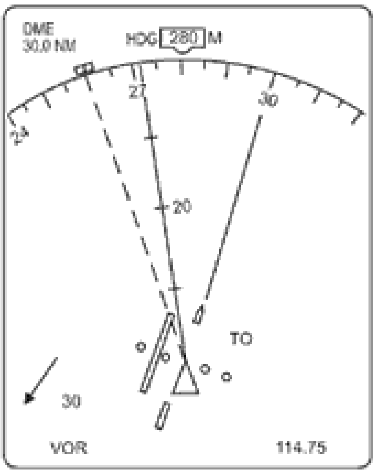


Figure 4

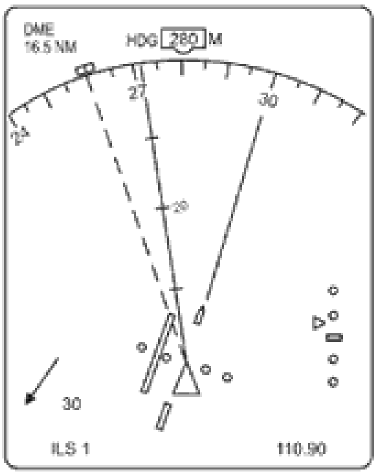


Figure 5

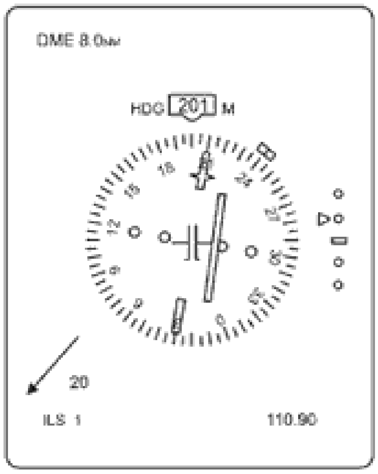


Figure 6

Annex 062-9909A

Which of the figures depicts an Electronic Flight Instrument System (EFIS) display in FULL VOR/ILS mode with an ILS frequency selected ?

a) Figure 6
b) Figure 3
c) Figure 2
d) Figure 5

Answer ;
a) Figure 6

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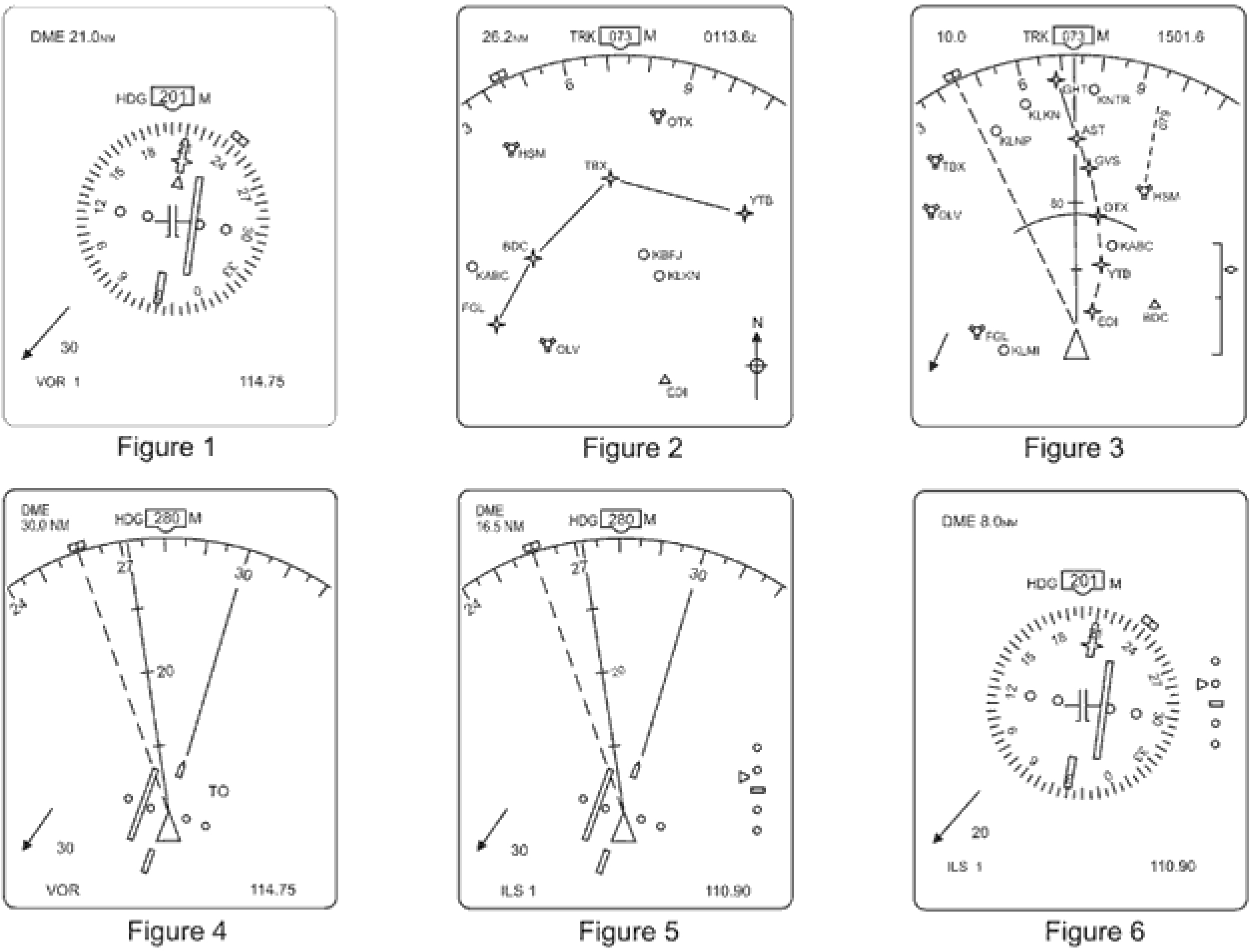
486. (For this question use annex 062-9909A)



Annex 062-9909A

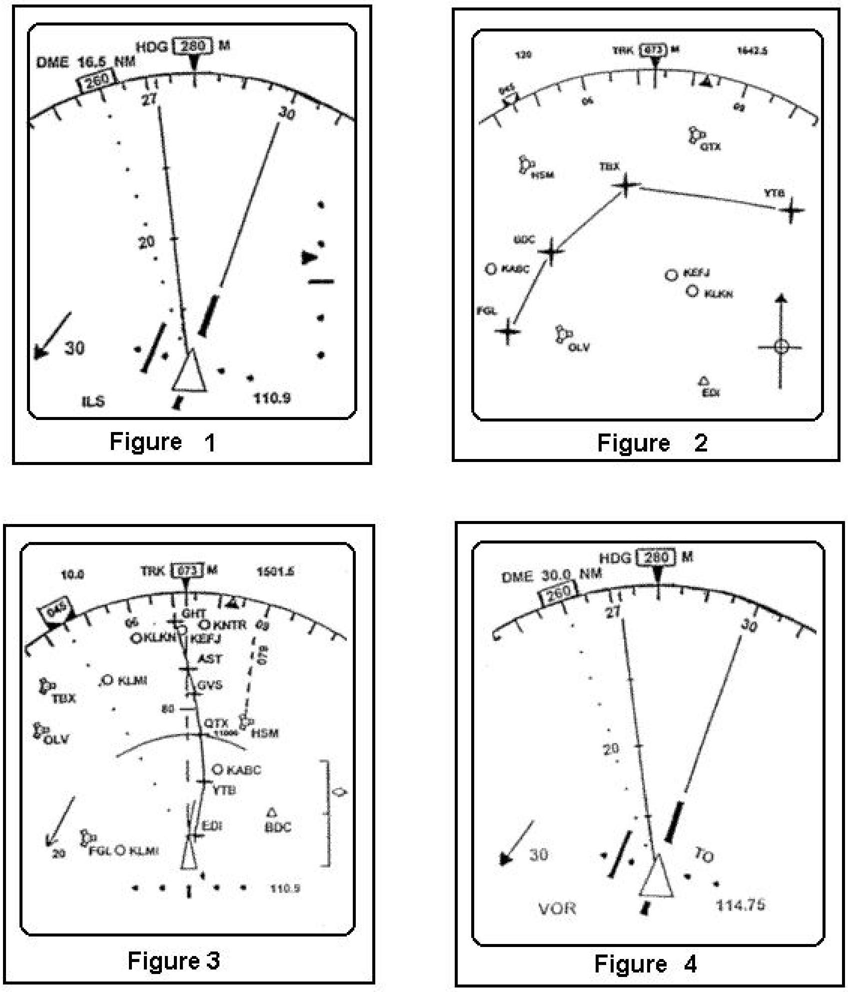
Which of the figures depicts an Electronic Flight Instrument System (EFIS) display in FULL VOR/ILS mode with an VOR frequency selected ?

- **a)** Figure 4
- **b)** Figure 5
- **c)** Figure 1
- **d)** Figure 6

Answer ;

c) Figure 1

0487. (For this question use annex 062-9905A)



Annex 062-9905A

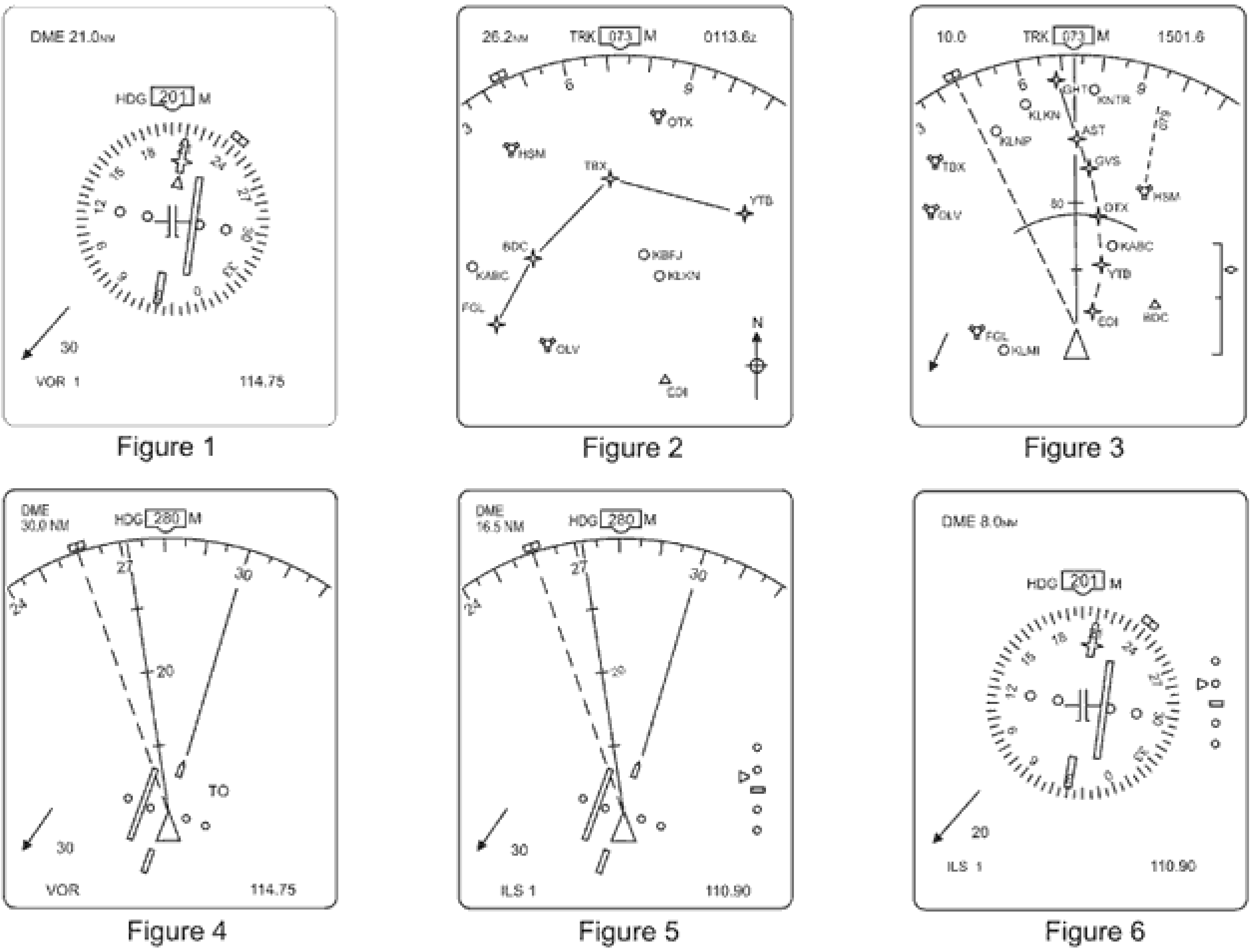
Which of the figures depicts an Electronic Flight Instrument System (EFIS) display in MAP mode ?

a) Figure 3
b) Figure 2
c) Figure 1
d) Figure 4

Answer ;

a) Figure 3

0488. (For this question use annex 062-9909A)



Annex 062-9909A

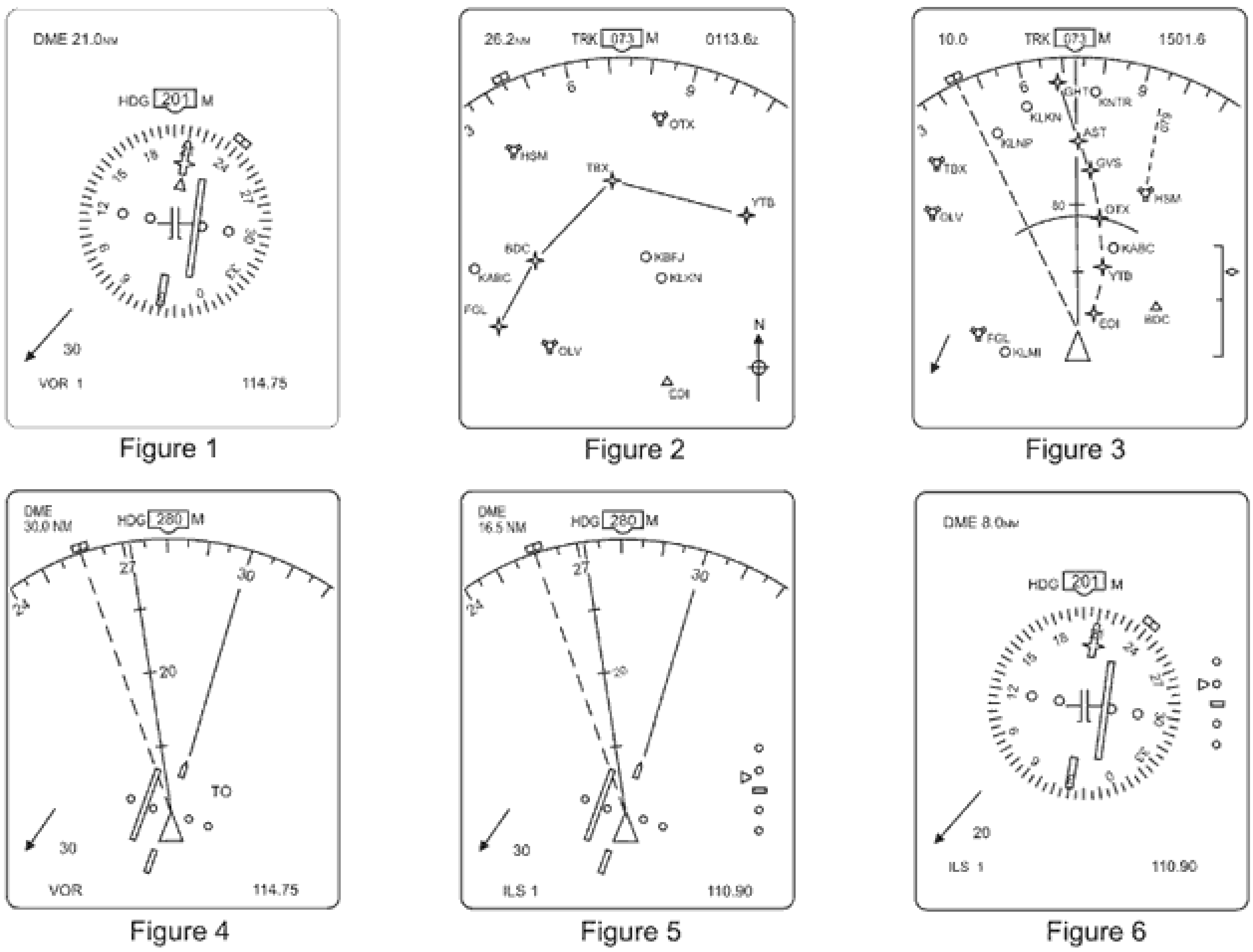
Which of the figures depicts an Electronic Flight Instrument System (EFIS) display in MAP mode ?

- **a)** Figure 4
- **b)** Figure 3
- **c)** Figure 5
- **d)** Figure 2

Answer ;

b) Figure 3

0489. (For this question use annex 062-9910A)



Annex 062-9910A

Which of the figures depicts an Electronic Flight Instrument System (EFIS) display in PLAN mode ?

a) Figure 6
b) Figure 4
c) Figure 3
d) Figure 2

Answer ;
d) Figure 2

0490. Which of the following affects VDF range ?

a) The height of the transmitter and of the receiver.
b) Strength of the pilot’s voice when transmitting.
c) Coastal refraction.
d) Sky wave propagation.

Answer ;
a) The height of the transmitter and of the receiver.

0491. Which of the following alternatives is correct regarding audio- and visual signals in cockpit when passing overhead a middle marker ?

a) Audio : 400 Hz , 2 dashes per second. Visual : Blue light flashes.
b) Audio : 75 MHz , 2 dashes per second. Visual : Blue light flashes.
c) Audio : 1300 Hz , alternating dots and dashes. Visual : Amber light flashes.
d) Audio : 3000 Hz , alternating dots and dashes. Visual : Amber light flashes.

Answer ;
c) Audio : 1300 Hz , alternating dots and dashes. Visual : Amber light flashes.

0492. Which of the following are stored in the navigation database of the Flight Management System (FMS) ?

1. waypoints.
2. details of radio navigation aids.
3. optimum altitudes.
4. company routes.
5. landing reference speeds.

a) 2 , 4 , 5.
b) 1 , 2 , 4.
c) 1 , 2 , 3.
d) 1 , 2 , 3 , 4.

Answer ;
b) 1 , 2 , 4.

0493. Which of the following cloud types is most readily detected by airborne weather radar ?

a) stratus
b) altostratus
c) cumulus
d) cirrocumulus

Answer ;
c) cumulus

0494. Which of the following cloud types is most readily detected by airborne weather radar when using the "weather beam" ?

a) altostratus
b) cirrocumulus
c) stratus
d) cumulus

Answer ;
d) cumulus

0495. Which of the following combinations is likely to result in the most accurate Area Navigation (RNAV) fixes ?

a) DME/DME
b) NDB/VOR
c) VOR/DME
d) VOR/VOR

Answer ;
a) DME/DME

0496. Which of the following combinations of satellite navigation systems provide the most accurate position fixes in air navigation ?

a) NNSS-Transit and GLONASS.
b) NAVSTAR/GPS and GLONASS.
c) NAVSTAR/GPS and NNSS-Transit.
d) GLONASS and COSPAS-SARSAT.

Answer ;
b) NAVSTAR/GPS and GLONASS.

0497. Which of the following coordinate systems is used by the NAVSTAR/GPS receiver to calculate position (latitude , longitude and altitude) ?

a) PZ 90
b) ED 87
c) WGS 84
d) ED 50

Answer ;
c) WGS 84

0498. Which of the following correctly describes the Instrument Landing System (ILS) localiser radiation pattern ?

a) Two overlapping lobes on the same UHF carrier frequency.
b) A pencil beam comprising a series of smaller beams each carrying a different modulation.
c) Two overlapping lobes on the same VHF carrier frequency.
d) Two overlapping lobes on different radio carrier frequencies but with the same modulation.

Answer ;
c) Two overlapping lobes on the same VHF carrier frequency.

0499. Which of the following correctly gives the principle of operation of the Loran C navigation system ?

a) Differential range by phase comparison.
b) Differential range by pulse technique.
c) Frequency shift between synchronised transmissions.
d) Phase comparison between synchronised transmissions.

Answer ;
b) Differential range by pulse technique.

0500. Which of the following data can be found in the navigation database ?

1. SIDs and STARs.
2. radio aids.
3. optimum flight level.
4. company routes.
5. cruise speeds.

a) Only 1 , 2 and 4.
b) Only 2 , 3 and 5.
c) Only 2 , 3 and 4.
d) 1 , 2 , 3 , 4 and 5.

Answer ;
a) Only 1 , 2 and 4.

0501. Which of the following data , in addition to the Pseudo Random Noise (PRN) code , forms part of the so called "Navigation Message" transmitted by NAVSTAR/GPS satellites ?

a) time , positions of the satellites.
b) data to correct receiver clock error , almanac data.
c) almanac data , satellite status information.
d) time , data to impair the accuracy of the position fix.

Answer ;
c) almanac data , satellite status information.

0502. Which of the following equipments uses primary radar principles ?

a) Distance Measuring Equipment (DME).
b) Airborne weather radar (AWR).
c) Secondary Surveillance Radar (SSR).
d) Global Positioning System (GPS).

Answer ;
b) Airborne weather radar (AWR).

0503. Which of the following equipments works on the interrogator/transponder principle ?

a) Global Positioning System (GPS).
b) Aerodrome Surface Movement Radar.
c) Airborne Weather Radar (AWR).
d) Secondary Surveillance Radar (SSR).

Answer ;
d) Secondary Surveillance Radar (SSR).

0504. Which of the following errors is associated with the use of VOR ?

a) Night effect.
b) Coastal refraction.
c) Quadrantal error.
d) Scalloping.

Answer ;
d) Scalloping.

0505. Which of the following frequencies is within the DME frequency band ?

a) 10 MHz
b) 100 MHz
c) 100 GHz
d) 1000 MHz

Answer ;
d) 1000 MHz

0506. Which of the following frequency-bands is used by the Loran C navigation system ?

- **a)** 978 - 1213 MHz
- **b)** 90 - 110 kHz
- **c)** 10.2 - 13.6 kHz
- **d)** 1750 - 1950 kHz

Answer ;

b) 90 - 110 kHz

0507. Which of the following geometric satellite constellations provides the most accurate NAVSTAR/GPS position fix ?

- **a)** 4 satellites with an azimuth of 90° from each other and an elevation of 45° above the horizon.
- **b)** 3 satellites with an azimuth of 120° from each other and an elevation of 45° above the horizon.
- **c)** 3 satellites with a low elevation above the horizon and an azimuth of 120° from each other together with a fourth directly overhead.
- **d)** 4 satellites with an azimuth of 90° from each other and a low elevation above the horizon.

Answer ;

c) 3 satellites with a low elevation above the horizon and an azimuth of 120° from each other together with a fourth directly overhead.

0508. Which of the following gives the best information about the progress of a flight between 2 en-route waypoints from a RNAV equipment ?

- **a)** Elapsed time on route
- **b)** ATA
- **c)** ETD
- **d)** ETO

Answer ;

d) ETO

0509. Which of the following , if any , is a prerequisite if a receiver of a NAVSTAR/GPS satellite navigation system is to be used in combination with a multi sensor system ?

- **a)** The prescribed IFR-equipment must be in working correctly and the navigation information continuously displayed.
- **b)** The RAIM-function of the GPS receiver must be able to monitor all prescribed navigation systems.
- **c)** The prescribed IFR-equipment must be installed and operational.
- **d)** Multi-sensor systems are not certificated for flights under IFR conditions.

Answer ;

c) The prescribed IFR-equipment must be installed and operational.

0510. Which of the following is a complete list of airborne weather radar antenna stabilisation axes ?

- **a)** pitch and yaw.
- **b)** roll and pitch.
- **c)** roll , pitch and yaw.
- **d)** roll and yaw.

Answer ;

b) roll and pitch.

0511. Which of the following is an advantage of Ground/DF (VDF) let-down ?

- **a)** It does not require any special equipment to be fitted to the aircraft.
- **b)** It only requires a VHF radio to be fitted to the aircraft.
- **c)** It is pilot interpreted and does not require the assistance of ATC.
- **d)** It does not require any special equipment , apart from a VHF radio , to be installed in the aircraft or on the ground.

Answer ;

b) It only requires a VHF radio to be fitted to the aircraft.

0512. Which of the following is an ILS localiser frequency ?

- **a)** 109.15 MHz
- **b)** 108.25 MHz
- **c)** 110.20 MHz
- **d)** 112.10 MHz

Answer ;

a) 109.15 MHz

0513. Which of the following is correct regarding false beams on a glide path ?

a) False beams will only be found below the correct glide path.
b) False beams are only present when flying a back-beam ILS approach.
c) False beams will only be found more than 10 degrees to the left or to the right of the localiser centreline.
d) False beams will only be found above the correct glide path.

Answer ;

d) False beams will only be found above the correct glide path.

0514. Which of the following is correct regarding the range of an NDB ?

a) The range of an NDB will most likely increase at day time compared to night time.
b) The transmitter power of the NDB station has no affect on the range.
c) Aircraft height is not limiting for the reception of signals from the NDB.
d) The range is limited to the line of sight.

Answer ;

c) Aircraft height is not limiting for the reception of signals from the NDB.

0515. Which of the following is likely to have the greatest effect on ADF accuracy ?

a) Frequency drift at the ground station.
b) Interference from other NDBs , particularly at night.
c) Interference from other NDBs , particularly during the day.
d) Mutual interference between aircraft aerials.

Answer ;

b) Interference from other NDBs , particularly at night.

0516. Which of the following is one of the functions of the Computer in a basic RNAV system ?

a) It transfers the information given by a VOR/DME station into tracking and distance indications to any chosen Phantom Station/waypoint.
b) It checks the ground station accuracy using a built-in test programme.
c) It calculates cross track information for NDB approaches.
d) It automatically selects the two strongest transmitters for the Area-Nav-Mode and continues working by memory in case one of the two necessary station goes off the air.

Answer ;

a) It transfers the information given by a VOR/DME station into tracking and distance indications to any chosen Phantom Station/waypoint.

0517. Which of the following is the datum for altitude information when conducting flights under IFR conditions on airways using the NAVSTAR/GPS satellite navigation system ?

a) The average of GPS altitude and barometric altitude.
b) GPS altitude.
c) Barometric altitude.
d) GPS altitude if 4 or more satellites are received otherwise barometric altitude.

Answer ;

c) Barometric altitude.

0518. Which of the following is the ICAO allocated frequency band for ADF receivers ?

a) 300 - 3000 kHz.
b) 200 - 2000 kHz.
c) 255 - 455 kHz.
d) 190 - 1750 kHz.

Answer ;

d) 190 - 1750 kHz.

0519. Which of the following list use the VHF band ;

1. Locator.
2. Localiser.
3. Outer Marker.
4. Glide path.

The combination that regroups all the corrects statements is ;

a) 1 , 2 , 3 and 4.
b) 2 and 4.
c) 1 and 3.
d) 2 and 3.

Answer ;

d) 2 and 3.

0520. Which of the following lists all the parameters that can be determined by a GPS receiver tracking signals from 4 different satellites ?

a) Latitude , longitude and time.
b) Latitude , longitude , altitude and time.
c) Latitude , longitude and altitude.
d) Latitude and longitude.

Answer ;
b) Latitude , longitude , altitude and time.

0521. Which of the following lists all the stages of flight when is it possible to change the route in the active flight plan on an FMS equipped aircraft ?

a) Only once the aircraft is airborne.
b) At any time before take-off and throughout the flight.
c) Only before the flight plan is activated.
d) Only before take-off.

Answer ;
b) At any time before take-off and throughout the flight.

0522. Which of the following lists are all errors that affect the accuracy and reliability of the Satellite-Assisted Navigation system (GNSS/GPS) ?

a) Satellite mutual interference , satellite ephemeris , atmospheric propagation.
b) Satellite clock , satellite ephemeris , atmospheric propagation.
c) Satellite mutual interference , frequency drift , satellite to ground time lag.
d) Satellite to ground time lag , atmospheric propagation , satellite clock.

Answer ;
b) Satellite clock , satellite ephemeris , atmospheric propagation.

0523. Which of the following lists information required to input a waypoint or “Phantom Station” into a basic VOR/DME-based Area Navigation System ?

a) Radials from a minimum of two VORs to the waypoint or “Phantom Station”.
b) Magnetic track and distance to a VOR/DME from the waypoint or “Phantom Station”.
c) Radial and distance from a VOR/DME to the waypoint or “Phantom Station”.
d) Magnetic track and distance from the aircraft to the waypoint or “Phantom Station”.

Answer ;
c) Radial and distance from a VOR/DME to the waypoint or “Phantom Station”.

0524. Which of the following lists the phenomena least likely to be detected by radar ?

a) Wet snow and turbulence in cloud that has precipitation.
b) Turbulence in cloud that has precipitation.
c) Precipitation.
d) Clear air turbulence.

Answer ;
d) Clear air turbulence.

0525. Which of the following Nav Aids will provide an RNAV system with position ?

a) VOR/DME
b) ADF
c) NDB
d) VDF

Answer ;
a) VOR/DME

0526. Which of the following NAVSTAR/GPS satellite navigation system codes can be processed by “unauthorised” civil aviation receivers ?

a) P and Y
b) P
c) C/A and P
d) C/A

Answer ;
d) C/A

0527. Which of the following procedures must be adopted if , on a flight under IFR conditions using a NAVSTAR/GPS satellite navigation system receiver , the position fix obtained from the GPS receiver differs from the position of conventional navigation systems by an unacceptable amount ?

- **a)** The pilot must determine the reason for the deviation and correct the error or switch off the faulty system.
- **b)** It must be continued under VFR conditions.
- **c)** It may be continued using NAVSTAR/GPS , prior to the next flight all systems must be checked.
- **d)** It may be continued using conventional navigation systems.

Answer ;

d) It may be continued using conventional navigation systems.

0528. Which of the following radar equipments operate by means of the pulse technique ?

1. Aerodrome Surface Movement Radar.
2. Airborne Weather Radar.
3. Secondary Surveillance Radar (SSR).
4. Aerodrome Surveillance (approach) Radar.

- **a)** 1 , 2 , 3 and 4.
- **b)** 2 , 3 and 4 only.
- **c)** 2 and 4 only.
- **d)** 1 , 2 and 4 only.

Answer ;

a) 1 , 2 , 3 and 4.

0529. Which of the following satellite navigation systems has Full Operational Capability (FOC) and is approved for specified flights under IFR conditions in Europe ?

- **a)** GLONASS
- **b)** NNSS-Transit
- **c)** NAVSTAR/GPS
- **d)** COSPAS-SARSAT

Answer ;

c) NAVSTAR/GPS

0530. Which of the following statements about the accuracy that can be obtained with the differential technique (D-GPS) of the satellite navigation system NAVSTAR/GPS is correct ?

- **a)** Only D-GPS allows position fixes accurate enough for “Non Precision Approaches”.
- **b)** The increase in accuracy of position fixes is independent of the receiver position in relation to a D-GPS ground station.
- **c)** A D-GPS receiver can detect and correct for SA providing a more accurate position fix.
- **d)** The nearer a receiver is situated to a D-GPS ground station , the more accurate the position fix.

Answer ;

d) The nearer a receiver is situated to a D-GPS ground station , the more accurate the position fix.

0531. Which of the following statements about the accuracy that can be obtained with the LAAS (local area augmentation system) of the satellite navigation system NAVSTAR/GPS is correct ?

- **a)** A LAAS cannot correct for satellite timing and orbital position error.
- **b)** A LAAS corrects the position of the aircraft by relaying the information via a geo-stationary satellite.
- **c)** The increase in accuracy of position fixes is independent of the aircraft position in relation to the LAAS ground reference station.
- **d)** The closer the receiver is to a LAAS ground reference station , the more accurate is the aircraft position fix.

Answer ;

d) The closer the receiver is to a LAAS ground reference station , the more accurate is the aircraft position fix.

0532. Which of the following statements about the scalloping (path deflection) of VOR-radials , in relation to the accuracy of navigation using a VOR/DME RNAV-system , is correct ?

- **a)** Scalloping has a positive effect on the accuracy of navigation.
- **b)** Scalloping has a negative effect on the accuracy of navigation.
- **c)** Scalloping has no effect on the accuracy of navigation because it only results in the movement of the needle of the Course Deviation Indicator.
- **d)** Scalloping has no effect on the accuracy of navigation because this accuracy is independent of VOR or DME measurements.

Answer ;

b) Scalloping has a negative effect on the accuracy of navigation.

0533. Which of the following statements about the "visibility" of NAVSTAR/GPS satellites is correct ?

a) It is the same throughout the globe.
b) It is greatest at the poles.
c) It varies , depending on the time and observer's location.
d) It is greatest at the equator.

Answer ;
c) It varies , depending on the time and observer's location.

0534. Which of the following statements concerning LORAN-C is correct ?

a) It is a navigation system based on simultaneous ranges being received from a minimum of four ground stations.
b) It is a hyperbolic navigation system that works on the principle of differential range by pulse technique.
c) It is a navigation system based on secondary radar principles , position lines are obtained in sequence from up to eight ground stations.
d) It is a hyperbolic navigation system that works on the principle of range measurement by phase comparison.

Answer ;
b) It is a hyperbolic navigation system that works on the principle of differential range by pulse technique.

0535. Which of the following statements concerning the L1 and L2 NAVSTAR/GPS transmission frequencies and codes is correct ?

a) The lower frequency is used to transmit both the C/A and P codes.
b) The higher frequency is used to transmit both the C/A and P codes.
c) The higher frequency is only used to transmit the P code.
d) C/A and P codes are transmitted at different times on both frequencies.

Answer ;
b) The higher frequency is used to transmit both the C/A and P codes.

0536. Which of the following statements concerning the variable , or directional , signal of a conventional VOR is correct ?

a) The receiver adds 30 Hz to the variable signal before combining it with the reference signal.
b) The transmitter varies the amplitude of the variable signal by 30 Hz each time it rotates.
c) The transmitter changes the frequency of the variable signal by 30 Hz either side of the allocated frequency each time it rotates.
d) The rotation of the variable signal at a rate of 30 times per second gives it the characteristics of a 30 Hz amplitude modulation.

Answer ;
d) The rotation of the variable signal at a rate of 30 times per second gives it the characteristics of a 30 Hz amplitude modulation.

0537. Which of the following statements regarding B-RNAV is correct ?

a) B-RNAV is only applicable when flying in TMA's.
b) In case of B-RNAV , RNP1 is required.
c) For 95% of the flight time , the track keeping accuracy must not exceed 5 NM.
d) B-RNAV can be used up to FL95. Above FL95 P-RNAV is required.

Answer ;
c) For 95% of the flight time , the track keeping accuracy must not exceed 5 NM.

0538. Which of the following will give the most accurate calculation of aircraft ground speed ?

a) An ADF sited on the flight route.
b) A DME station sited across the flight route.
c) A DME station sited on the flight route.
d) A VOR station sited on the flight route.

Answer ;
c) A DME station sited on the flight route.

0539. Which of these markers has the highest audible frequency ?

a) Airways
b) Middle
c) Outer
d) Inner

Answer ;
d) Inner

0540. Which one is the most correct statement regarding the range of the DME system ?

- **a)** Range within “line of sight” , and maximum 200 NM.
- **b)** Has unlimited range due to ground wave propagation.
- **c)** Operates on the principle of phase comparison.
- **d)** Operates on VHF.

Answer ;

a) Range within “line of sight” , and maximum 200 NM.

0541. Which one of the following correctly lists the major ground based components of a Microwave Landing System (MLS) ?

- **a)** Separate azimuth and elevation transmitters , outer and middle marker beacons.
- **b)** Combined azimuth and elevation transmitter , DME facility.
- **c)** Separate azimuth and elevation transmitters , DME facility.
- **d)** Combined azimuth and elevation transmitter , outer and inner marker beacons.

Answer ;

c) Separate azimuth and elevation transmitters , DME facility.

0542. Which one of the following disturbances is most likely to cause the greatest inaccuracy in ADF bearings ?

- **a)** Coastal effect.
- **b)** Local thunderstorm activity.
- **c)** Quadrantal error.
- **d)** Precipitation interference.

Answer ;

b) Local thunderstorm activity.

0543. Which one of the following errors can be compensated for by a NAVSTAR/GPS receiver comparing L1 and L2 frequencies ?

- **a)** Receiver noise
- **b)** Multipath
- **c)** Tropospheric
- **d)** Ionospheric

Answer ;

d) Ionospheric

0544. Which one of the following inputs to an Area Navigation System (R-NAV) comes from an external , not on-board system ?

- **a)** Pressure altitude.
- **b)** VOR/DME radial/distance.
- **c)** Magnetic heading.
- **d)** Inertial Navigation System (INS) position.

Answer ;

b) VOR/DME radial/distance.

0545. Which one of the following is an advantage of a Microwave Landing System (MLS) compared with an Instrument Landing System (ILS) ?

- **a)** It is insensitive to geographical site and can be installed at sites where it is not possible to use an ILS.
- **b)** It does not require a separate azimuth (localiser) and elevation (azimuth) transmitter.
- **c)** There is no restriction on the number of ground installations that can be operated because there is an unlimited number of frequency channels available.
- **d)** The installation does not require to have a separate method (marker beacons or DME) to determine range.

Answer ;

a) It is insensitive to geographical site and can be installed at sites where it is not possible to use an ILS.

0546. Which one of the following is an advantage of a multi-sensor system using inputs from a global navigation satellite system (GNSS) and an inertial navigational system (INS) ?

- **a)** The GNSS can be used to update a drifting INS.
- **b)** The only advantage of coupling both systems is double redundancy.
- **c)** The activation of “Selective Availability” can be recognised by the INS.
- **d)** The average position calculated from data provided by both systems increases overall accuracy.

Answer ;

a) The GNSS can be used to update a drifting INS.

0547. Which one of the following is an advantages of a multi-sensor system using inputs from a global navigation satellite system (GNSS) and an inertial reference system (IRS) ?

- **a)** The activation of "Selective Availability" can be recognised by he INS.
- **b)** The GNSS can be used to update the FMS position from a drifting IRS.
- **c)** The GNSS is never used as an input in a multi-sensor system.
- **d)** The FMS always uses average position calculated from the GNSS and the IRS.

Answer ;

b) The GNSS can be used to update the FMS position from a drifting IRS.

0548. Which one of the following is an advantage of a secondary radar system when compared to a primary radar system ?

- **a)** The required power of transmission from the ground equipment is reduced.
- **b)** Is not limited to line of sight.
- **c)** The relatively small ground antenna transmits no side lobes , thus eliminating the danger of false replies from the airborne transponder.
- **d)** Possibility of obtaining speed information for aircraft within range.

Answer ;

a) The required power of transmission from the ground equipment is reduced.

0549. Which one of the following lists information given by a basic VOR/DME-based Area Navigation System ?

- **a)** True airspeed , drift angle.
- **b)** Wind velocity.
- **c)** Crosstrack distance , alongtrack distance , angular course deviation.
- **d)** Aircraft position in latitude and longitude.

Answer ;

c) Crosstrack distance , alongtrack distance , angular course deviation.

0550. Which one of the following lists information given by a basic VOR/DME-based Area Navigation System when tracking inbound to a phantom waypoint ?

- **a)** Aircraft position in latitude and longitude.
- **b)** Crosstrack distance , alongtrack distance.
- **c)** True airspeed , drift angle.
- **d)** Wind velocity.

Answer ;

b) Crosstrack distance , alongtrack distance.

0551. Which of the following lists information required to input a waypoint or "Phantom Station" into a basic VOR/DME-based Area Navigation System ?

- **a)** Radials from a minimum of two VORs to the waypointor "Phantom Station".
- **b)** Magnetic track and distance from the aircraft to the waypoint or "Phantom Station".
- **c)** Radial and distance from a VOR/DME to the waypoint or "Phantom Station".
- **d)** Magnetic track and distance to a VOR/DME rom the waypoint or "Phantom Station".

Answer ;

c) Radial and distance from a VOR/DME to the waypoint or "Phantom Station".

0552. Which one of the following errors can be compensated for by a NAVSTAR/GPS receiver comparing L1 and L2 frequencies ?

- **a)** Ionospheric
- **b)** Multipath
- **c)** Tropospheric
- **d)** Receiver noise

Answer ;

a) Ionospheric

0553. Which one of the following methods is used by a Microwave Landing System (MLS) to indicate distance from the runway threshold ?

- **a)** Timing the interval between the reception of sequential secondary radar pulses from the MLS station to the aircraft.
- **b)** Measurement of the frequency shift between the MLS azimuth and elevation transmissions.
- **c)** Timing the interval between the transmission and reception of primary radar pulses from the aircraft to MLS station.
- **d)** A precision facility DME.

Answer ;

d) A precision facility DME.

0554. Which one of the following sensors/systems is self-contained ?

a) VOR/DME
b) Inertial Navigation System
c) GPS
d) Basic RNAV system

Answer ;
b) Inertial Navigation System

0555. Which one of the statements below is correct regarding the DME ?

a) Two lines of position obtained from two different DME's give an unambiguous fix.
b) The DME operating frequencies are in the UHF frequency band.
c) The indicated distance is the ground distance measured from the aircraft's projected position on the ground to the DME ground installation.
d) The DME ground station is always co-located with a VOR station.

Answer ;
b) The DME operating frequencies are in the UHF frequency band.

0556. Which range facility associated with the ILS may be identified by a two-letter identification group ?

a) Glide path.
b) Inner marker.
c) Outer marker.
d) Locator.

Answer ;
d) Locator.

0557. Which statement about RNAV-routes is correct ?

a) RNAV-routes are only used in terminal areas in order to make more efficient use of the approach and landing facilities of an airport.
b) A method of navigation which permits aircraft operation on any desired flight path.
c) All waypoints of RNAV-routes are called phantom stations.
d) In general RNAV-routes coincide with standard routes in order to make optimum use of the available VOR/DME-stations.

Answer ;
b) A method of navigation which permits aircraft operation on any desired flight path.

0558. Which statement about the errors and effects on NDB radio signals is correct ?

a) Lightning during atmospheric disturbances may cause a reduction of the signal strength that may result in only slight bearing errors.
b) Shore line effects may cause a huge bearing error due to reflection of the radio signal onto steep coasts.
c) Night effect is a result of interference of the surface wave and the space wave causing a reduction in range.
d) The mountain effect is caused by reflections onto steep slopes of mountainous terrain which may cause big errors in the bearing.

Answer ;
d) The mountain effect is caused by reflections onto steep slopes of mountainous terrain which may cause big errors in the bearing.

0559. Which statement about the interrogation by the DME-interrogator is correct ?

a) The interrogation does not start before pulse-pairs of the tuned DME-station are received.
b) The interrogation starts when the interrogator has been warmed up sufficiently , whether pulse-pairs are received or not.
c) The interrogation can only take place if the Echo Protection Circuit has been locked.
d) The interrogation starts directly after the correct DME-frequency has been selected on the frequency-selecting-panel by the pilot.

Answer ;
a) The interrogation does not start before pulse-pairs of the tuned DME-station are received.

0560. Which statement is correct for homing towards an NDB in an area with constant wind and constant magnetic variation ?

a) The Relative bearing of the NDB should be equal (in magnitude and sign) to the applied Wind Correction Angle.
b) The Relative bearing of the NDB should be kept 000º.
c) The Relative bearing of the NDB should be equal (in magnitude and sign) to the experienced Drift Angle.
d) The Relative bearing of the NDB should be equal to the QDM.

Answer ;
b) The Relative bearing of the NDB should be kept 000º.

0561. Which statement is correct for tracking towards an NDB in an area with constant wind and constant magnetic variation ?

- **a)** The Relative bearing of the NDB should be equal to the QDM.
- **b)** The Relative bearing of the NDB should be equal (in magnitude and sign) to the experienced Drift Angle.
- **c)** The Relative bearing of the NDB should be equal (in magnitude and sign) to the applied Wind Correction Angle.
- **d)** The Relative bearing of the NDB should be kept 000º.

Answer ;

b) The Relative bearing of the NDB should be equal (in magnitude and sign) to the experienced Drift Angle.

0562. Which statement is correct with respect to selections on the transponder control panel ?

- **a)** With ALT selected a reply in the modes A , C and S can be transmitted.
- **b)** The correct ICAO 24 bits aircraft address code must be inserted on the control panel.
- **c)** With the selector knobs the four digits of the squawk can be selected using the decimal symbols 0 through 9.
- **d)** With ON selected a reply in the modes A , C and S can be transmitted.

Answer ;

a) With ALT selected a reply in the modes A , C and S can be transmitted.

0563. Which statement is correct with respect to the different types of VOR ?

- **a)** A CVOR is primarily used for instrument approaches.
- **b)** A TVOR has a limited range.
- **c)** A DVOR is less accurate than a CVOR.
- **d)** A VOT is located along an airway with the purpose to provide an in-flight check of the airborne equipment.

Answer ;

b) A TVOR has a limited range.

0564. Which statement is correct with respect to the range of an NDB ?

- **a)** With propagation over sea the range will be greater than the range with propagation over land.
- **b)** In order to double the range of an NDB , the transmission power should be increased with a factor 16.
- **c)** The range depends on the altitude of the aircraft.
- **d)** During the night the range of an NDB will decrease due to the interference of the direct- and earth reflected wave.

Answer ;

a) With propagation over sea the range will be greater than the range with propagation over land.

0565. Which statement is true about the use of the Doppler effect in a Doppler VOR ?

- **a)** The Doppler effect is used to create a signal which is received by the aircraft's VOR-receiver as an amplitude modulated signal.
- **b)** By using the Doppler effect it is also possible to determine the aircraft's approach speed to the VOR.
- **c)** By using the Doppler effect it is possible to determine the range of the aircraft from the VOR station more accurately.
- **d)** The Doppler effect is used to create a signal which is received by the aircraft's VOR-receiver as a frequency modulated signal.

Answer ;

d) The Doppler effect is used to create a signal which is received by the aircraft's VOR-receiver as a frequency modulated signal.

0566. Which statement is true regarding a compass when directly overhead the north magnetic pole ?

- **a)** the compass tip will point directly down.
- **b)** the horizontal component of the earth's magnetic field is horizontal.
- **c)** the compass tail points down.
- **d)** the magnetic variation is 90º.

Answer ;

a) the compass tip will point directly down.

0567. Which statement relating to the stabilization of airborne weather radar antenna is true ?

- **a)** They are stabilized with respect to the pitch , roll , and yaw axis.
- **b)** The pilot can choose the axes of stabilization with the system's stabilization selector switch.
- **c)** They are stabilized with respect to the pitch and roll axis but not with respect to the yaw axis.
- **d)** They are stabilized with respect to the yaw axis , but not with respect to the pitch and roll axis.

Answer ;

c) They are stabilized with respect to the pitch and roll axis but not with respect to the yaw axis.

0568. Which statement with respect to interrogation modes is correct ?

a) Mode S Broadcast will transmit information to all mode S transponders.
b) Mode S Selective will trigger replies of all mode S transponders. Mode A/C transponders are suppressed.
c) Mode S Only all-call will trigger transponder replies of all mode A/C/S transponders.
d) Intermode A/C/S all-call and Intermode A/C only all-call , differ in the pulse spacing used between the P1 and P3 pulse.

Answer ;
a) Mode S Broadcast will transmit information to all mode S transponders.

0569. Why is a secondary radar display screen free of storm clutter ?

a) A moving target indicator facility suppresses the display of static or near static returns.
b) The frequencies employed are too low to give returns from moisture sources.
c) The principle of "echo" return is not used in secondary radar.
d) The frequencies employed are too high to give returns from moisture sources.

Answer ;
c) The principle of "echo" return is not used in secondary radar.

0570. With mode-S each aircraft can be uniquely interrogated and identified by using the ;

a) 24 bits address code , giving more than 16 million possible codes.
b) Special Position Identification (SPI) pulse.
c) long (30.25 msec) P3 pulse.
d) twelve-pulse train giving 212 possible codes.

Answer ;
a) 24 bits address code , giving more than 16 million possible codes.

0571. With regard to the range of NDB's and the accuracy of the bearings they provide can be stated that in general at night ;

a) the range and the accuracy both increase.
b) the range decreases and the accuracy increases.
c) the range increases and the accuracy decreases.
d) the range and the acuracy both decrease.

Answer ;
c) the range increases and the accuracy decreases.

0572. With respect to the principle of distance measurement using DME can be stated that ;

1. the interrogation signal is transmitted on the same frequency as the reply signal.
2. the DME station always has a transponder delay of 50 milliseconds.
3. the time between the pulse pairs of the interrogation signal is at random.
4. In the search mode more pulse pairs per second are transmitted than in the tracking mode.

Which of the above given statements are correct ?

a) 2 , 3 and 4.
b) 1 , 2 , 3 and 4.
c) 3 and 4.
d) 1 and 2.

Answer ;
c) 3 and 4.

0573. You are flying along an airway which is 10 NM wide (5 NM either side of the centreline). The distance to the VOR/DME you are using is 100 NM. If you are on the airway boundary , how many dots deviation will the VOR needle show if one dot represents 2 degrees ?

a) 4.5
b) 6.0
c) 1.5
d) 3.0

Track Error Angle (TEA) = (Distance Off-Track x 60) : Distance Along Track
formülünü kullanarak ;
TEA = (5 x 60) : 100 = 300 : 100 = 3° ,
1 dot 2°yi temsil ettiğine göre , 3° 1.5 için dot gereklidir.

Answer ;
c) 1.5

0574. You are on a compass heading of 090° on the 255 radial from a VOR. You set the course 190° on your OBS. The deviation bar will show ;

a) Full scale deflection right with a "to" indication.
b) Full scale deflection left with a "from" indication.
c) Full scale deflection left with a "to" indication.
d) Full scale deflection right with a "from" indication.

Answer ;
b) Full scale deflection left with a "from" indication.

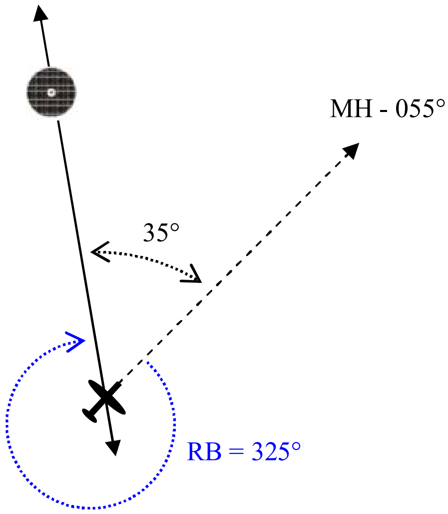
0575. You are on a magnetic heading of 055° and your ADF indicates a relative bearing of 325°. The QDM is ;

a) 020°
b) 055°
c) 235°
d) 200°

Answer ;

a) 020°

Çözüm ;



QDM = RB + MH
QDM = 325° + 055°
QDM = 380° = 380° – 360° = 020°

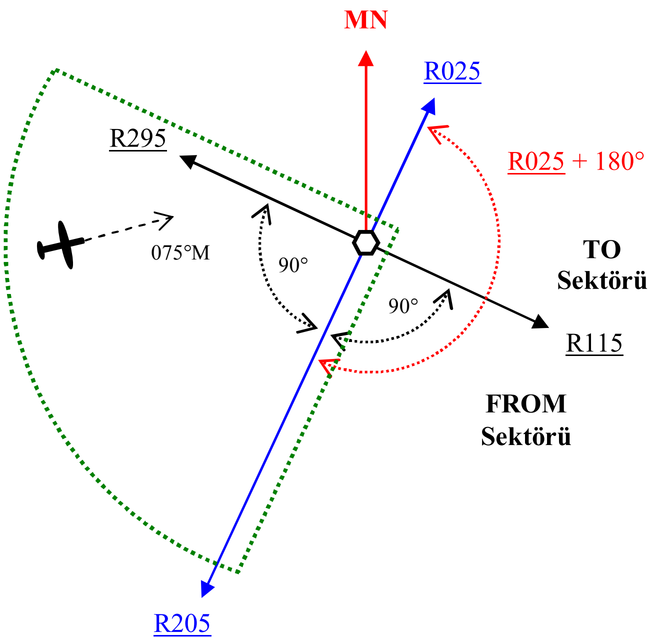
0576. Your aircraft is heading 075°M. The OBI is set to 025°. The VOR indications are "TO" with the needle showing right deflection. Relative to the stations , you are situated in a quadrant defined by the radials ;

a) 205° and 295°
b) 115° and 205°
c) 295° and 025°
d) 025° and 115°

Answer ;

a) 205° and 295°

Çözüm ;



OBI'de (Omni Bearing Selector) Course 025° seçilidir ve TO göstergesi ile birlikte ibrenin pozisyonun tam sağdadır. Burada uçuş başının bir önemi yoktur. Eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile aynı sektörde ise CDI'da görülecek ibare FROM olacaktır. Diğer durumda eğer uçağın bulunduğu radyal , Course olarak seçmiş olduğu radyal ile farklı sektörde ise CDI'da görülecek ibare bu sefer TO olacaktır. Görülen TO olduğuna göre uçağımız 205 radyal tarafında yer almaktadır.
İbre sağda gözüktüğüne göre halihazırda bulunduğu radyal seçmiş olduğu radyal + 180° alanı içinde değildir , yani uçak 205 radyalden daha büyük değerdeki bir radyaldedir , ancak bu da bulunduğu sektör itibarı ile maksimum 295 radyal olabilir.
Cevap ; 205° and 295°
